

Rotational Diagonal Kalman Linear Attention: Model, Lessons from Mamba-2/Mamba-3, and Production Implementation

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Abstract

Mamba-2 supplies a strong selective state-space baseline, but its SISO state transition is a scalar radial contraction: it cannot rotate a two-dimensional latent subspace. The installed Mamba-3 SISO implementation adds rotation economically by accumulating a bounded angular rate and applying the same rotary transform to queries and keys. Its most useful lesson is therefore not a particular state-space write rule, but a representation: pairwise rotation can be moved out of the recurrent state update and into a phase-rotated query/key basis.

Rotational diagonal Kalman linear attention (RDiag KLA, E2 clock) adopts that representation while retaining KLA’s diagonal Kalman covariance and innovation write. It pair-ties radial transition and process-noise parameters, uses a detached live radial statistic to set a bounded token/head clock, scans phase in a documented negative gauge, and isolates the phase VJP from radial dynamics. This note defines the model, the precise finite-precision implementation contract, and the tests used for promotion. The sealed E2 baseline implementation is commit `c3346feaeae1486cdbc71f82b4edeacc4198950`; its immutable production snapshot passed the complete one-H200 correctness and performance gate (117 core RDiag tests plus 54 E2 tests). Distributed training-system checks and the multi-node soak/attestation remain before the planned 750M DiagKLA production submission. The one-H200 result concerns implementation correctness and measured kernel/layer performance, not the eventual language-model quality of a long training run. A newer shared-DT investigation, including its unresolved production-qualification status, is reported in Section 10.

1 Scope and notation

Let B, T, H, K, V, D denote batch size, sequence length, number of heads, key dimension per head, value dimension per head, and model width. The rotary prefix has even width $K_r = 2P$, where P is its number of adjacent pairs. Tensors use the following shapes:

$$\begin{aligned} \mathbf{x} &\in \mathbb{R}^{B \times T \times D}, & \mathbf{q}, \mathbf{k}, \boldsymbol{\alpha}, \boldsymbol{\omega} &\in \mathbb{R}^{B \times T \times H \times K}, \\ \mathbf{v} &\in \mathbb{R}^{B \times T \times H \times V}, & \mathbf{e} &\in \mathbb{R}^{B \times T \times P}, \\ \boldsymbol{\tau} &\in \mathbb{R}^{B \times T \times H}, & \boldsymbol{\delta}, \boldsymbol{\phi} &\in \mathbb{R}^{B \times T \times H \times P}. \end{aligned}$$

The raw angle e_{btj} is shared across heads; the clock τ_{bth} makes the resulting increment head-specific. For an adjacent pair $(2j, 2j + 1)$, define

$$R(\vartheta) = \begin{bmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{bmatrix}, \quad R(\vartheta)^\top R(\psi) = R(\psi - \vartheta). \quad (1)$$

All equations below are per batch element unless a batch index is useful. “SISO” follows the Mamba implementation’s terminology; a head can still carry multiple value channels that share the same scalar state-space coefficients.

2 What Mamba-2 has—and what it lacks

For head h , value channel p , and SSM state coordinate n , the Mamba-2 selective SISO update can be written

$$d_{th} = \text{softplus}(\tilde{d}_{th} + b_h^d), \quad a_{th} = \exp(A_h d_{th}), \quad A_h < 0, \quad (2)$$

$$X_{t,h,p,n} = a_{th} X_{t-1,h,p,n} + d_{th} B_{t,n} u_{t,h,p}, \quad y_{t,h,p} = \sum_n C_{t,n} X_{t,h,p,n} + D_h u_{t,h,p}. \quad (3)$$

The input projection is ordered as $[z, x, B, C, dt]$. It emits one dt row per token and head; B and C are token-dependent state vectors. The factor a_{th} is a scalar radial contraction over the head state, and $d_{th} B_t u_t$ is its discretized injection. There is no angle projection, phase state, adjacent-pair structure, sine/cosine operation, or 2×2 rotation in this path. Mamba-2 therefore supplies selective decay and injection, but not oscillatory transport. A pair of real coordinates within a head shares the same scalar decay, but that factor is $a_{th} I$: it has no skew component, phase state, or mechanism for rotating one coordinate into the other.

3 The installed Mamba-3 SISO rotational path

This section describes the code shipped in the installed `mamba-ssm 2.3.2.post1` package, not an inferred variant. Its input projection is ordered

$$[z, x, B, C, dd_{dt}, dd_A, \text{trap}, \text{angle}]. \quad (4)$$

The dd_{dt} , dd_A , and trapezoid fields each have H rows. In contrast, the angle field has only P_M rows, where P_M is half of the selected half- or full-rotary state prefix. The raw angle rows are expanded across heads. Thus Mamba-3 separates a token/head clock from token/pair angular content:

$$DT_{th} = \text{softplus}(dd_{dt,th} + b_h^d), \quad (5)$$

$$A_{th} = \min\{-\text{softplus}(dd_{A,th}), -A_{\text{floor}}\}, \quad \rho_{th} = \exp(A_{th} DT_{th}), \quad (6)$$

$$\nu_{tj} = \pi \tanh(e_{tj}), \quad (7)$$

$$\Delta\theta_{thj} = DT_{th} \nu_{tj}, \quad (8)$$

$$\theta_{thj} = \theta_{0,hj} + \sum_{s=1}^t \Delta\theta_{shj} \text{pmod} 2\pi. \quad (9)$$

The cumulative phase is positive. Its Triton loop accumulates in FP32, but the combined training wrapper casts raw angles to BF16 and materializes phase and final carry in that input dtype; this is not an FP32-storage guarantee. For every selected adjacent pair, the SISO kernel applies the same $R(+\theta_{thj})$ to Q_t and K_t . Consequently the same-token dot product is unchanged, whereas a cross-token term obtains the relative rotation

$$(R(\theta_t) q_t)^\top R(\theta_s) k_s = q_t^\top R(\theta_s - \theta_t) k_s. \quad (10)$$

If S_t denotes a state in a fixed physical frame, this is equivalent to the moving-frame recurrence

$$S_t^- = \rho_t R(-\Delta\theta_t) S_{t-1}, \quad (11)$$

pair by pair, with scalar radial decay ρ_t . The identity follows from $R(\theta_t)^\top R(\theta_{t-1}) = R(-\Delta\theta_t)$. It is the key computational idea: no dense rotational state transition is needed when radial coefficients are pair-isotropic.

Mamba-3 additionally uses a trapezoidal SSM write. With

$$\gamma_t = DT_t \text{sigmoid}(\text{trap}_t), \quad \eta_{t+1} = DT_{t+1}(1 - \text{sigmoid}(\text{trap}_{t+1})), \quad (12)$$

the offline kernel scales token t 's rotated key by $\gamma_t + \eta_{t+1}$, while its same-token contribution uses γ_t . At a streaming boundary, the cached previous key/value contributes the missing η_t term. This is a two-endpoint discretization of the Mamba write; it is not the KLA innovation update defined below.

The angle clock is also coupled in backward. The VJP of the positive cumulative phase is a reverse cumulative sum. It produces both

$$\bar{e}_{tj} = \sum_h \bar{\Delta\theta}_{thj} DT_{th} \pi (1 - \tanh^2(e_{tj})), \quad (13)$$

$$\overline{DT}_{th}^{\text{angle}} = \sum_j \bar{\Delta\theta}_{thj} \pi \tanh(e_{tj}). \quad (14)$$

Here the sum over h is the backward of the head expansion of the shared raw angle rows. The combined implementation explicitly adds $\overline{DT}^{\text{angle}}$ to the direct trapezoidal-write DT cotangent; ordinary autograd subsequently adds the $A \overline{ADT}$ contribution through $ADT = A DT$. This is a coherent choice for a shared physical discretization clock, but it is intentionally not the RDiag E2 gradient contract. The Mamba-2/Mamba-3 comparison here is architectural, not a matched causal rotation ablation: Mamba-3 also removes Mamba-2's convolutional path, makes A token-dependent, normalizes and biases B/C , and changes the write discretization.

4 RDiag KLA with the E2 bounded clock

4.1 Pair-tied radial dynamics

Let the radial preactivation be f_{bthk} with bias b_{hk}^d , and let

$$d_{bthk} = \text{softplus}(f_{bthk} + b_{hk}^d). \quad (15)$$

For rotary pair j , define the pair step and tied process noise

$$\bar{d}_{bthj} = \frac{1}{2}(d_{bth,2j} + d_{bth,2j+1}), \quad (16)$$

$$\bar{\omega}_{bthj} = \frac{1}{2}(\omega_{bth,2j}^{\text{in}} + \omega_{bth,2j+1}^{\text{in}}), \quad (17)$$

$$\alpha_{bth,2j} = \alpha_{bth,2j+1} = \exp\left[-\exp(A_h^{\log})\bar{d}_{bthj}\right], \quad (18)$$

$$\omega_{bth,2j} = \omega_{bth,2j+1} = \bar{\omega}_{bthj}. \quad (19)$$

Only the selected prefix is pair-tied; tail lanes retain the diagonal-KLA frontend's original α and ω . Pair tying is not cosmetic. On each pair it makes $\text{diag}(\alpha_t) = \alpha_{tj} I_2$ and the diagonal process covariance $\text{diag}(\omega_t) = \omega_{tj} I_2$, so the *prediction* terms commute with $R(\vartheta)$. It does not force posterior information or the Kalman gain to be pair-isotropic; an observation generally splits them. The diagonal posterior approximation is defined in the current moving frame, as made explicit below.

4.2 Detached live radial clock and persistent anchor

Set the open upper clock bound to $C = 0.1$. At construction, snapshot one FP32 anchor per head,

$$a_h = \text{mean}_k \text{softplus}(b_{hk}^d), \quad 0 < a_h < C. \quad (20)$$

The anchor is a persistent buffer, not a parameter. At each token compute the detached live radial mean

$$m_{bth} = \text{mean}_k \text{softplus}\left(\text{stopgrad}(f_{bthk} + b_{hk}^d)\right), \quad (21)$$

and a learnable token/head residual

$$z_{bth}^{\text{clk}} = \frac{\langle w_h, x_{bt} \rangle}{\sqrt{D}} + \beta_h, \quad (22)$$

evaluated in FP32. The E2 clock is

$$\boxed{\tau_{bth} = C \text{ sigmoid}\left[\log m_{bth} - \log(C - a_h) + \tanh(z_{bth}^{\text{clk}})\right]} \quad (C = 0.1). \quad (23)$$

The bounded residual is strictly in $(-1, 1)$ and the sigmoid keeps a finite valid clock strictly in $(0, C)$. The implementation does not clamp or use a straight-through estimator: invalid, nonfinite, endpoint, or dead-derivative conditions are made visible to health checks, with invalid clock outputs replaced by NaNs.

Detachment is a model choice, not merely an optimization. The current radial timescale determines the forward phase scale, while the phase loss cannot change f , b^d , $A^{\log}$, α , or ω through this clock. The persistent anchor fixes the initialization reference across save/resume; recomputing it from a trained b^d would change the model after restart.

4.3 Angle, sign convention, and rotating basis

One projection shared across heads produces e_{btj} . The head-specific increment is

$$\delta_{bthj} = \pi \tanh(e_{btj}) \tau_{bth}, \quad |\delta_{bthj}| < 0.1\pi. \quad (24)$$

RDiag uses the negative inclusive phase scan

$$\phi_{bthj} = \phi_{b0,hj} - \sum_{s=1}^t \delta_{bshj}, \quad (25)$$

implemented in FP32 and reduced to $[0, 2\pi)$ at the bounded-clock consumer. The same $R(+\phi_{thj})$ is applied to the corresponding pairs of q_t and k_t . Hence, for $s < t$,

$$(R(\phi_t)q_t)^\top R(\phi_s)k_s = q_t^\top R\left(\sum_{u=s+1}^t \delta_u\right)k_s. \quad (26)$$

Under the column-vector convention of Eq. (1), the negative scan therefore represents $R(+\delta_t)$ transport. Mamba-3’s positive scan represents $R(-\Delta\theta_t)$. This sign is a gauge choice because replacing the zero-initialized angle projection by its negative maps one convention to the other. It is nevertheless an implementation invariant: forward, streaming carry, and the negative suffix-scan VJP must use the same sign.

More explicitly, let $\mathcal{R}_t$ be block diagonal with the pair rotations $R(\phi_{thj})$ and an identity tail. The corresponding physical transition and process covariance are

$$D_t = \mathcal{R}_t^\top \text{diag}(\alpha_t) \mathcal{R}_{t-1}, \quad \Omega_t^{\text{phys}} = \mathcal{R}_t^\top \text{diag}(\omega_t) \mathcal{R}_t. \quad (27)$$

On a tied pair, these reduce to $D_{tj} = \alpha_{tj} R(+\delta_{tj})$ and $\Omega_{tj}^{\text{phys}} = \omega_{tj} I_2$. With moving-frame state $\bar{S}_t = \mathcal{R}_t S_t$ and addresses $\bar{k}_t = \mathcal{R}_t k_t$, $\bar{q}_t = \mathcal{R}_t q_t$, the mean-state recurrence becomes purely radial and can reuse the diagonal-KLA memory kernel.

4.4 Diagonal Kalman gain and innovation memory

Let the phase-rotated addresses be $\tilde{q}_t = \bar{q}_t$ and $\tilde{k}_t = \bar{k}_t$. With posterior diagonal information $c_{t-1,hk} > 0$, observation noise $r_{th} > 0$, and information scale $s_h > 0$, RDiag KLA uses the moving-frame state (written as S_t below for brevity)

$$p_{thk}^- = \frac{\alpha_{thk}^2}{c_{t-1,hk}} + \omega_{thk}, \quad (28)$$

$$\kappa_{thk} = \frac{p_{thk}^- \tilde{k}_{thk}}{r_{th} + \sum_i p_{thi}^- \tilde{k}_{thi}^2}, \quad (29)$$

$$c_{thk} = \frac{1}{p_{thk}^-} + s_h \frac{\tilde{k}_{thk}^2}{r_{th}}. \quad (30)$$

For associative state $S_{th} \in \mathbb{R}^{K \times V}$,

$$S_{th}^- = \text{diag}(\alpha_{th}) S_{t-1,h}, \quad (31)$$

$$S_{th} = S_{th}^- + \kappa_{th} \left(v_{th} - \tilde{k}_{th}^\top S_{th}^- \right)^\top, \quad (32)$$

$$o_{th} = S_{th}^\top \left(K^{-1/2} \tilde{q}_{th} \right). \quad (33)$$

This is the KLA estimator. It is not a Mamba SSM write: κ_t is an adaptive anisotropic Kalman gain and the write is the current innovation. The diagonal information recurrence can be computed by its existing associative Möbius scan, followed by the existing chunked affine memory recurrence. In physical coordinates, c_t represents the model’s diagonal projection of $\mathcal{R}_t P_t^{-1} \mathcal{R}_t^\top$; it is not a claim that rotating an arbitrary anisotropic posterior covariance remains diagonal. In particular, c , κ , k , and q are never pair-tied.

4.5 VJP isolation

The computational graph has two intentionally separate branches:

$$(f, b^d, A^{\log}, \omega^{\text{in}}) \longrightarrow (\alpha, \omega) \longrightarrow \text{Kalman dynamics},$$

$$\text{stopgrad}(f + b^d), a, (W, \beta), e \longrightarrow \tau, \delta, \phi \longrightarrow \text{rotated } q, k.$$

In its radial-only specialization, the direct-pair backward has no angle data dependency or load. Conversely, the clock’s m and a arguments are detached, so the phase VJP returns no gradient to the radial gate, radial bias, or $A^{\log}$. Unlike Mamba-3, RDiag does not add an angular-clock contribution to the dynamical radial-step gradient. Among the new phase-specific parameters, gradients reach the angle projection and clock weight/bias; they also return to the shared hidden states through both projections. The isolation claim is specifically that no clock edge reaches the radial parameters.

5 What to take from Mamba-3

The useful transfer from Mamba-3 is structural: encode rotation as a relative query/key phase with a small shared angle projection and a token/head clock. The estimator-specific pieces should remain KLA-specific.

Aspect	Mamba-2 SISO	Mamba-3 SISO	RDiag KLA E2
Transition geometry	Scalar radial decay; no rotation	Radial SSM plus relative pair rotation	Pair-tied radial KLA plus relative pair rotation
Clock	$DT_{th} > 0$	$DT_{th} > 0$ shared by decay and angle	$0 < \tau_{th} < 0.1$, based on detached live radial mean plus residual
Angle rows	None	Token/pair rows shared across heads	Token/pair rows shared across heads; head-specific τ
Phase gauge	None	Positive cumulative phase	Negative cumulative phase; sign documented by Eq. (26)
Write	Euler-like $DT B x$	Trapezoidal two-endpoint SSM write	Kalman innovation with anisotropic gain
Clock gradient	Only SSM dynamics	Angle dDT is added to dynamical dDT	Phase/radial VJPs intentionally isolated
Covariance	No Kalman posterior covariance	No KLA diagonal information update	Diagonal information in moving frame; pair-isotropic prediction parameters

Concretely, RDiag should take the following from Mamba-3:

1. represent rotation by the same adjacent-pair transform on queries and keys, avoiding a dense recurrent rotation;
2. use a bounded angular rate $\pi \tanh(e)$ and one token/head clock broadcast over pair modes;
3. share the compact angle projection across heads, while permitting head-specific timing;
4. use FP32 phase arithmetic and, unlike the installed training wrapper, retain the materialized phase and streaming carry in FP32 under a specified modular reduction contract;
5. treat every dependency of a shared clock explicitly in backward and test cancellation-heavy reductions against the executable reference; and
6. retain half- or full-prefix rotary layouts rather than forcing every key lane to rotate.

It should intentionally *not* copy the following:

1. Mamba-2’s scalar-only transition as a purported rotational mechanism;
2. Mamba-3’s trapezoidal SSM write, because KLA’s innovation update defines a different estimator;
3. Mamba-3’s angle-to- DT gradient coupling, because E2 deliberately uses a detached radial timescale;
4. approximate softplus, trigonometric, or reassociated reduction sequences without proving the required finite-precision parity;
5. independent radial coefficients inside a rotated pair while claiming a literal scalar-damped rotation; they instead define a more general rotated anisotropic contraction; or
6. hard clock clamps or straight-through boundary gradients, which conceal saturation instead of making it observable.

6 Implementation contract

6.1 Layout and forward pipeline

The production-oriented route is restricted to contiguous CUDA tensors, BF16 projected $q, k, v, f_{\text{raw}}, z_{\text{raw}}, y_{\text{raw}}$, FP32 frontend parameters, equal key/value head counts, and $K \in \{64, 128\}$. The clock preactivation z^{clk} is a separate FP32 quantity. The rotary fraction is 1/2 or 1, so K_r is positive and even. Adjacent memory lanes $(0, 1), (2, 3), \dots, (K_r - 2, K_r - 1)$ form the rotary pairs; no split-half pairing is permitted.

The forward pipeline is:

1. project q, k, v and the diagonal-KLA radial/noise quantities;
2. compute the standard frontend’s frozen α^{in} and ω^{in} ;
3. call the unchanged direct-pair operator with a detached angle, retain its pair-tied α, ω , and log-decay prefix, and discard its legacy phase output;
4. independently materialize the precise detached FP32 $[B, T, H, K]$ clock steps and form the exact native-framework radial mean m described below;
5. keep Wx as the existing FP32 GEMM, then fuse the pointwise clock chain and Eq. (24), using the exact CUDA-eager sigmoid quotient;
6. perform the bounded negative phase scan in chunks of 64, retaining an FP32 carry;
7. normalize q, k in FP32, round the memory-facing copies as specified, and rotate adjacent pairs by the same phase; and
8. run the diagonal information/gain kernel and the S3 chunked KLA memory recurrence.

The gain-facing key remains FP32. The memory-facing query/key route is rounded through BF16 in the same place as its executable reference before rotation/final rounding; changing that boundary is a numerical model change for parity purposes.

6.2 The precise detached-clock mean

The actual radial dynamics retain the FLA softplus used by the qualified baseline. The detached clock uses a separate precise FP32 copy

$$\text{softplus}_{20}(x) = \begin{cases} x, & x > 20, \\ \log 1p(\exp x), & x \leq 20, \end{cases} \quad (34)$$

evaluated with the precise device intrinsics. To reproduce the eager reference, the optimized path must materialize

$$\text{clock_dt} \in \mathbb{R}^{B \times T \times H \times K} \quad (35)$$

in FP32 and then invoke the framework reduction

$$\text{radial_clock} = \text{torch.mean}(\text{clock_dt}, \text{dim}=-1, \text{out}=\text{radial_clock}). \quad (36)$$

A mathematically equivalent Triton lane sum is not an exact substitute. It changes addition order and was shown to leave small residual differences that can be amplified by long phase scans and cancellation-heavy angle gradients. The precise clock copy is detached; the FLA-softplus radial outputs and their VJP remain untouched. The temporary costs exactly $4BTHK$ bytes. At

$B = 4, T = 4096, H = 16$, that is 64 MiB for $K = 64$ and 128 MiB for $K = 128$; at $H = 18, K = 128$ it is 144 MiB. This cost belongs in the end-to-end memory and speed gate rather than being hidden by a numerically different reduction.

6.3 Reusing the direct-pair radial path safely

The production wrapper does not maintain a second implementation of Eq. (19). It calls the established direct-pair transform with `angle_raw.detach()`, ignores the returned legacy delta, and reuses only g , α , and ω . This keeps the radial forward arithmetic and its VJP identical to the already-qualified direct-pair path.

Discarding the legacy delta makes its incoming cotangent absent. The direct-pair backward turns that fact into the compile-time specialization `HAS_DDELTA=false`. In this specialization the generated kernel does not load the detached angle, evaluate its tanh, or form the phase term $\pi \tanh(e) \bar{\delta}$. The guard is semantically important: merely setting $\bar{\delta} = 0$ would not isolate a nonfinite discarded angle, because IEEE arithmetic gives $0 \times \text{NaN} = \text{NaN}$. The phase-enabled `HAS_DDELTA=true` algebra is unchanged, while dedicated $\text{NaN}/\pm\infty$ tests establish that a discarded phase cannot contaminate radial outputs or gradients.

6.4 Exact sigmoid and bounded-delta forward

Let

$$\ell_{bth} = \log m_{bth} - \log(C - a_h) + \tanh(z_{bth}^{\text{clk}}). \quad (37)$$

For FP32 CUDA inputs, PyTorch’s native sigmoid kernel evaluates $1/(1 + \exp(-\ell))$. The generic Triton `tl.sigmoid` uses a different approximate lowering and produced one-ULP clock differences even when ℓ was bitwise identical. The production kernel therefore spells out the eager expression as

$$\tau_{bth}^{\text{raw}} = 0.1 \text{tl.div_rn}(1, 1 + \text{libdevice.exp}(-\ell_{bth})), \quad (38)$$

then applies the fail-visible validity predicate to obtain τ . The angle nonlinearity is computed once by native ATen as `torch.tanh(angle_raw.float()).contiguous()` and passed to the fused clock kernel, which forms $\delta = \pi \tanh(e) \tau$. This boundary makes the actual-logit τ and δ bitwise equal to the eager oracle, rather than merely close in real arithmetic.

6.5 Phase scan and backward order

The phase forward is an inclusive FP32 scan. Each chunk computes $\phi = \text{carry} - \text{cumsum}(\delta)$, wraps valid values to $[0, 2\pi)$, and passes the wrapped final value as the next carry. Before reduction, $|\phi|$ must not exceed 65,536; nonfinite, out-of-contract, or noncanonical residues become sticky NaNs. Its VJP is the global negative suffix scan

$$\bar{\delta}_t = - \sum_{u=t}^T \bar{\phi}_u, \quad (39)$$

with the final-carry cotangent subtracted from every token when present.

For the fused clock/delta backward, operation order is part of the contract. Let $g_{thj} = \bar{\delta}_{thj}$. The

eager-compatible order is

$$\bar{\tau}_{th} = \sum_j [g_{thj}(\pi \tanh(e_{tj}))], \quad (40)$$

$$\bar{z}_{th}^{\text{clk}} = \text{mask}_{th}(\bar{\tau}_{th}) \tau_{th}^{\text{raw}} \left(1 - \frac{\tau_{th}^{\text{raw}}}{C}\right) \left(1 - \tanh^2 z_{th}^{\text{clk}}\right), \quad (41)$$

$$\bar{e}_{tj} = \left[\left(\sum_h g_{thj} \tau_{th} \right) \pi \right] (1 - \tanh^2 e_{tj}). \quad (42)$$

Here $\text{mask}_{th}(g)$ is g when the forward validity predicate selected τ_{th}^{raw} and zero when it selected the detached NaN branch. The subsequent multiplication order is retained, so a NaN local derivative remains fail-visible even after a zero cotangent mask. Equation (40) must be evaluated as the displayed broadcast multiply followed by the framework’s sum over P ; pulling π outside the sum is not accepted as “equivalent.” For Eq. (42), multiply by τ , reduce over H , multiply by π , and only then apply the tanh derivative. Concretely, production saves the native-ATen tanh output from the forward and evaluates

```
torch.ops.aten.tanh_backward( (d_delta * tau[... , None]).sum(dim=2) * pi, angle).
```

The result is cast to `angle_raw`’s dtype (BF16 on the production route) before the angle projection VJP. No Triton angle-partial tensor or head-reduction kernel remains. The validity mask is applied to the incoming clock cotangent before the sigmoid/tanh clock chain, matching the eager `where(valid, raw_tau, detached_nan)` VJP. Weight and hidden-state gradients then flow through the ordinary linear operations.

6.6 Persistent state and FSDP

The anchor a_h must satisfy all of the following:

1. FP32, contiguous, shape $[H]$, finite, and strictly inside $(0, 0.1)$;
2. registered as a persistent buffer and present in the model state dictionary;
3. absent from named parameters and optimizer groups;
4. identical across distributed ranks after construction, FSDP wrapping, save, and resume;
5. restored from the checkpoint rather than regenerated from the current radial bias.

Production enforces this with `bounded_rdiag_fsdp_mixed_precision`: it retains the installed BF16-mixed parameter/reduction policy but overrides `buffer_dtype=torch.float32`. Resume tests compare the anchor bytes before and after round trips and verify that fresh construction cannot silently overwrite a checkpointed anchor.

6.7 Initialization and health signals

The angle projection weight and bias, and the bounded-clock residual weight and bias, are initialized to zero. Rotation therefore starts at the *pair-tied* diagonal-KLA function: $e = 0$ gives $\delta = 0$ even when τ is nonzero, while the pair averaging in Eq. (19) remains active. This is preferable to initializing the clock at zero, which would place it on a boundary and kill useful clock gradients once angles become live.

The audited bounded RDiag named configurations use $r_{\min} = 0.01$, process-noise floor $\omega_{\min} = q_{\min} = 0$, and `gain_init=iso`. The last setting zeroes the output map of the process-noise projection,

so ω starts uniform across key channels; it does *not* initialize ω itself to zero, because the positive softplus remains. These are recipe and initialization choices rather than changes to Eqs. (28)–(33).

At every checked optimizer boundary, diagnostics should aggregate across microbatches, layers, and ranks:

- nonfinite counts for the radial mean, base logit, residual, logit, raw clock, valid clock, derivative, phase, parameters, and gradients;
- exact endpoint and near-boundary counts, using 10^{-4} and $0.1 - 10^{-4}$ as the observed lower and upper near-boundary bands;
- minimum positive clock Jacobian and a near-dead count below 10^{-8} ;
- token variation of the residual projection after bootstrap;
- phase magnitude, delta magnitude, and canonical wrapped-range checks; and
- schema/count agreement across ranks, including exactly one observation per expected microbatch/layer pair.

The kernel itself fails visibly at invalid endpoints. Promotion screens may impose stricter distributional thresholds than the continuous health controller; both classes of gate are needed.

7 Reference pseudocode

The following pseudocode fixes the mathematical order without prescribing launch geometry.

1. Radial and clock preparation.

```

 $x^d = f + \text{dt\_bias},$ 
 $d^{\text{radial}} = \text{FLA\_softplus}(x^d),$ 
 $\text{clock\_dt} = \text{softplus}_{20}(\text{stopgrad}(x^d)) \quad \text{materialized FP32 } [B, T, H, K],$ 
 $m = \text{torch.mean}(\text{clock\_dt}, \text{dim}=-1, \text{out}=\text{radial\_clock}).$ 

```

Invoke the unchanged direct-pair transform with a detached angle, pair-average d^{radial} and ω on the rotary prefix, and construct α from Eq. (19). Discard its legacy delta and do not use `clock_dt` for radial dynamics.

2. Bounded clock and phase.

```

 $z^{\text{clk}} = \text{linear}(x, W) / \sqrt{D} + \beta,$ 
 $\ell = \log m - \log(0.1 - a) + \tanh z^{\text{clk}},$ 
 $\tau = 0.1 \text{div\_rn}(1, 1 + \text{libdevice.exp}(-\ell)),$ 
 $\hat{e} = \text{torch.tanh}(e.\text{float}()),$ 
 $\delta = \pi \hat{e} \tau,$ 
 $\phi_t = \text{wrap}_{2\pi}(\phi_{t-1} - \delta_t).$ 

```

Reject invalid open-clock or phase-contract values by producing observable NaNs.

- ### 3. Addresses and Kalman recurrence.
- Rotate the same adjacent prefix of normalized q and k by $R(\phi)$, preserve tail lanes, then apply Eqs. (28)–(33). The chunked implementation must match this token recurrence under the documented BF16/FP32 boundary.

4. **Backward.** Run Eq. (39); use Eqs. (40)–(42) exactly; evaluate the angle derivative with native `aten.tanh.backward` on the saved $\hat{e}$; compile the discarded direct-pair phase branch with `HAS_DDELTA=false`; return no clock gradient to m or a ; and run the independent radial/Kalman VJPs.

8 Lessons from the E-screen and performance/numerical debugging

An open interval is not a distributional guarantee. An earlier bounded-clock E screen completed 519 optimizer updates and produced no exact endpoints or nonfinite clocks, yet the step-520 evaluator rejected it. In layer 0, the lower-near-boundary fraction was 0.00205326, above the preregistered 10^{-3} threshold; the observed clock range was approximately $[8.65 \times 10^{-5}, 0.09813]$. Thus a smooth sigmoid bound alone did not prevent probability mass from accumulating near a boundary. The design response was the detached live radial base plus a bounded token/head residual, not a harder clamp.

The redesigned E2 screen reached its step-520 evaluation after 519 completed optimizer updates (520 measured gradients), with zero lower and upper near-boundary counts in the reported gate and an observed clock range of approximately $[0.01060, 0.08181]$. That result supports the clock design and justified integration work. It does not prove long-run training quality, kernel parity under every shape, distributed resume correctness, or a production speedup.

Correctness-first composition exposed the actual performance target. At $B = 4, T = 4096, H = 16, K = V = 128, D = 2048$, the unfused rotational-v1 wrapper measured 24.706 ms for combined mixer forward/backward versus 13.735 ms for its frozen control, a $1.799\times$ ratio (79.9% overhead). The identical-input frozen S3 stage was $0.999\times$ its control, localizing the cost to the eager second- dt , pair tying, phase/trigonometry, three rotations, and prefix rebuild. This motivated the direct radial kernel, fused normalization/rotation, and fused clock/delta path while retaining the established gain and S3 kernels. These are historical isolation numbers for the v1 geometry, not a speed claim for the final E2 composition. The later final A/B measurements, including the exact-mean FP32 scratch, are reported in Table 1.

A tiny local error can become a large gradient discrepancy. The first optimized clock path reused the FLA radial softplus. Against eager PyTorch, the initial radial-mean discrepancy was only about 6.33×10^{-8} in maximum absolute value and 1.54×10^{-6} in relative L_2 . After the logit, long phase scan, BF16 rotation boundary, and cancellation across heads, the angle-weight gradient showed about 3.81×10^{-3} relative L_2 error in the focused diagnostic. Near-zero reference entries also made elementwise relative metrics misleading, so absolute, normwise, and cancellation-stress gates are all required.

Using Eq. (34) reduced the clock error substantially, but a Triton reduction still did not reproduce eager bitwise. Materializing the FP32 $[B, T, H, K]$ precise values and using the exact framework mean was bitwise identical in the focused $K = 64$ and $K = 128$ experiment while leaving the FLA radial α , log-prefix, and ω outputs bitwise unchanged. Likewise, algebraically reassociated all-Triton backward reductions failed the focused VJP gate; the explicit framework reduction of Eq. (40) and the factor order of Eq. (42) passed it. The lesson is narrow but important: the executable eager graph, including reduction order, is the reference for this mixed-precision path.

Elementary functions are also part of the executable reference. After the exact mean fix, a full $K = 64$ trace found bitwise-identical radial clock, base logit, bounded residual, and final logit, yet the fused clock still differed from eager at 61,655 of 262,144 FP32 τ entries (maximum absolute error 7.45×10^{-9}). Those one-ULP clock differences propagated to 929,205 of 4,194,304 delta entries and were amplified by the length-4096 phase scan. The cause was `tl.sigmoid`, not the model equations. Replacing it with Eq. (38), matching PyTorch’s CUDA source expression and rounded division, removed the actual-logit discrepancy. Computing the angle tanh and its backward with native ATen then removed the remaining reduction/transcendental-order drift. This is why the final path uses selective eager operations at two small boundaries rather than insisting that every operation be fused into Triton.

9 Invariants and qualification status

The following are invariants, not tunable approximations:

1. rotary prefix width is positive and even; pairing is adjacent;
2. α and ω are exactly equal within each selected pair;
3. q and k receive the same phase and same rotation matrix;
4. the phase scan is negative and inclusive, and its backward is a negative suffix scan;
5. m and a are detached FP32 quantities; phase gradients do not enter radial dynamics;
6. the valid clock and its local derivative are finite and strictly positive, with $0 < \tau < 0.1$;
7. the anchor is persistent FP32 checkpoint state, not optimizer state;
8. precise clock softplus plus `torch.mean` is separate from FLA radial softplus;
9. the bounded sigmoid uses Eq. (38), not `tl.sigmoid`;
10. the displayed forward and backward reduction orders are preserved, with the angle tanh and its VJP evaluated by native ATen;
11. a discarded direct-pair delta selects `HAS_DDELTA=false`, eliminating every angle access and phase product from the radial backward; and
12. the unrotated tail is an exact pass-through of diagonal KLA.

The one-H200 kernel/layer checklist is closed by the cumulative qualification campaign:

1. pure-PyTorch token recurrence versus optimized forward for zero, half, and full rotation;
2. first-order VJPs for every input/parameter, including adversarial cancellation and partial chunks, at $K = 64$ and $K = 128$;
3. exact checks for radial outputs and detached clock mean where exactness is part of the contract, plus justified tolerances elsewhere;
4. segmented/streaming equivalence, phase carry, wrap-boundary, and sticky-NaN tests;
5. zero-angle equivalence to pair-tied diagonal KLA and angle-sign/gauge unit tests;
6. multi-step health checks covering bootstrap-zero angle, subsequently live clock gradients, token variation, and no nonfinite/endpoints;

7. exact full-layer forward/VJP comparison with the sealed E2 candidate at $K = 64$, $K = 128$, and the 18-head 1.3B-style geometry; and
8. interleaved A/B timing against both the sealed E2 candidate and the legacy pair clock.

The authoritative one-H200 job 1777777 completed with exit code 0 and reported 117 passing core RDiag tests followed by 54 passing E2 tests. Table 1 gives the paired runtime ratios for an exact RDiag-layer forward plus backward; values below one are faster than the named reference. Every upper endpoint of the bootstrap 95% confidence interval is below the 1.05 promotion limit.

Table 1: Final H200 paired A/B timing from job 1777777, with 40 measured rounds after 20 warmups. “Accepted” is the sealed eager E2 candidate and “legacy” is the production pair-*dt* path.

Geometry (H, D, K)	E2/accepted	95% CI high	E2/legacy	95% CI high
(16, 1024, 64)	0.969512	0.978247	1.014135	1.020172
(16, 2048, 128)	0.950407	0.956546	1.008590	1.013666
(18, 2304, 128)	0.960111	0.964929	1.013622	1.014984

Thus the optimized E2 implementation is bitwise faithful where exact parity is required, faster than the sealed accepted implementation in all three measured geometries, and within roughly 1–1.4% of the simpler legacy clock. This closes the E2 one-H200 kernel/layer gate, but not a full production-submission gate. Train-smoke, distributed checkpoint/resume, and multi-node soak/attestation remained for that baseline at the time of sealing. The next section records the later shared-DT candidate and its separate, still-open qualification.

10 Shared-DT engineering status as of September 6, 2026

This section records a newer production candidate and does not revise the closed E2 result above. The candidate starts from commit `6472a0711eb61177c3ef91845855fffa82202290` and replaces E2’s detached bounded clock with a shared-DT transition. Its qualification is *not complete*. Four gates passed in the first campaign, whose gradient gate exposed both a defective acceptance rule and real rotational-gradient amplification; that soak was cancelled before allocation. The subsequent reviewed head-scaled commit passed all five non-soak gates, but its 3300-update soak failed closed with an aggregate count at boundary 269. A divergent diagnostic replay independently exposed rejection of the finite-input exact-zero decay endpoint. The narrow lower-endpoint correction then passed all five non-soak gates and 675 soak updates before failing closed at boundary 676 with an unrecorded key. A second divergent replay independently exposed rejection of the finite-input exact-one endpoint; the reviewed upper-endpoint correction passed its fresh numerical H200 suite, but its evidence could not be sealed because a pre-existing asynchronous log-finalization race omitted the terminal PASS marker from the retained log. The narrow source-gate logging correction was reviewed, committed, and published. Its fresh campaign passed and sealed the H200 gate and all four short distributed gates, then its soak crossed both old boundaries and failed closed at update 790 on a rank-equal pre-clip global norm that triggered the combined nonfinite, negative, or greater-than-10 rejection predicate. The failed run did not retain which branch fired. No soak seal or aggregate authorization exists; the diagnostic replay and threshold-provenance analysis are recorded below.

10.1 Relation to Mamba-2 and Mamba-3

The source-level comparison in Sections 2 and 3 found no discrepant transition equation or kernel sign bug in the shared-DT path. Its radial factor is the Mamba-2 form

$$\alpha_{th} = \exp[-\exp(A_{\log,h})DT_{th}], \quad (43)$$

and its bias-free pair-angle projection follows the installed Mamba-3 construction with an additional versioned head scale. For the b37 v2 candidate,

$$e_{tj} = (W_{\text{angle}}x_t)_j, \quad e_{tj}^{\text{eff}} = 0.5H^{-1/2}e_{tj}, \quad \delta_{thj} = \pi \tanh(e_{tj}^{\text{eff}})DT_{th}, \quad \phi_{thj} = -\sum_{s \leq t} \delta_{shj}, \quad (44)$$

with one angle value per token/pair broadcast over heads, a headwise DT , a cumulative phase scan, and the corresponding negative reverse-suffix backward. This is the literal RDiag source convention; installed Mamba-3 uses the positive scan. The two parameterized model families are related by the gauge transform $W_{\text{angle}} \mapsto -W_{\text{angle}}$; exact continuation of an Adam checkpoint also negates that parameter’s first moment. This rules out a structural sign error, but flipping only the scan sign at one fixed, untransformed checkpoint can change outputs and gradient norms. The relevant difference from the E2 model is instead that the angle and DT paths remain live and the same shared DT controls radial decay and rotation. Writing $s_H = 0.5H^{-1/2}$ for v2 and bars for cotangents, the exact phase branch contributes

$$\bar{\delta}_{thj} = -\sum_{u \geq t} \bar{\phi}_{uhj}, \quad (45)$$

$$\bar{e}_{tj} = s_H \pi \operatorname{sech}^2(s_H e_{tj}) \sum_h DT_{th} \bar{\delta}_{thj}, \quad (46)$$

$$\overline{DT}_{th}^{\text{phase}} = \pi \sum_j \tanh(s_H e_{tj}) \bar{\delta}_{thj}. \quad (47)$$

The first expression exposes the reverse suffix accumulation, the second the head reduction, and the third the phase contribution added to the radial DT gradient. Existing aggregate category norms do not separately observe those three effects.

10.2 Issue, diagnosis, and completed work

Two distinct problems were found. First, the original per-microbatch global-gradient limit of 10 was not a valid gate: all 24 measured model/batch observations exceeded 10, including stock DiagKLA at 14.001–14.232. Stock DiagKLA and the bounded-clock layer were also architecture-mismatched controls. Second, matched interventions localize genuine amplification to the live shared-angle cumulative-rotation branch. Its output is broadcast over recurrent heads and its VJP contains both a head reduction and a reverse temporal suffix sum, but the retained evidence does not yet separately measure head coherence, suffix growth, or the phase contribution to the shared-DT gradient. This is evidence for a model-parameterization or optimization issue, not for a faulty radial or phase kernel, while the exact causal decomposition remains open.

On the same eight real FineWeb batches, the unmodified shared-DT model produced global norms 34.596–45.300 (median 38.758), versus a DiagKLA median of 14.096. Disabling rotation returned the median to 14.180, while detaching only the angle path gave 18.035 and zero angle initialization still gave 25.593. A phase-only DT cap of 0.1 was insufficient. Scaling the effective angle preactivation by $1/\sqrt{16} = 0.25$ gave median 16.765 and shared/Diag median and 95th-percentile

nearest-rank ratios 1.191 and 1.261, with live angle and DT gradients. These interventions localize amplification to head-shared cumulative rotation, but do not alone identify which of its head sum, suffix accumulation, angle VJP, or phase-to- DT VJP is sufficient for the late instability.

Completed engineering work. The Mamba-2/Mamba-3 source audit, exact H200 measurement, rotation/gradient ablations, matched- backbone control, and scale/cap sweep are complete. The replacement design and implementation plan received an independent review with no remaining Important findings. The layer-side $H^{-1/2}$ change and its nonpersistent-state host contracts are now implemented in the isolated checkout below: 41 host tests passed. Development H200 job 1827839 passed the focused H=18 all-parameter oracle/VJP case and H=16, T=4096 BF16 boundary/backward case; job 1827905 then passed the full 109-test production-H200 suite. These runs exercised the mutable development checkout, so they are correctness evidence rather than immutable qualification seals.

The same checkout now also contains the complete gradient-v3 control closure: both arms construct the same 220M configuration, strictly load one full reference state, prove matching configuration, parameter-inventory, and full-state digests, and reset CPU/CUDA randomness symmetrically. The no-rotation arm installs connected-zero hooks only on the twelve angle projections, checks all twelve angle-weight gradients are present, finite, and exactly zero for every batch, checks the shared radial gradient norms remain finite and positive, and proves hook removal. The v3 recipe, catalog digest, result attestor, stale-schema rejection, and aggregate-v3 recursive requirement are integrated. The focused gate-runner host suite passed 38 tests; the complete stable host sweep is also complete: the qualification-control suite passed 71 tests with one filesystem-capability skip (the skipped case passed separately on local temporary storage), and the broader thirteen-file shared-DT host sweep passed 418 tests with three expected skips.

In campaign 9870d362bcbb4ec3b9b3cafb13459ef8, production-H200 (job 1824028), train-smoke (1824365), checkpoint-fault (1824367), and authenticated-resume (1824368) produced recursively valid seals with SHA-256 prefixes ecc03259, 9ca6354e, 06c1f60a, and 656c9412, respectively. Gradient job 1824370 failed the invalid absolute-10 rule and produced no seal; the soak was cancelled before allocation. These seals characterize the unmodified base commit only and do not qualify the proposed head-scaled change.

Qualification outcome. The implementation commit passed separate exact-SHA specification/math and code-quality reviews without a finding and was pushed to the branch named below. Its immutable snapshot contains 4227 files and has source-manifest SHA-256 5b6b80bf8c38f4c959ac1f908a10fc01a25ff6986e77f0ee1c0a35de6843fbd9. Fresh campaign a65918f1771446e7acfbffcc3e5c7dc3 is closed incomplete. Production-H200 gate job 1828250 passed all 109 tests and attestor job 1828252 published seal SHA-256 prefix a05c99d0544a721a. The other four short gates and their attestors passed. Gradient A/B job 1828811 measured shared/no-rotation median and 95th-percentile ratios 1.179 and 1.262, inside the 1.25/1.50 limits; attestor 1828820 published seal SHA-256 prefix 0eabab83a02ac397. Checkpoint-fault jobs 1828813/1828823, authenticated-resume jobs 1828814/1828821, and train-smoke jobs 1828815/1828822 published seals with SHA-256 values prefixes a3558a17796d427c, 1ea7a7da40576d20, and c2e19737cff63305, respectively. They are valid partial evidence but cannot authorize the commit without a valid soak seal.

The production soak did *not* pass. Job 1828812 trained normally through completed update 268 (loss 5.356, global gradient norm 1.101, and zero reported health failures), then failed closed at the pre-optimizer boundary for update 269 with aggregate health-failure count 158720. It therefore

emitted neither a result nor a seal, and dependent attestor 1828824 was cancelled. The exception predated the detailed failed-counter breakdown. Its stage excluded observation/parameter/gradient coverage, gradient-nonfinite, and bootstrap-liveness failures, but could not distinguish the remaining forward-domain predicates from parameter non-finiteness or collective-structure disagreement. That ambiguity motivated the fail-closed diagnostic replays below.

The first diagnostic replay identified the candidate predicate. Non-authorizing job 1829026 ran on a different H200 node pair and observed its first exact FP32 zero earlier, at boundary 241, so the exact-boundary harness correctly exited nonzero rather than labeling the run a reproduction. Every one of the 16 ranks nevertheless reported the same sole failed key, `alpha_nonpositive_count=128`: one scalar $[B, T, H]$ decay site materialized over the $K = 128$ key width. All other failure counters were zero; the aggregate diagnostics included $\min \alpha = 0$, $DT_{\max} = 7.3498993$, and $\max |\tanh(e^{\text{eff}})| = 0.9377124$. The retained scheduler log is `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_sdt_soak269_diag_1829026.log`, with SHA-256 `5dc12cd3ea0ea52765495763b28bd0185b038c306bad6c333fabb537eb67e6ee`. This is evidence for the same candidate failure mode, not an exact reproduction of the original boundary-269 trajectory. The original aggregate count $158720 = 1240 \times 128$ is consistent with the same width replication, but its exception did not retain a per-key breakdown. Intermediate job 1829217, from commit `28531cf959304ab26b22789f2ea4ce082adbed13`, remained healthy through boundary 269 on yet another node pair and correctly failed closed without a summary when its search condition was not met. Its source-manifest and scheduler-log SHA-256 values are respectively `0dee86b8f758f14aabd78047e2d0d530475be2d29f99d27587c256233644c73f` and `142120433da0eddb643d6d6160974c756397ed730fe802c009fbd1eaed1b3924`.

Final non-authorizing search job 1829305 ran from diagnostic commit

`06eaf237f5be9f79ee8d00075b79dcc0e7a9e96a`.

It completed 277 optimizer updates and stopped before clipping or update 278 with status `SAME_FAILURE_MODE` and `authorizing=false`. All 16 ranks agreed on the sole production-predicate count `alpha_nonpositive_count=128`; removing the repeated key-width dimension leaves one unique scalar decay site. Every recorded $A_{\log}$, $\exp(A_{\log})$, and $\exp(A_{\log})DT$ value was finite. Their global ranges were respectively $[0.0472397, 2.7735384]$, $[1.0483732, 16.0152016]$, and $[0.000105659, 104.639008]$. The largest exponent exceeds the FP32 round-to-nearest zero threshold of approximately 103.972; 87.3365 is only the normal-to-subnormal onset. Thus finite inputs to $\exp[-\exp(A_{\log})DT]$ can produce the observed stored zero.

The retained summary is

`diagnostic_runs/soak269_replay_06eaf237f5be_1829305/
artifacts/replay-summary.json`,

with SHA-256 `558a1dde9c842b95d6afaafb16c54994f8e75aae68b7fed0bfeef11cfa4718fb5`; its trajectory has SHA-256 `eeb607f58f40a156e4fb42df899ef27f724a17c3ede5f85906395a4917692614`. The scheduler log has SHA-256 `c46c802ac2b73fa875a40699ce2ac96b2f32bff61c1e128a54625e8916264b03`. The replay compared 268 original updates, retained nine additional updates, and did not exactly match the original prefix; it therefore establishes the floating-point endpoint and candidate failure mode, not an exact boundary replay. The harness temporarily exempted only `alpha_nonpositive_count` so that it could retain this evidence and still stopped before the update; its logged zero health-failure count is not a production pass.

The evidence classifies the reproduced zero-alpha endpoint as a health-policy false positive rather than a model or kernel math defect; because the original exception lacks a per-key breakdown, it does

not prove that job 1828812 had identical causality. At original update 268, 583512 distinct scalar decay sites were already below the diagnostic 10^{-6} floor while loss and gradients remained finite. The S3 prefix clamps this range to 10^{-6} and assigns the below-floor branch zero cotangent; the Kappa recurrence is also defined at raw $\alpha = 0$. The narrow correction keeps `alpha_nonpositive_count` as telemetry, adds `alpha_negative_count` as the fatal domain predicate, and leaves the model’s alpha and both recurrences unchanged. At commit 83118d2..., alpha nonfinites and $\alpha \geq 1$ remained fatal.

One login-namespace preparation attempt failed closed on the intentionally strict live-data identity check, and the first compute-side preparation attempt (job 1828245) failed before manifest publication because its three mutable roots had not been pre-created. Preparation was rerun in the Slurm user namespace with those private roots fixed at mode 0700; neither failed attempt executed a numerical gate.

Diagnostic evidence. The primary metric artifacts are

- `diagnostic_runs/gradient_norm_6472a071_20260902T170943Z/metrics.json`, SHA-256 9c6a8a96a297799a;
- `diagnostic_runs/gradient_rotation_ablation_6472a071_20260902T183817Z/metrics.json`, SHA-256 274abd1902e7c7d3;
- `diagnostic_runs/gradient_matched_control_6472a071_20260902T185000Z/metrics.json`, SHA-256 d908637092fdcf3e; and
- `diagnostic_runs/gradient_angle_scale_6472a071_20260902T185100Z/metrics.json`, SHA-256 e503ab386a9de79d.

All are rooted at `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1`.

10.3 Approved implementation and remaining qualification

Chosen change. For H recurrent heads, the development layer derives the nonpersistent scalar $H^{-1/2}$ and forms

$$e_{tj}^{\text{eff}} = H^{-1/2}(W_{\text{angle}}x_t)_j, \quad \delta_{thj} = \pi \tanh(e_{tj}^{\text{eff}})DT_{th}. \quad (48)$$

The projection remains bias-free and its stored weight is unchanged. In the BF16 production path, the layer performs the multiply on the BF16 projection output before the frontend’s FP32 cast and FP32 `tanh`; the frontend must not scale a second time. The scale is neither a parameter nor a buffer, so old state dictionaries remain strictly loadable. This preserves the transition family and both learning paths while normalizing the initial head-summed VJP.

The replacement gradient gate is an exact same-model/same-state two-arm intervention. The `shared_dt` arm runs normally; `shared_dt_no_rotation` replaces only each angle- projection output by the connected zero `output * 0.0`. Config, full state, parameter inventory, and radial shared-DT path must be identical. Acceptance requires shared/no-rotation nearest-rank median ≤ 1.25 and 95th percentile ≤ 1.50 , both inclusive. The microbatch runaway sentinel becomes an inclusive 100; the late-training soak limit of 10 is unchanged. Only the gradient sample/result/recipe/seal closure advances to schema v3. Aggregate-v3 retains its schema name but must recursively require the v3 gradient seal.

Head-scale checkout and plans. Implementation is isolated from the main checkout in this worktree:

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_headscale_20260902.
```

The observed checkout is branch `codex/rdiag-shared-dt-angle-headscale-20260902`, based on the commit above. The clean reviewed and published implementation snapshot is

```
a76dc7b930ae4d85157391d444c40ef7db14ef86,
```

with tree `4cdb6287080365089349f56b14f1d782202d6157`. The reviewed design/plan snapshot is commit

```
9e621b5ede53529df50619804f0c9d8d32707284.
```

Its documents are

- `docs/plans/2026-09-02-rdiag-shared-dt-angle-headscale-design.md`; and
- `docs/plans/2026-09-02-rdiag-shared-dt-angle-headscale-plan.md`.

The primary implementation and qualification paths are

- `lit_gpt/rdiag_shared_dt_kla.py`;
- `lit_gpt/kla_ops/rdiag_shared_dt.py` and `lit_gpt/kla_ops/rdiag_bounded_phase.py`;
- `scripts/tests/rdiag/test_rdiag_shared_dt_production_host.py` and `scripts/tests/rdiag/test_rdiag_shared_dt_production_h200_v1.py`;
- `scripts/tests/rdiag/test_rdiag_shared_dt_training_health_host.py`;
- `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab.py` and `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab_run.py`;
- `scripts/train/rdiag_kla/rdiag_recipe.py`, `scripts/tests/rdiag/attest_rdiag_shared_dt_gate.py`, and `scripts/train/rdiag_kla/rdiag_shared_dt_qualification.py`;
- `scripts/tests/rdiag/prepare_rdiag_shared_dt_source_gate.py` and `scripts/tests/rdiag/prepare_rdiag_shared_dt_qualification.py`;
- `scripts/tests/rdiag/rdiag_shared_dt_production_soak.py`, `scripts/tests/rdiag/run_rdiag_shared_dt_production_soak.sbatch`, `scripts/tests/rdiag/attest_rdiag_shared_dt_production_soak.py`, and `scripts/tests/rdiag/attest_rdiag_shared_dt_qualification.py`; and
- `scripts/train/rdiag_kla/submit_rdiag_shared_dt.py`, `scripts/train/rdiag_kla/rdiag_job_control.py`, and `scripts/train/rdiag_kla/rdiag_batch_supervisor.py` for the separately snapshotted production run and its mandatory evaluation tail.

The immutable qualification source is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/  
rdiag_shared_dt_auth_v3_qualification_  
a76dc7b930ae4d85157391d444c40ef7db14ef86,
```

and the campaign root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
campaigns/  
rdiag_shared_dt_auth_v3_a76dc7b930ae4d85157391d444c40ef7db14ef86.
```

The failed-soak investigation is isolated in sibling worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_soak269_diag_20260903
```

on branch `codex/rdiag-shared-dt-soak269-diagnostic-20260903`, based exactly on `a76dc7b930ae4d85157391d444c40ef7db14ef86`. The principal diagnostic code is `lit_gpt/rdiag_shared_dt_training_health.py`; the retained journal is under `production_soak/output/evidence/qual_rdiag_shared_dt_soak3300_a65918f1771446e7acfbffcc3e5c7dc3`, and the job logs are under `production_soak/logs` within the campaign root. The completed replay plan retained in that diagnostic branch is

```
docs/plans/2026-09-03-rdiag-shared-dt-soak269-diagnostic-plan.md.
```

Its replay entry points are `scripts/analyses/replay_rdiag_shared_dt_soak269.py` and `scripts/analyses/run_rdiag_shared_dt_soak269_replay.sbatch`.

The diagnostic line progressed through initial and early-mode commits

```
f82fa86f52c0f661d0ddbc74197d1dbc82a9e8f2,  
28531cf959304ab26b22789f2ea4ce082adbed13,
```

before the final search commit and tree

```
06eaf237f5be9f79ee8d00075b79dcc0e7a9e96a,  
f1ecf305edc04824ce14042fe75dbbba106f21c0.
```

The final immutable replay source is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/  
rdiag_shared_dt_alpha_zero_search_  
06eaf237f5be9f79ee8d00075b79dcc0e7a9e96a,
```

with source-manifest SHA-256

```
81f40cb7ccd697483d7b2757e0c728da18c4ee3a99b85b54be2f63e9af51cffe.
```

Its replay job `1829305` wrote only below `diagnostic_runs/soak269_replay_06eaf237f5be_1829305` and produced neither a qualification result nor a seal.

The then-current qualification checkout, containing the minimal lower-endpoint production-policy correction, is isolated from both the main checkout and diagnostic harness in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_alpha_zero_fix_20260903
```

on branch `codex/rdiag-shared-dt-alpha-zero-fix-20260903`. It is based exactly on the reviewed head-scaled commit. The production-policy change itself is commit and tree

```
46ba8ec3105177a9caba2f2c153e9b71c8b62cac,  
e38fdab82c68300a361613e4ee3a8c784b7c1a12.
```

The executable plan is `docs/plans/2026-09-03-rdiag-shared-dt-alpha-zero-policy-plan.md`. Aside from adding that plan, commit `46ba8ec...` changes only `lit_gpt/rdiag_shared_dt_training_health.py`, its focused host test `scripts/tests/rdiag/test_rdiag_shared_dt_training_health_host.py`, and one existing case in `scripts/tests/rdiag/test_rdiag_shared_dt_production_h200_v1.py`; it does not change model or kernel math.

The first qualification attempt exposed a test-construction issue rather than a kernel failure. Its new subcase forced $DT = 120$ at all 33 tokens, roughly 10.9 times the reproduced maximum, and therefore also imposed phase stress outside the replay-observed range. Immutable job 1829420 passed 108 of 109 tests; the modified full-layer case passed its output comparison and finite elementwise/relative- L_2 hidden-gradient checks, but its hidden-gradient cosine was $0.996079 < 0.999$. The gate failed closed, result SHA-256 was `d64821ee25f44c06aad4b6fb8d27b1433af328381be9a61d08e6e244e0a117bc`, and no seal was created. This failed snapshot and campaign are retained as evidence and are not reused.

Correction commit and tree

```
83118d2d96325c9b9b555d60ad4316c950c5a12d,
8c05ae8f2b5b205a17ba64042c4fad337a1d4d45
```

restore the ordinary strict output/all-VJP oracle case unchanged. A separate subrun within the same catalog item uses replay-bounded $DT = 6.53125$ and rate 16, giving exponent 104.5 and exact FP32 $\alpha = 0$. It proves that Kappa receives raw zero, S3 receives the finite BT32 prefix of $\log_2(10^{-6})$, the full forward matches the literal oracle, all model gradients remain finite, and live Kappa/S3 cotangents are finite; no tolerance or 109-test inventory entry changed. The related host sweep passed 131 tests. Development H200 job 1829576 passed the corrected item in 117.26 seconds; its stdout SHA-256 is `271d1ceb10772e4516cb9fe897e67cb7989e9af07939e329120a3d394f36f3a2`. Two independent exact-SHA reviews approved this final commit without blockers, and the branch is published.

The superseded 4228-file qualification snapshot for `46ba8ec...` is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/
rdiag_shared_dt_auth_v3_qualification_
46ba8ec3105177a9caba2f2c153e9b71c8b62cac,
```

with source-manifest SHA-256

```
2507dede155e9e50df876a493170d8ccfc2523413bab115e7a3f0e3369f21b93.
```

Failed campaign `2f02ccb07eac4837bd6416546b17112c` is rooted at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
campaigns/
rdiag_shared_dt_auth_v3_
46ba8ec3105177a9caba2f2c153e9b71c8b62cac.
```

Compute-side preparation job 1829417 passed after the login namespace correctly rejected a distinct live-data inode identity. Job 1829420 ran from execution-manifest SHA-256 `b23ea908ac550bc14c76267ff8e295e06f620d21412e37c766787f8ecfd72eb2`.

The now-superseded immutable qualification manifest for the corrected lower-endpoint test covers 4228 source files at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/  
rdiag_shared_dt_auth_v3_qualification_  
83118d2d96325c9b9b555d60ad4316c950c5a12d,
```

with source-manifest SHA-256

```
52a82f97849f400730dfa2496472017916ca84e36dd785a7bac9e2da629936df.
```

That now-superseded campaign, a262363313d141e9bda423b0ce243940, is rooted at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
campaigns/  
rdiag_shared_dt_auth_v3_  
83118d2d96325c9b9b555d60ad4316c950c5a12d.
```

Compute-side preparation job 1829624 passed. Production-H200 gate job 1829629 then completed on H200 node h200-028-028 with scheduler exit 0:0 and all 109 tests passing. It ran from execution-manifest SHA-256 98488ffdb2db208f015e34df678ef57a4003cc8ac8c29337aa7648755509e0df. Its retained result has SHA-256 32c5b2eed517f2add1e327d841c43f012858454edcf975bb5681403ec45e8675; the JUnit report and run log have SHA-256 values 507bfa8de347b1967d573b41578c3c3760a23b42e8b412f4076fe7b443dcba99 and c6357f6d46f284dda75280ae94c53d3668e0e78bdce92abd6b5c051e83fed30d, respectively. Compute-side attestor job 1829796 completed successfully and published the read-only H200 bootstrap seal with SHA-256 6f900869b3c13e6ef41e8920a52f9115bc31f7e69f7f8169a8b4014dc55b05f8. Five compute-side preparers then authenticated that seal and emitted read-only execution manifests for train-smoke, checkpoint-fault, authenticated-resume, gradient A/B, and the 3300-update soak, with SHA-256 values respectively

```
0d1e63001f57615c21a494e11eacfda8f04f464d258eb6d3c10207557f70fdc8,  
9287308502d9e2bbfffc6ff0bb2e71506c4ea9d14758a703edbef8519914157d2,  
34280cd404e1680b3bc13a94bea26678cd4b24fd37bc49c4d64ad5be817ed281,  
8f4fd8e74f56c3770c0464a385faac255561933b510d741d3820ddcae4ef7da7,  
5f3e7a352e529666aebfbbd841c06413310b7db0a00525f45184880772e0f8ed.
```

Only their emitted commands were submitted, as jobs 1829974–1829978. Train-smoke, checkpoint-fault, authenticated-resume, and gradient A/B jobs 1829974–1829977 passed and were independently attested. Their seal SHA-256 values are respectively

```
472ebcf70fa24a50870b68b25024c74df296d6f35f1e218eeeb36965458dd13a,  
b4f44aa38f2e11397f300f88f50b03881008ebe4d19ea7afa688ce9565ea416f,  
153c83d8be341bf7600990adeea69bb35e792b441459159e6fdc928316036a12,  
164138bf3a3c2809d8ed4fb7dd93a7a829fa42714fd123ddd8a634b6e3bc2d11.
```

Gradient A/B again passed with shared/no-rotation median and 95th-percentile ratios 1.179 and 1.260. Production-soak job 1829978 ran on nodes h200-207-042 and h200-211-043; its immutable-source and runtime-identity stages completed successfully. It crossed the original failure point: updates 268, 269, and 270 all completed with zero health failures. At update 269, loss was 5.3710, global gradient norm was 1.2287, and 11211776 alpha elements were below the S3 floor but remained diagnostic-only. The run then completed 675 healthy updates. Its final retained record has loss 4.0055, global gradient norm 1.4537, and zero health failures.

The next pre-optimizer check, boundary 676, failed closed with aggregate failure count 256 on all 16 ranks; update 676 was not applied. The Python step exited 1 and the surrounding Slurm job

terminated with exit 15:0. The ordinary soak evidence deliberately does not retain the failed-key breakdown, so at that stage the responsible predicate was not identified. No result or seal was emitted. The 675-line update journal, stdout, and stderr have SHA-256 values respectively

```
d26a63a89430a738c689d3e1fddc2d554ce30f3c400b1258c63c2a3fd2ef31cc,  
958939cf35074c8f9ca8611a19ec3dabd2830bd3090ee2e2bb104a12627dfd93,  
202e1bf22ac711421f006ef01c4c9820ac1522726cb4569480f635e7217c5a28.
```

Boundary-676 diagnostic replay. The non-authorizing investigation is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_boundary676_diag_20260903
```

on branch `codex/rdiag-shared-dt-boundary676-diagnostic-20260903`. Its executable plan is `docs/plans/2026-09-03-rdiag-shared-dt-boundary676-diagnostic-plan.md`. Diagnostic-only health rendering was introduced at commit `ef4ee3c053bc1d34243f4086edd88e62b85f73d4`; the reviewed replay harness is commit and tree

```
a9ce0db738dc65c3d72e4bcd85b955815952b163,  
bfb397412c1977acaef4742850738f69e948fc87.
```

The relevant code paths are `lit_gpt/rdiag_shared_dt_training_health.py`, `scripts/analyses/replay_rdiag_shared_dt_boundary676.py`, `scripts/analyses/run_rdiag_shared_dt_boundary676_replay.sbatch`, and `scripts/tests/rdiag/test_rdiag_shared_dt_boundary676_replay_host.py`. The focused replay suite passed 29 tests, the broad shared-DT host sweep passed 522 tests with four skips, and three independent reviews approved the exact staged tree that became the replay commit.

The immutable 4232-file diagnostic source is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_boundary676_diagnostic_  
a9ce0db738dc65c3d72e4bcd85b955815952b163,
```

with source-manifest SHA-256

```
6f7e52aa691c684543be51aeda0b6ab3b8d5d43c90b753b306070af68209da45.
```

Replay job 1830570 ran from that snapshot with the pinned failed-run data manifest and the exact original nodes `h200-207-042`, `h200-211-043`. It completed with scheduler exit 0:0; every retained record carries `authorizing:false`, and no qualification result or seal was emitted. The replay stopped naturally before optimizer boundary 649 after 648 completed updates. Its sole fatal key was `alpha_at_or_above_one_count=640`, exactly five shared scalar sites expanded over width 128. Global `alpha_max` was exactly one. All DT , $A_{\log}$, $\exp(A_{\log})$, and decay-exponent values were finite; $DT_{\min} = 1.1172518910 \times 10^{-8}$ and the smallest positive decay exponent was $3.4472314780 \times 10^{-8}$. Parameter and gradient nonfinite counts were zero. Thus the captured failure is rejection of the finite FP32 identity-decay endpoint, and no monitored nonfinite predicate fired. The replay alone is not a Kappa/S3 oracle check; the later targeted $K = 128, T = 33$ H200 subrun supplies that forward/VJP evidence. The observed exponent is slightly above the ideal round-to-nearest-even crossover $-\log(1 - 2^{-25}) = 2.9802322832 \times 10^{-8}$; CUDA's transcendental implementation need not be correctly rounded, so the retained output value and bitwise endpoint test, rather than the ideal crossover alone, govern the implementation claim.

The replay is deliberately classified `DIFFERENT_FAILURE`: it did not reproduce the old trajectory, boundary, or count; instead it failed at boundary 649 with `alpha_at_or_above_one_count=640`, while the original boundary-676 key remains unknown. Across the 648 comparable updates, maximum absolute loss and gradient-norm differences were 0.1493 and 1.3539. This evidence does not identify the old boundary-676 key retroactively, but it independently demonstrates that the replayed 83118d2... upper-endpoint policy rejected a valid production-shaped state. The retained artifact root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/boundary676_replay_a9ce0db738dc_1830570.
```

The summary, trajectory, invocation, inner stdout, inner stderr, and artifact-inventory SHA-256 values are respectively

```
9a8c357b37731924aed61e02b7bf8ea599b943f7508cb93f91fb7bd572c89157,
7eb3ab52f083079f85059522c7b0a219e22cacdd3923c3b55df135d59b851ac7,
82848cb2fb25f4b448cebb2e87dddf197d320c7c962dbc6946e654ce6a531fa7,
859d3c15b6223acafabddecc491afd896ebaf22368b5fc54823af481a05c18f4,
645326b62f8847a96dcc47cb8ddf9b39ed214dde65d6b731131c3f8b90e8b01c,
af23c4fef6684ae97102f3d81bb4cb6edc46e116623cc5e9f23fca88da78e370.
```

All 24 inventoried artifact hashes independently revalidated. The terminal outer scheduler log and empty scheduler error file have SHA-256 values respectively

```
2034750abd9cf1baea2074a0a8a28f2e6196838a1beac4fa84ad984d9ddbc151,
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.
```

The resulting narrow correction is isolated from both qualification and replay evidence in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_alpha_one_fix_20260903
```

on branch `codex/rdiag-shared-dt-alpha-one-fix-20260903`, based exactly on 83118d2.... Its plan is `docs/plans/2026-09-03-rdiag-shared-dt-alpha-one-policy-plan.md`. The reviewed and published correction commit and tree are

```
e5d07ad0f49f74e90d89a0b49ab8279b39152546,
2573a97d33244e9125ba94557b428868816360f4.
```

That commit changes exactly the plan and three implementation/test files

- `docs/plans/2026-09-03-rdiag-shared-dt-alpha-one-policy-plan.md`;
- `lit_gpt/rdiag_shared_dt_training_health.py`;
- `scripts/tests/rdiag/test_rdiag_shared_dt_training_health_host.py`; and
- `scripts/tests/rdiag/test_rdiag_shared_dt_production_h200_v1.py`.

It retains `alpha_at_or_above_one_count` as telemetry, adds `alpha_above_one_count`, and makes only strict $\alpha > 1$ fatal; negative and nonfinite checks are unchanged. The 109-case H200 catalog gains no test node: an existing full-layer case now includes a positive-exponent exact-one subrun covering raw FP32 bits, Kappa, the BT32 S3 prefix, literal recurrence agreement, and live finite VJPs. Host policy tests passed 36/36, the broad host sweep on the final pre-commit worktree

passed 492 tests with four skips, and targeted H200 job 1830711 passed in 114.16 seconds. This was non-authorizing mutable-worktree development evidence, run before the final commit; the immutable full H200 gate is job 1830943 below. The targeted job's stdout and stderr SHA-256 values are

```
689c9451b67e5e9593fb15bc858723f4d4e7e89d3b47fdb90ebbce5044501442,  
57ac2d0201a2657267a97ef3371038e486fe85eac5d21983843138591659f847.
```

Three independent exact-SHA reviews approved the correction without blockers.

The superseded e5d07ad... source snapshot manifest covers 4229 source files; the snapshot directory also contains SOURCE_MANIFEST.json. That snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_auth_v3_qualification_  
e5d07ad0f49f74e90d89a0b49ab8279b39152546,
```

with source-manifest SHA-256

```
3f6a8ae9176ddb9e9aee323febf6cb5d9dc22bc88d8933540be727c23a604832.
```

Its corresponding campaign 7a24d88d497d4ad28b572b4ee885ac90 is closed and incomplete; it is rooted at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
campaigns/rdiag_shared_dt_auth_v3_  
e5d07ad0f49f74e90d89a0b49ab8279b39152546.
```

Compute-side preparation job 1830934 passed and emitted production-H200 execution-manifest SHA-256 0d9277ab08b69f8d9e673e96cdd2d91ce8c753ce1d495879850edf0355377426. Its exact control path is

```
production_h200/control/  
production_h200_7a24d88d497d4ad28b572b4ee885ac90.manifest.json
```

relative to the campaign root above. The corresponding scheduler logs are under production_h200/logs/, and retained gate artifacts are written below production_h200/artifacts/production_h200_7a24d88d497d4ad28b572b4ee885ac90/. The resulting H200 gate, job 1830943 on h200-004-175, completed with scheduler exit 0:0 and 32:21 wall time; pytest reported all 109 cases passed in 405.96 seconds after the source/runtime preflight. Its result, JUnit report, and run log have SHA-256 values respectively

```
6ac6a4ce2775230cfcfd929aa77638c968ca6841b4a863a85824e479019f67da,  
b6132c80118ec46578178ab567cae9a76361d30d8476d403b8071a3c154b65be,  
5301d528c8b215fddca189209c17f8f3efefe7cbd831204e141ec29aed067c41.
```

Its outer stdout has SHA-256 e33b84600aa1d7896723fcac68d40d342c5264914c73a052b1c608863d303f2f; the empty stderr has SHA-256 e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.

The numerical result cannot authorize downstream work because its retained run.log is the exact 2706-byte prefix of the 2728-byte scheduler stdout: only the final 22-byte PASS:production_h200 line is absent. At e5d07ad..., both source-gate runners redirected through an asynchronous process-substitution tee, froze run.log, printed PASS, and exited without waiting for that writer. The race affected both production-H200 and train-smoke; the attestor's requirement that both closed logs contain PASS was correct and remains unchanged.

The first attester allocation, job 1830944, had an operationally insufficient 20-minute wall limit. After 5:43 it had read 34496.23 MiB of the runtime closure; the corresponding prior successful attester read 270926.75 MiB in 42:47, projecting this attempt well beyond its limit. It was therefore manually cancelled before publishing a seal; dependent preparer 1830946 never allocated and was also cancelled. Replacement attester job 1831239 used the byte-identical command and evidence pins with a one-hour limit. It completed the full closure read and then correctly failed 2:0 after 57:28 on the missing retained-log marker; it emitted no seal. Dependent preparer 1831240 never allocated. A redundant fail-safe attester, job 1831513, was cancelled after the deterministic artifact defect was identified rather than spending another full validation cycle.

The logging correction is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_source_gate_log_fix_20260903
```

on branch `codex/rdiag-shared-dt-source-gate-log-fix-20260903`, based exactly on `e5d07ad`. ... Its executable plan is

```
docs/plans/
2026-09-03-rdiag-shared-dt-source-gate-log-finalization-plan.md.
```

Working-location index. The worktree above is the historical qualification-source checkout, on branch `codex/rdiag-shared-dt-source-gate-log-fix-20260903` at the exact reviewed commit `1fce1b9`...; current diagnostic and control development is isolated in the generation-zero, forward-determinism, RMSNorm-control, trigger-capture, and matched-intervention worktrees named below. The main checkout is not the source of any qualification job. Within the historical source worktree, the shortest paths into the implementation and its controls are:

- `lit_gpt/rdiag_shared_dt_kla.py` for the layer parameterization and the $H^{-1/2}$ angle head scale;
- `lit_gpt/kla_ops/rdiag_shared_dt.py` and `lit_gpt/kla_ops/rdiag_shared_dt_reference.py` for the composed GPU path and reference recurrence, with phase/clock helpers beside them;
- `lit_gpt/rdiag_shared_dt_training_health.py` for the alpha endpoint telemetry and fatal-domain policy;
- `scripts/tests/rdiag/` for H200 cases, distributed smoke/soak/fault gates, runners, and attestors; and
- `scripts/train/rdiag_kla/` for recipe pinning, campaign/job control, authorization, and the production submitter.

The design and implementation plans for the numerical correction are, relative to that worktree,

- `docs/plans/`
`2026-09-02-rdiag-shared-dt-angle-headscale-design.md`; and
- `docs/plans/`
`2026-09-02-rdiag-shared-dt-angle-headscale-plan.md`.

The two subsequently discovered finite-precision endpoint policies are recorded in the corresponding `alpha-zero-policy` and `alpha-one-policy` plans dated 2026-09-03. The boundary-269 and

boundary-676 plans live only in their respective historical diagnostic worktrees. Those filesystem-writable worktrees are retained for investigation but are not qualification sources; the current immutable source snapshot and campaign root are named below.

The only intended runtime changes are `scripts/tests/rdiag/run_rdiag_shared_dt_production_h200_v1.sbatch` and `scripts/tests/rdiag/run_rdiag_shared_dt_train_smoke.sbatch`; regression coverage is in `scripts/tests/rdiag/test_rdiag_shared_dt_source_gate_host.py`. The corrected lifecycle is PASS emission, closure of both pipe writers, an explicit wait for the captured `tee` PID, and only then read-only log publication. It preserves any earlier nonzero status and makes a logger or `chmod` failure fatal. No result, seal, aggregate, or attestor schema is weakened.

The reviewed and published logging correction is `commit/tree`

```
1fce1b9a8e4a80d4da82a689dfb69983109aad0c,  
297b4b1f3f4ba7e4ac64ac35df0510c9bdea69be.
```

It changes exactly the two runners and the source-gate host test named above, plus the tracked plan. A focused selection over `test_rdiag_shared_dt_source_gate_host.py`, `test_rdiag_shared_dt_production_h200_v1_host.py`, `test_rdiag_shared_dt_train_smoke_host.py`, and `test_rdiag_shared_dt_gate_runners_host.py` passed 134 tests. The full 22-file RDiag host sweep passed all 823 tests, and independent collection still reports exactly 109 H200 cases. Two earlier broad attempts were deliberately excluded: concurrent test artifacts filled the 512 MiB `/var/tmp` filesystem in one, while placing temporary trees on Lustre changed the filesystem-specific test branch in the other. The final sweep used a spacious local filesystem and had no failures. Three independent exact-SHA reviews approved the commit without blockers.

The replacement immutable source snapshot contains 4230 tracked regular files and excludes only the approved paper gitlink. It is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_auth_v3_qualification_  
1fce1b9a8e4a80d4da82a689dfb69983109aad0c,
```

with source-manifest SHA-256

```
429348ceb16350782bc0eb2b4a34affabc5eb73f1d3ac997e5da42a2d1949067.
```

Fresh campaign `ae2041e3d1fd4b1ca7c1d609184c2642` is rooted at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
campaigns/rdiag_shared_dt_auth_v3_  
1fce1b9a8e4a80d4da82a689dfb69983109aad0c.
```

Compute-side preparation job 1832038 completed with execution-manifest SHA-256

```
a1759ea8f8b058b4ab49b235d7ce0d083d7760cc41447d0d9447e01515ee925e.
```

The manifest is

```
production_h200/control/  
production_h200_ae2041e3d1fd4b1ca7c1d609184c2642.manifest.json
```

relative to that campaign root. Its authenticated emitted command was checked as an exact argv sequence inside H200 allocation 1832039 and submitted as job 1832040. That gate completed 0:0 on h200-052-125 in 31:54; all 109 cases passed in 423.33 seconds. The result, JUnit report, and retained log have SHA-256 values respectively

```
e39ebe47ce0e1bd43f23c312c75269a5a6300b6f367f6f43df7906568bc8d057,  
a654c778be9062d4cc9f41b62b0f779732f09bf0c5e9c27521538384c1c9a732,  
5335a009a8407fcf2965cc67eb09f02124d562d378c98019e05119cf57e1401d.
```

The mode-0400 retained log is byte-identical to scheduler stdout, includes exactly one terminal PASS:production_h200 marker, and the scheduler error file is empty (SHA-256 e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855). This directly exercises and resolves the former publication race. The two-hour external attestor is job 1832042; it completed 0:0 in 52:16 and published a mode-0400 bootstrap seal with SHA-256

```
a6c63dae54416ad4085631de5e7d27ee97423ad881a5d1f75596d68d4dc35cff.
```

Five manifest-only preparation jobs were dependency-bound to that attestor: production soak 1832100, train smoke 1832101, checkpoint fault 1832102, authenticated resume smoke 1832103, and gradient A/B 1832104. All five completed 0:0; their stderr logs are empty and their mode-0400 execution-manifest SHA-256 values, in that order, are

```
2039a4b702586e5549c6a92bcfb0d0b3054151df8af290086e153439674c9873,  
36c6b0a2597dd6ee31d95b048d9434eb6a021f2f8a65a26736fa4fca6822e0df,  
f2ae3776688b6a3c06e6fd7b2f6e1e71b303e50663fb2b296bebf799797fb63f,  
5ca194e4c54c7342c498689000d377acbf283e1e5a0be9075e7b9116c6e9a15c,  
db126e731f9bae7cdd1172fa4815a966dc49084ecb21310ca0026dfa30e5d119.
```

Control allocation 1832771 independently rehashed each manifest, loaded it through the pinned control code, rebuilt the scheduler argv inside an H200 allocation, required exact equality with the preparer-emitted argv, and found no existing job with any of the five unique scheduler comments. This compute-side step was required because the login node intentionally has a different data-root device identity. It submitted the exact workloads and dependency-bound external attestors as the following pairs:

```
production-soak1832774->1832775,  
train-smoke1832776->1832777,  
checkpoint-fault1832778->1832779,  
authenticated-resume1832780->1832781,  
gradient-A/B1832782->1832783.
```

The four short workloads and attestors then completed successfully:

- train-smoke jobs 1832776->1832777, seal SHA-256
d04cbec0b679496a8d4e05c22435c919ce1dbce455ea1e08dd48bbc04abad556;
- checkpoint-fault jobs 1832778->1832779, seal SHA-256
d382d2ef31dbb49588316a7850f8c022ee3796681ee6aa418e10a53f0c0902aa;
- authenticated-resume jobs 1832780->1832781, seal SHA-256
dc1ab93029eca699c27d4d1a8ac23a946cc6ff2c76c95d30de77edd82d897e23; and
- gradient-A/B jobs 1832782->1832783, seal SHA-256
a17fb2ecc2f116c41c7b33853571a8665b4811a23685b8530fea071b1e12e2b.

All eight scheduler exits are 0:0; every external-attestor stderr is empty, and each independently rehashed seal is an owner-held, single-link, mode-0400 regular file. The new train-smoke retained log is byte-identical to scheduler stdout and contains exactly one terminal PASS marker, providing

a second end-to-end exercise of the log-finalization fix. Checkpoint-fault and authenticated-resume emitted the same framework initialization, FSDP deprecation, and process-group shutdown warnings seen in the earlier successful campaign; neither emitted a Python traceback, and the external attestors accepted their exact results.

The two-node soak allocation 1832774 started at 2026-09-03 20:15:33 UTC on h200-232-233 and h200-233-046. Slurm emitted two bootstrap warnings because it probed the exported job-private TMPDIR before the first per-node bootstrap step had created that directory, and therefore used /tmp for that step; the runner then created and used its intended mode-0700 cache trees. This is retained scheduler stderr, not the training failure.

The journal crossed both historical failure boundaries with zero aggregate health failures. Its exact update-269 row records loss 5.395138889551163, global gradient norm 1.213560938835144, and 26,092,800 alpha values below the diagnostic S3 floor out of 17,179,869,184. Its update-676 row records loss 4.053754523396492, global gradient norm 1.0578112602233887, and 102,754,304 below-floor values out of the same denominator. The workload then failed closed at the pre-optimizer boundary for update 790. All 16 ranks reported **SoakEvidenceError:globalgradientexceedsthesoaklimit**; job 1832774 ended **FAILED1:0**, and dependency-bound attestor 1832775 was cancelled without allocation. No soak result, final checkpoint, seal, or aggregate exists.

The retained journal contains exactly 789 contiguous accepted rows and has SHA-256 4814e6c0cedbdc4efdceefb4b944ca2e8b1df357d4f7a49118857a64f8da286f. Its median and lower-empirical 95th and 99th percentile pre-clip norms are respectively 1.14646, 2.27364, and 3.35581. The maximum is 7.970221519470215 at update 789; excluding that row, the maximum is 4.06596 at update 24. Updates 783–789 measure 3.39357, 1.82029, 1.61033, 2.27467, 2.26284, 3.04553, and 7.97022, followed by the unretained update-790 rejection. Loss at the rejected boundary was finite, and every accepted row has zero health failures.

The name `after_clip` is temporal rather than semantic. Lightning delegates this call to FSDP, which returns the accumulated distributed norm *before* multiplying gradients in place by approximately $\min(1, 1/(\text{norm}+1\text{e-}6))$. Thus update 789 was subsequently clipped to approximately one; update 790 had a rank-equal pre-clip norm that was nonfinite, negative, or above 10 and was rejected before `optimizer.step`. Its exact scalar and predicate branch are irrecoverable from this run: the harness checks the combined predicate before constructing or appending a record, and named-gradient sampling begins only at update 3000. The run retained only its generation-zero checkpoint.

This is a genuine failure of the frozen qualification contract, but not yet proof that the 750M model is unstable. The limit 10 was copied from the older bounded-E2 soak. That accepted E2 screen observed a sampled maximum 1.888 over updates 300–500, but it did not calibrate the tail of a completed 3300-update shared-DT run. Conversely, the separate 220M gradient-v3 gate uses 100 only as a gross fail-fast ceiling around a matched shared/no-rotation ratio experiment; it cannot justify changing the 750M soak threshold to 100. The late rise makes attribution mandatory before either a model change or a versioned gate-policy change.

The non-authorizing investigation is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_soak790_diag_20260903
```

on branch `codex/rdiag-shared-dt-soak790-diagnostic-20260903`, branched from `1fce1b9...`. The reviewed and published execution commit is `47bb7812da1bd5809d974ea9c5504209f3a357f3`, tree `2fc10ee5320a418a72960a86145083916a3d739c`. Post-execution evidence hardening is the current branch head:

```
commit 4091b2572f19f8cefaab8932b2d035969f663ec0,  
tree bf8abd733df27ba03331fadf35560107fadf1e8b.
```

It does not alter the executed source. Relative to that worktree, the completed replay plan is `docs/plans/2026-09-03-rdiag-shared-dt-soak790-diagnostic-plan.md`; its entry points are `scripts/analyses/replay_rdiag_shared_dt_soak790.py` and `scripts/analyses/run_rdiag_shared_dt_soak790_replay.sbatch`. The replay keeps the production recipe, ordered data, seed, optimizer schedule, 2-by-8 topology, FP32 FSDP buffers, clipping, and strict limit under measurement-only instrumentation. This is a same-recipe replay, not bit-exact production reproduction: read-only scans, copies, and reductions can perturb GPU scheduling, and “same boundary” denotes only the boundary number. It revalidates the failed run’s complete runtime fingerprint, computes per-rank target-token hashes and pre-clip category and top-parameter diagnostics at every boundary, and retains the hashes for a gradient-rejection boundary. It stops non-authorizingly on a natural production-health failure, a finite norm above 10, a nonfinite returned norm or post-health named gradient, or exhaustion through update 820. Nonfinite values use canonical JSON-safe tokens, finite-only norms are labeled as such, and every offending parameter remains separately listed. The validator requires the exact 32-field PRE-health inventory, the full-update alpha denominator 17,179,869,184, and name-to-category agreement for every retained contributor. Physical-family nonfinites remain owned by the earlier production-health failure path.

The final pre-execution replay, health, and pretrain-integration host suite passed 87 tests. Independent runner, schema, and mathematical-contract reviews approved execution commit 47bb7812.... A Git-object snapshot was materialized at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/  
rdiag_shared_dt_soak790_diag_47bb7812da1bd5809d974ea9c5504209f3a357f3
```

with source-manifest SHA-256 a948eb8a0540a2ff7625f344372a2d04389865bb5e3a977ad7ceff123e4a742f. The manifest closes 4,234 tracked regular files; its `SOURCE_MANIFEST.json` is stored at the snapshot root. Two-node H200 replay job 1833926 ran on h200-101-019 and h200-208-076 from 2026-09-03 22:40:57 through 23:02:14 UTC. It and all Slurm steps completed 0:0. The rank-complete result is `NO_GRADIENT_LIMIT_EXCEEDANCE_THROUGH_820`: all 820 returned norms were finite, nonnegative, and no larger than 10; all named-gradient nonfinite counts and all health-failure counts were zero. Update 790 measured norm 1.0757828950881958 and loss 4.024734705686569, update 820 measured norm 4.972261428833008, and the replay maximum was 7.394214630126953 at update 817. Thus the replay did not reproduce the original update-790 rejection.

The replay is not a matched trajectory. Of the 789 comparable records, 673 differ in at least one shared field; floating-point norms differ at update 1, alpha-below-floor counts first differ at update 3, and every update from 139 through 789 differs. Original versus replay update 789 is respectively 7.970221519470215 versus 1.4658762216567993. Gradient norms over 1–789 have Pearson correlation 0.5998 and first-difference correlation 0.0389; over 1–99 those values are 0.999565 and 0.99191, while update 100 starts the first five-update run with absolute norm differences above 0.25. Loss remains tightly same-step coupled (Pearson 0.999511), showing the same broad recipe but not the same numerical state. No original update-790 input hashes exist, and this no-rejection outcome retained no replay batch hashes, so data equality at that boundary is unproved.

The late replay excursion is broad rather than an isolated radial failure. Across updates 783–820, squared gradient-norm mass is 45.57% ordinary/backbone, 32.91% angle, 21.50% DT weight, 0.026% DT bias, and 0.00085% `A_log`. At the maximum, update 817, category norms are 5.2138

ordinary/backbone, 4.5117 angle, 2.6695 DT weight, 0.0909 DT bias, and 0.0271 A_log. This localizes co-amplification but is not a causal decomposition; it supports neither removing rotation nor changing the frozen limit.

The finalized artifacts are under

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/soak790_replay_47bb7812da1b_1833926/artifacts.
```

The retained evidence digests are:

```
summary: 9620387f55d38bfde32e5763b1ecf72c1c4fe23223ea6d01d051acb246a718f5,  
trajectory: d8fbb3c1fdea2bc1f63f089cb457b9ee832372cd828465fbee2070327801a7fc,  
invocation: ff2a6a83950942e2d7ba12d8ac1bac163264e628f6d34a97e010ab27ad644634.
```

An independent audit found one packaging defect: the invocation named Slurm’s ephemeral spool copy of the runner, which disappeared after cleanup. Its recorded SHA-256 and size exactly match the immutable snapshot runner, and substituting that path regenerates the summary byte-for-byte, so the result is corroborated but is not self-validating as originally packaged. Commit 4091b2572f19f8cefaab8932b2d035969f663ec0 fixes future packages by retaining a mode-0400 artifact-local `runner.sbatch`, binding it to the retained source-manifest entry, and rejecting external or writable runner records. Its focused suite passes 48 tests. There is no evidential reason to rerun job 1833926 merely to repair this historical packaging defect.

The shortest code paths for this boundary are `pretrain.py` (data fetch, health, clip, optimizer, and checkpoint lifecycle), `data.py` (stateful worker/buffer order), `lit_gpt/rdiag_checkpoint.py` and `lit_gpt/fsdp_strategy.py` (full model/optimizer and per-rank sidecars), and `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab_run.py` (the existing connected-zero rotation intervention). Replay-specific instrumentation remains confined to the helper and runner named above.

Remaining work. The head scale and gradient-v3 closure remain frozen and published at a76dc7b930ae4d85157391d444c40ef7db14ef86; the alpha-zero policy source is frozen and published at 83118d2d96325c9b9b555d60ad4316c950c5a12d. That source crossed the former boundary-269 failure but did not pass the full soak. The alpha-one correction is frozen at e5d07ad0f49f74e90d89a0b49ab8279b39152546; the failed source head is its log-finalization child 1fce1b9a8e4a80d4da82a689dfb69983109aad0c. Every existing seal is source-bound historical evidence; the five successful non-soak seals cannot authorize a source or policy change.

The formerly proposed production root for 1fce1b9... remains deliberately absent:

```
/lustre-storage/checkpoints/ngocbh/hdn/production/  
rdiag_shared_dt_auth_v2_1fce1b9a8e4a80d4da82a689dfb69983109aad0c.
```

It will not be created from a failed qualification. A future production root, if authorized from a later source commit, will have fixed children `source_snapshot`, `manifests`, `control`, `output`, and `logs`. The separate production snapshot must be created atomically by the immutable launcher’s `prepare-fresh` mode; the qualification snapshot is not reused as the training snapshot. The four recipe digests listed below are historical pins for 1fce1b9..., not reusable authorization for changed source:

- `rdiag_kla_shared_dt_v1_220M`:
8f4d2d21f38ee93bda045e74d80bf6c5c1dd1feef9102e3ab413583b1542d9e1;
- `rdiag_kla_shared_dt_v1_750M`:
7a07b7af54a9df20736fc85c2eb615900c969c8a977ca12836cee210f301b2c4;

- `rdiag_kla_shared_dt_v1_1.3B:`
`f11e601a6a60fe67dd069ec5fc72954c2f32396bda4d910a32b13ed34f377834`; and
- `swa_rdiag_kla_shared_dt_v1_1.3B:`
`225f1cd89233f0a290c527ad5c59ce4b71f9d884dfcddb1a6569044a93bd96b3f`.

The launcher credential prerequisite has been materialized through the no-argument/no-stdout credential path at `/storage/home/ngocbh/.wandb-key`; it is an owner-controlled, single-link regular file with mode 0400. No separate production-training snapshot, execution manifest, output directory, or training job has been created. Because update 790 was not reproduced, the next investigation first locates the earliest divergence from the exact original generation-zero checkpoint. It is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_gen0_repro_diag_20260903
```

on branch `codex/rdiag-shared-dt-gen0-repro-diagnostic-20260903`. Its executable plan is `docs/plans/2026-09-03-rdiag-shared-dt-gen0-reproducibility-plan.md`. It pins the original 2.9-GB generation-zero model with SHA-256

```
8a37e667678d2743c5bdcfb3b808c909f980c50b906e2ff9e97e29b02074b581
```

and all 16 rank sidecars. It runs three six-microbatch loader controls: no snapshot (the 1833926 path), snapshot-and-continue on the same loader (the failed fresh-production path), and new-loader sidecar restore. Four common-anchor training arms run in counterbalanced `minimal-A1/instrumented-B1/instrumented-B2/minimal-A2` order for three updates on the same allocation and GPU mapping. Complete local parameter/optimizer Merkle inventories and exact input hashes are prerequisites. Historical process RNG was not checkpointed, so this test localizes only the first observed divergence among captured stages; it cannot alone identify the unique historical cause. This experiment is diagnostic only. The generation manifest and its sidecar/ordered-file composites are pinned respectively to

```
6649a1a68f83b80df9f5600864dc52e51639d19b42a827b78c9eeacef9e9f8f6,
162ff05e5940a427504c0075c82002aa8894091a8bd4f1c22c5bb5e0cb69172f,
6a184a861f445a0eb9ed383e029d726f9cec78f97285df3c350fda95bc52fd86.
```

The validator requires exact zero counters, modes, identities, and the 17-file model-plus-sidecar closure before any arm runs.

Implementation and review are complete in the worktree above. The reviewed execution identity is

```
commit 4804be477a09f19adbd88d5ca773be721608d9f3,
tree b3d882ee859038366f6d18cbd612ad91aa2d31d3.
```

It is based on diagnostic-evidence head `4091b2572f19f8cefaab8932b2d035969f663ec0`. Relative to the worktree, its executable entry points and host contract are

- `scripts/analyses/reproduce_rdiag_shared_dt_gen0.py`;
- `scripts/analyses/run_rdiag_shared_dt_gen0_repro.sbatch`; and
- `scripts/tests/rdiag/test_rdiag_shared_dt_gen0_repro_host.py`.

Pre-execution review found and corrected four material harness defects: the post-run snapshot verifier omitted its required private-copy argument; the terminal completion marker preceded verified node-local cleanup; an instrumented-arm wrapper discarded ordinary loss capture; and the classifier could ignore some global-loss, learning-rate, named-gradient, rank-agreement, and loader-to-training input discrepancies. The corrected helper now rederives every field used for classification, binds the restored L2 input matrix to all training arms, requires finite rank-equal pre-clip norms, distinguishes minimal- and instrumented-pair collective differences, and validates an exact 214-file artifact closure before completion. The exact reviewed helper, runner, and test SHA-256 values are respectively

```
37b0e2061a145b2b1935a74a362235a1d462457119ed42e4e1efd3d36b7be7bc,  
f9d5c8a04f7384b360eb04435c32d0a2abeb177e6f6534e91f4910e5b7c1bb14,  
750d0bae60ca24c7ae7fecffc0a52c2decba915854b5f282910b038bd088bfdc.
```

The focused generation-zero suite plus the retained soak-replay suite passes 94 tests; Python compilation, batch-shell parsing, staged whitespace checks, and a fresh exact validation of all 17 generation-zero inputs also pass. Independent artifact/runner, mathematical-contract, and Lightning/FSDP runtime reviews found no remaining Critical or Important issue on these exact bytes. The broader RDiag host sweep first reached 898 passes before 19 late fixture writes failed because the login node's 1-GB /tmp filled; these were ENOSPC errors rather than assertion failures. Re-running both affected files with a diagnostic-Lustre temporary root passed 82/82, closing all 917 distinct broad-suite tests. The two disposable test-owned temporary trees were then removed.

Commit 4804be477... is pushed on the branch named above. Its immutable Git-object snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/  
rdiag_shared_dt_gen0_repro_diag_4804be477a09f19adbd88d5ca773be721608d9f3.
```

The snapshot contains 4,238 tracked regular files, excludes only the external `paper` gitlink, and was independently revalidated against source-manifest SHA-256

```
1bbf00a65b30ffd26bd01926e09b7c306c2836054f40d83aab9963904e16c893.
```

Two-node H200 diagnostic job 1834859 ran directly from that snapshot on h200-091-173 and h200-101-011 from 2026-09-04 01:23:02 through 01:28:07 UTC and failed 15:0 in the first loader probe. Source staging, both runtime identities, invocation capture, and the complete generation-zero input preflight succeeded. Before any training update or rank result was written, all ranks rejected the first batch at `batch_record`: the live Transformers collator returns a `BatchEncoding`, which implements `Mapping` and contains `input_ids`, `attention_mask`, and `labels`, but the diagnostic required the concrete type `dict`. No summary, artifact inventory, completion marker, or scientific result exists. The retained scheduler log and stderr SHA-256 values are respectively

```
29939733705f0f3e49dd2fe2e1210484f68a3d6b90f6fafa361fb992cc7718a2,  
4f5e6fb8ab374b4bce06213cab824ed588b5ee4ae5c6ad818ce4b2f10ee34326.
```

The failure is a fail-closed diagnostic type-contract bug, not model evidence.

The corrected execution identity is

```
commit f1cfd305cfe76039cf8a0d62881e84ec5faee6d7,  
tree 5c558d3e470b3c5f47aa7912f8ffed71bcbd5255.
```

It replaces that concrete-type check with `isinstance(batch, Mapping)`, documents that the retained identity is the two-field training-input projection, and tests an actual three-field `BatchEncoding` plus negative non-mapping/missing-field cases. The focused generation-zero plus replay suite now passes 97 tests. An independent review reproduced the real collator type and found no analogous live-API type blocker. The corrected helper SHA-256 is

```
9e9c20f71fb5810ae39da45e29dddaa2c7b130e4b5b7740e819a9a2664cbb63c.
```

The runner remains byte-identical. The replacement 4,238-file immutable snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/  
rdiag_shared_dt_gen0_repro_diag_  
f1cfd305cfe76039cf8a0d62881e84ec5faee6d7,
```

independently revalidated against source-manifest SHA-256

```
524164d0e3b75e91b9cf3013ac581b0d0a12b8288aeec8ec33579efb6eb87a55.
```

Replacement two-node H200 job 1835101 ran directly from that snapshot on h200-081-214 and h200-199-061 from 2026-09-04 01:57:03 through 02:11:54 UTC and failed 15:0. Unlike job 1834859, all three loader modes completed: 48 rank records prove that the two-field training-input projections of all six microbatches match exactly between no-snapshot, snapshot-and-continue, and new-loader sidecar-restore paths on every rank. The L1 and L2 canonical loader-state hashes are also identical. Thus neither calling `state_dict` nor restoring its captured state is the first observed loader fork in this environment.

The first training arm then failed on all ranks before model construction or any update because the shared-DT frontend import loads TileLang, whose TVM initialization intentionally prepends its Python directory to the in-process `PYTHONPATH`. The harness validated the pristine launch environment after that import and therefore rejected the expected library mutation. No training-arm record, summary, artifact inventory, completion marker, or scientific classification exists. The retained artifact root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/gen0_repro_f1cfd305cfe7_1835101/artifacts.
```

Its scheduler log, stderr, invocation, and pre-run checkpoint-inventory SHA-256 values are respectively

```
1d1770bc94f23f14a4d3d97083827130031eec773131a7512fd4ce2abe350c8f,  
ccb4628c9a94d73a0f324f748c106c38bc44b846b3e797eef809febb6d3ac9b9,  
b13029c37f098209ec5eb046e78ecd4652c38af6998fbe9e88fe8caf7be8b45c,  
20bbc0c2a345eaa46671d670c0d865b6f9a73266c9310bd81a1f43243ed120fd.
```

The second correction moves the unchanged launch-environment validation before heavyweight model imports; it does not weaken the check. It is reviewed and pushed as

```
commit e2b5e317b750e88437d30e95eb6a89ac88cb11c2,  
tree f625647a419c9ded1dd02a9b5367827981b00b18.
```

The helper and host-test SHA-256 values are now

```
fc3c44e57af6d0994934a7bd0c7433580904a481428966773e6863b77adf2217,  
4be33408b0e33743486be389d31968f3fd6337c83d58f823a9d1ce33860f803f.
```

The combined generation-zero/replay host suite passes 98 tests; an independent import-chain audit reproduced the TileLang mutation and found no earlier mutating import. A fresh 4,238-file Git-object snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_gen0_repro_diag_  
e2b5e317b750e88437d30e95eb6a89ac88cb11c2,
```

verified against source-manifest SHA-256

```
797858b93cf5b7c1e98581e00bd3fa58b0db28d1de3c87e4b23b3b5a4775b968.
```

Replacement job 1835588 ran on h200-046-019 and h200-171-121 from 2026-09-04 02:49:35 through 03:07:27 UTC and failed 15:0. The environment-order correction worked: all three loader phases and the three-update `minimal_a1` arm completed on every rank. Its 16 training records have finite, rank-equal pre-clip norms at all three boundaries and use the same input projection as the restored sidecar. The subsequent cache-inventory step failed before the next arm because the training path, unlike each loader probe, had not explicitly stopped its four persistent DataLoader workers per rank. A local reproduction showed that live workers create `TMPDIR/pymp-*/listener-*` AF_UNIX sockets; the strict cache inventory correctly rejects such a special entry. This is post-measurement harness cleanup evidence, not a model failure, but the job has no final summary, artifact inventory, completion marker, or admissible classification. The retained artifact root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/gen0_repro_e2b5e317b750_1835588/artifacts.
```

Its scheduler log, stderr, invocation, and rank-0 `minimal_a1` record SHA-256 values are respectively

```
a064b744df142a24957ae3ad3915da441d127d7c37c84c2b6ed91018024f2fe3,  
f9f48e95fb92bebe27c4f0ddb5ec44f27dc560854f5c28da92ca3c696d812424,  
356a404e648d383301ac25b79bd6680c9ce289dd49877538fc869bf7a62e9374,  
b358564f0dfb6f8a39f077258a620119d5030743ce49e4c2c10118a77cf2cf4c.
```

The third correction factors the already-used worker shutdown into one helper and invokes it from both loader probes and training-arm cleanup. Special cache entries remain fail-closed, and future errors now include their relative path, decoded kind, and mode. The reviewed, pushed identity is

```
commit 0e9cda6875e8709fa250110dcb35772d46d22817,  
tree dad58c03e607bf38304dd41ae7d6e8993d729720.
```

The helper and host-test SHA-256 values are respectively

```
a37fdea21c59757246c6fabff0dac15d8ca93ac3cfdc784d0549413544e30f70,  
92766a1a21c35be8a26314cf0c4e85b8d298eec88abc37467d6a5e69d632ffa4.
```

The combined focused suite passes 100 tests, including an AF_UNIX-socket rejection test, and an independent review confirmed that shutdown removes all four reproduced listener sockets without changing measured training math. Its fresh immutable snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_gen0_repro_diag_  
0e9cda6875e8709fa250110dcb35772d46d22817,
```

verified against source-manifest SHA-256

```
b228c5e0b44a9888d96d2d381c53940d7f6eb7264c71367ecde06f26535cf0e6.
```

Replacement job 1835824 ran directly from that snapshot on h200-046-019 and h200-094-010 from 2026-09-04 03:42:39 through 04:13:57 UTC. The job and every step .0-.18 completed 0:0. It produced the exact preregistered closure: 112 rank JSON records across seven 16-rank phases, 214 mode-0400 files recorded by `artifact-inventory.json`, and 215 final directory entries because the inventory does not list itself. Independent validation returned `VALID`; the log has exactly one non-authorizing completion marker and the scheduler stderr is empty. The retained artifact root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/gen0_repro_0e9cda6875e8_1835824/artifacts.
```

The summary, artifact inventory, invocation, scheduler log, and empty scheduler stderr SHA-256 values are respectively

```
aa466253b9bcdfeebf695ead220357b6693189c93aa5d089c573a3e887a71df6,  
7aef2fc9349c84876b0ea7d46ee114147a4dc52be2cd4b0c776c4d6245b08339,  
eba71feda5c6f650968a143f49c4881fa524ce22ac4dec31b713ca1e0bbc9534,  
8567a91cdacd6df770a49cc61f13ffe5be0431e479f30bbeae21afb50bdc6f40,  
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.
```

The result is valid diagnostic evidence but fails the scientific hard gate. All loader and training input matrices match the original sidecars; the reconstructed model, optimizer, loader, first-RNG anchor, rank topology, and corresponding-rank mapping also match. Nevertheless both the minimal A1/A2 pair and instrumented B1/B2 pair classify as `ORDINARY_RUNTIME_NONDETERMINISM`, making the overall training classification the same. The first classifier-stage difference is the boundary-1 local-gradient Merkle root after the first two microbatches. The earliest chronological scalar difference occurs even earlier at rank 1, microbatch 1: minimal A1 records loss token `0x1.50da340000000p+3`, whereas minimal A2 records `0x1.50ddc80000000p+3`, despite joint input SHA-256 `5dcab61b5c06aa429639e90ef660cba3caf85ed6161fd6cf36812696f369cf7a` and pre-forward RNG SHA-256 `7db887f05ff2b0af83e6e7c4b4c9d7ac1a71e2f172d903ea409ce54cf21bdabe` being identical. Because both minimal replicates diverge before clipping, this is neither a loader restore failure nor instrumentation-only or pre-clip-norm-collective divergence. The exact operator remains unidentified; trigger capture stays blocked pending a focused runtime/kernel repeatability diagnostic and a successful repeat of this hard gate.

That focused forward-determinism diagnostic was implemented in the isolated worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_forward_determinism_20260904
```

on branch `codex/rdiag-shared-dt-forward-determinism-20260904`, based on commit

```
0e9cda6875e8709fa250110dcb35772d46d22817.
```

Its reviewed executable plan is `docs/plans/`

`2026-09-04-rdiag-shared-dt-forward-determinism-plan.md`; its entry points are

- `scripts/analyses/reproduce_rdiag_shared_dt_forward_determinism.py`;
- `scripts/analyses/run_rdiag_shared_dt_forward_determinism.sbatch`; and
- `scripts/tests/rdiag/test_rdiag_shared_dt_forward_determinism_host.py`.

The design compares two ordinary-autotune and two diagnostic-only RMSNorm-warp-8 phases. Each process retains one minimally observed cold forward through the normal Fabric/FSDP and health path, stops after the exact ordinary cross-entropy reduction and before backward, then performs three equal-RNG warm traces plus a post-trace replay of all six existing RMSNorm forward configurations. Cold and warm evidence are reported on separate axes; every configuration replay must be stable across corresponding ranks before RMSNorm selection can be called strongly supported. This diagnostic cannot authorize trigger capture: any candidate execution control must still pass a fresh copy of the unchanged generation-zero `minimal=MATCHED` gate. Independent plan and runtime reviews found and closed three pre-launch defects: an invalid cross-scope FSDP weight-object identity requirement, conflation of the raw BF16 `lm_head` output with Fabric’s distinct FP32 return/loss input, and a missing final fail-together exchange after cleanup and rank-local result validation. The focused suite now passes 57 tests and the combined forward-diagnostic, generation-zero, and soak-790 host suites pass 157 tests. The reviewed and pushed diagnostic identity is

```
commit 70165b017cd28df832a9a8797a72a1f8234284d0,
tree c58e220299a54525c443f4f0dc204e22e1fb19b8.
```

The plan, helper, runner, and host-test SHA-256 values are respectively

```
7aff0bea0a85887e7744f57816239dd598426c5bc4623ffb140a01f410a98f26,
52583f1d8cfd571897a446ce00a69b598a72889e244be9e342498fd32010f,
eeb1704bdb8bb30a376ba5483747e1ad6022426f96fe6cab3e8627cf642fe0fd,
71a20bbcfce4c1109766263bca07e5e8099312f7ec6196d4b2e276aba62a3a65.
```

The 4,242-file immutable source snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_forward_determinism_
70165b017cd28df832a9a8797a72a1f8234284d0,
```

verified against source-manifest SHA-256

```
e1129c0caa1832c5bc5ff3d2f978a8c6a18a38a1c55f7d0027b1792ddb77e665.
```

H200 diagnostic job 1836996 ran from this snapshot on h200-079-003 and h200-079-202 from 2026-09-04 06:17:47 through 06:37:39 UTC. The allocation and every step .0–.12 completed 0:0. It produced the exact preregistered closure: 64 rank JSON records, 64 NCCL logs, eight node/phase cache inventories, 154 mode-0400 files recorded by `artifact-inventory.json`, and 155 final directory entries because the inventory does not list itself. Independent validation returned `VALID`; the log has exactly one non-authorizing completion marker and scheduler stderr is empty. The retained artifact root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/forward_determinism_70165b017cd2_1836996/artifacts.
```

The summary, artifact inventory, invocation, scheduler log, and empty scheduler stderr SHA-256 values are respectively

```
741e75362c6214b4a82644bbab2b12b61fc6ed827e80ff4ac4d3f86be94861ff,  
463b68fc3d70f85682fe03137df857e5e577d9c9aae009e7f7c13659f591931f,  
0179440c6f58ac1935626684916b4599bd2035820cde6ee4ccf1cc5ebcddfde5,  
45a7018391bd5ec572d48c2557d3a0239cadd679289cf465f8e241a444cfb29a,  
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.
```

The pre/post generation-zero input inventories are byte-identical, both with SHA-256 b5bc763929a966c54adbfef1d9a1d9d1c0ae891556860ad89449a333fec50cbe.

The preregistered classification is RMSNORM_AUTOTUNE_SELECTION_STRONGLY_SUPPORTED. Every initial-state and input comparability axis matches, every within-process warm repeat matches its cold result, and both warp-8 phases match exactly across processes. By contrast, the two ordinary cold processes diverge on ranks 3, 6, 12, and 14. Those are exactly the ranks whose selected forward RMSNorm warp count changes: respectively $8 \rightarrow 4$, $2 \rightarrow 4$, $4 \rightarrow 8$, and $4 \rightarrow 1$. Their first warm-trace difference is `block_00.norm_1`; their external logits, unreduced loss, and reduced loss then differ, while z-loss remains equal. Replaying each selected configuration reproduces its recorded warm output, and all six forced configurations are stable when compared at equal input, weight, rank, and configuration. Thus timing-dependent RMSNorm forward autotune selection is strongly supported as the source of the observed first-forward variance. This remains diagnostic rather than production authorization: it does not test backward repeatability, and a narrowly scoped fixed-warp production control must pass a fresh unchanged generation-zero hard gate.

The corrective implementation is now isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_rmsnorm_control_20260904
```

on branch `codex/rdiag-shared-dt-rmsnorm-control-20260904`, based exactly on diagnostic commit 70165b017cd28df832a9a8797a72a1f8234284d0. Its executable plan is `docs/plans/2026-09-04-rdiag-shared-dt-rmsnorm-determinism-plan.md`. The selected scope is the narrow forward-only control justified by job 1836996: for the four resolved shared-DT configurations only, require the cold RMSNorm forward autotuner's exact existing six-candidate closure and install its unique existing warp-8 object as a singleton. The backward RMSNorm tuner, every FLA tuner, precision policy, recurrence, parameters, and generation-zero helper/runner remain unchanged. Activation keys on the resolved internal config identity rather than the raw CLI name, because the generation-zero diagnostic deliberately uses a temporary CLI alias whose copied payload retains the production internal name. No environment selector, config/recipe field, or runtime-fingerprint field is added: the historical runtime, data, model, and sidecar identities remain fixed, while the new source-manifest SHA binds the control. The implementation must fail closed on warm, altered, or ambiguous tuner state, emit a canonical policy digest with rank consensus before model construction, cover the direct evaluation and gradient-gate constructors that bypass `pretrain.py`, and prove that the backward tuner is not mutated. The process-global pin is intentionally irreversible; every authenticated entry point runs one shared-DT model family per process, and a mixed-family caller must use a separate process rather than attempting to restore Triton global state.

The initial policy implementation and smoke harness were reviewed, committed, and pushed as

```
commit 28ea1c51a16825ae02d3d4c6520087e8b9b2ca42,  
tree 1f7dd60edbebe92b429e4a541fdea85e6ec3c182.
```

The plan, policy module, trainer integration, authenticated evaluation integration, and gradient gate integration SHA-256 values are respectively

```

91cc8fd7580ff953f71cedd5594a489eacc126168bce68904d7b17ffebbcbbb2,
74898f2dfa3c3073e491616eb7e847acdaa695c4b8b70596af5534e5c6591433,
ff9fa812d81266577c5d9b4404966e039e1bad205abf19009099ffc24651a584,
b78e03c7996933e5c68a2e08a46ab074a8a235b5ac783e2588a523e14497c5f8,
755fa87cc6598b21391a06de20a9b25ce1a9286b383e8e714a26764c141015e6.

```

The corresponding implementation entry points are

- `lit_gpt/rdiag_shared_dt_rmsnorm.py` for applicability checks, fail-closed tuner inspection, installation, and the canonical policy record;
- `pretrain.py`, `scripts/eval/eval_lm_harness.py`, and `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab_run.py` for authenticated training, evaluation, and gradient-gate activation; and
- `scripts/tests/rdiag/test_rdiag_shared_dt_rmsnorm_policy_host.py`, `scripts/tests/rdiag/test_rdiag_shared_dt_rmsnorm_integration_host.py`, and the policy smoke files below for host and H200 contracts.

The smoke helper and runner are `scripts/tests/kernels/test_rdiag_shared_dt_rmsnorm_policy.py` and `scripts/tests/kernels/run_rdiag_shared_dt_rmsnorm_policy.sbatch`, with SHA-256 values `aa58e4cfa0efb2f35e4cf4c38d86eb1492a1fbba359b644c7cf526c8c15e13cc` and `85f8ce9bf11cf61d006916ac44c3955fbeb1f7b1c070ac17453f4afd89f850c84`. It exercises the actual wrapped Triton forward at $M = 16384$ and widths 1024, 2048, and 2304, requires repeat and two-rank output/rstd byte equality, and benchmarks all six existing candidates with 40 order-balanced samples each. Warp 8 must remain within $1.25\times$ the empirical fastest candidate by both median and lower-quartile latency. Installation leaves the backward candidate objects unchanged; a later backward smoke only proves that the untouched path still executes. At the initial implementation revision, the focused policy/integration/smoke contract suite passed 49 tests. The combined existing generation-zero, forward-diagnostic, runtime-identity, gate-runner, and authorization suite passed 309 tests with three declared skips when pytest temporary files were redirected to Lustre; an initial attempt exhausted the already-full system temporary filesystem and is not a code failure. After the later smoke-helper and launcher-only revisions, the final focused runner/smoke-host suite passed nine tests. Independent final review reports no Critical, Important, or Minor finding. The generation-zero helper and runner remain byte-identical to commit `70165b0`, with SHA-256 values `a37fdea21c59757246c6fabff0dac15d8ca93ac3cfdc784d0549413544e30f70` and `f9d5c8a04f7384b360eb04435c32d0a2abeb177e6f6534e91f4910e5b7c1bb14`.

The 4,249-file immutable control snapshot is

```

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_rmsnorm_control_
28ea1c51a16825ae02d3d4c6520087e8b9b2ca42,

```

independently verified against source-manifest SHA-256

```

4207beb5e9eb0aeaaa11c7eb489d2295122ac257ecedfc77b5dfde21f35241c1.

```

Non-authorizing two-H200 smoke job 1837294 was submitted from exactly this snapshot and failed before Python after two seconds. The batch shell and its `srun` task steps receive different node-local `/tmp` namespaces on this cluster; the batch-created cache parent was therefore absent in both GPU tasks. No numerical test executed and no result JSON was published; the empty artifact directory is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/rmsnorm_policy_28ea1c51_ZxLw0VLa/artifacts.
```

The runner correction creates and removes a distinct private /tmp root inside each `srun` task, preserves the workload exit status through guarded cleanup, and has passed its seven host contracts, shell syntax check, and independent review. Because this changes the runner bytes, the reviewed and pushed replacement identity is

```
commit edf5bb3aa95e2d64beabc8f4196c56029851b037,  
tree fbcc46493836860a268d9fe56514ca8eadd38993.
```

The corrected plan, runner, and runner-host-test SHA-256 values are respectively

```
cb77a67a4e06e5d31c93547e4560064eb10bd045261a37a8bc5b971e7338fdb7,  
48463cff46aa3eae19a46514655dfac90b8886e02a8df08036a9df4ad12c66ba,  
2b06e25a26341d7eb108f06cabe7c087a7ea035d9ede29715eb36f801814b75.
```

Its independently verified 4,249-file snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_rmsnorm_control_  
edf5bb3aa95e2d64beabc8f4196c56029851b037,
```

with source-manifest SHA-256

```
2dc7fc4d3856c43f607e00df606a63c4c221fcbd5c5da5f461e6fab34fc59c98.
```

Replacement smoke job 1837353 reached Python from this corrected snapshot but failed on the first NCCL collective: task-local one-GPU visibility caused NCCL 2.27 to report `invaliddeviceordinal`. The irrecoverable allocation was cancelled after 2:08 to release its H200s. It also published no result; its empty artifact target is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/rmsnorm_policy_edf5bb3a_MBXIibLe/artifacts.
```

The second launcher correction follows a completed project H200 reference: both allocated GPUs are visible to both tasks via `--gpus-per-node=2--gpu-bind=none`, and ranks select CUDA devices by `LOCAL_RANK`. It also prints rank-local phase tracebacks before recovery and re-raises fatal CUDA/NCCL failures instead of attempting another collective on a poisoned CUDA context. The reviewed and pushed identity is now

```
commit fd9ca79e697b2fce14b8938459a7d2e63e434bea,  
tree b72796b9106c8e06d2b7e59051e5cab094ec8b40.
```

The current plan, smoke helper, runner, and runner-host-test SHA-256 values are respectively

```
61277a3f3db22a8495f5cc2cfbdd50fd038cc53664c219c04bcacad8e8f3c572,  
af2f17b7b178930291a1225345373713c3fc1357165a3812afa9f237c494d31e,  
87652ee3fdb1fd54cf17cfdbcb32e65877aa1fa27b8c146af96b860e0fe9d38f,  
cfb736f894504e8bc2f941074452579dbb87c825ce5fd7cb54c36e8674821d69.
```

Its independently verified 4,249-file snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_rmsnorm_control_  
fd9ca79e697b2fce14b8938459a7d2e63e434bea,
```

with source-manifest SHA-256

```
0d49e9471d3e6a3b48bc21838b6366cf26d7b8f9418639b70a5f012dec866f20.
```

Smoke job 1837393 ran from this snapshot on h200-129-161 from 2026-09-04 07:51:39 through 07:52:08 UTC. The allocation, batch, extern, and sole srun step all completed 0:0. It published exactly one retained artifact,

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/rmsnorm_policy_fd9ca79e_gz8Nu1Ly/artifacts/result.json.
```

That mode-0400, single-link, root-owned result is 17,835 bytes and has SHA-256

```
f39137e97c07d7cafc9d930b4060dce798b352a439aba9a22201d106107aeb30.
```

The scheduler log SHA-256 is

```
a6228b2518a21a27e627c69c2f0b61d96613396230833eaa7acd2c757b288490.
```

Independent validation returned VALID. The result is explicitly `authorizing=false` and `production_gate=false`. Both ranks agreed on policy record SHA-256

```
cd31a0f1bc60aa5421cb7074f5236a14ff4fb66d59632e587e0920f4e343ecac.
```

At widths 1024, 2048, and 2304, the first forward used the exact pre-existing warp-8 object, the forward cache remained empty, repeat outputs and reciprocal standard deviations were byte-exact, and the corresponding cross-rank values were byte-exact. Across all rank/width pairs, the maximum warp-8-to-fastest ratio was 1.04614 by median and 1.04703 by lower-quartile latency, well within the preregistered $1.25\times$ bounds.

The deliberately untouched backward smoke also exposed the next concrete risk. Rank 0 selected warp 16 while rank 1 selected warp 8; equal forward outputs and weight-gradient hashes accompanied different input-gradient hashes:

```
f364b0deea41662c4571b76f4ae57c9d45ac5923b47b0d0685f357ca7f83647,  
ae9fca3f6523a4a59d0670dd4af9bdbf041c6992f5df43eb37907baa1a9be3d3.
```

Backward cross-rank equality was not a preregistered requirement of this forward-control smoke, so the bounded smoke validly passes; the observation nevertheless makes RMSNorm backward autotuning the leading suspect if the forward-only control is insufficient. The unchanged replacement generation-zero run discriminates that sufficiency, not backward causality. That gate, not installation or this smoke, decides whether trigger capture can begin. If it does not return `minimal=MATCHED`, focused backward-RMSNorm localization is the next priority, and a separately reviewed narrow control is warranted only if that localization implicates it. No trigger or production authorization is presently claimed.

The unchanged replacement generation-zero job 1837446 then ran from the same fd9ca79e... snapshot on h200-004-175 and h200-071-008 from 2026-09-04 07:57:40 through 08:27:18 UTC. The allocation, batch, extern, and all steps .0-.18 completed 0:0. Its retained artifact root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/gen0_repro_fd9ca79e697b_1837446/artifacts.
```

Independent validation returned **VALID**: the directory contains the exact 214-file inventoried closure plus **artifact-inventory.json**, for 215 mode-0400, single-link files. The summary, inventory, invocation, scheduler log, and empty scheduler stderr SHA-256 values are respectively

```
c69adb84b265f7c20560ad1be54a7654fcda963ffe068cda252f398c942d9c54,  
97a46cae0bc31162351ffa1d39acf6f55b048d4f8ccc22677b549082a3c64a14,  
2bf614563fa4f82662220809cfaf2c1b9d23f512e11400e87fa11db40644f179,  
fb0cbcf3c30f950f5b899095bb603caf1110cb369eb25eff6bdba682d396b1e90,  
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.
```

The log has exactly one completion marker. Its pre/post checkpoint-input inventories are byte-identical, both with SHA-256

```
20bbc0c2a345eaa46671d670c0d865b6f9a73266c9310bd81a1f43243ed120fd.
```

Both node runtime-identity records retain SHA-256

```
135ea68e47d1e9728f1295e480dd22518566f818619d06b6853bed343b00639d.
```

The hard gate passes. The reconstructed common anchor, rank topology, training inputs, and sidecar inputs match; all three loader arms match over their first six rank-local microbatches. Both preregistered replicate-pair classifications are **MATCHED**, including the required **minimal=MATCHED**; global loss and pre-clip norm are exact at each of the first three update boundaries. The overall training classification is **MATCHED_RECONSTRUCTED_FIRST_THREE_UPDATES**. This remains non-authorizing and does not make the reconstructed process RNG historically comparable to the failed production run, whose RNG was not retained. It does establish that the forward-only control is sufficient for the generation-zero repeatability gate. The backward-smoke observation remains a risk note rather than evidence warranting a second intervention, so no backward pin is introduced. This result released revision, review, and non-authorizing execution of the trigger-capture diagnostic; it does not authorize production.

The gated follow-on capture work is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_trigger_capture_20260904
```

on branch `codex/rdiag-shared-dt-trigger-capture-20260904`; its plan is `docs/plans/2026-09-04-rdiag-shared-dt-trigger-capture-plan.md`. Its earlier plan-only identity was

```
commit 8074daf35bbd3dac839f0fc2601361d7cf069d50,  
tree b27a3f6ab3269df09daf02f08507f178a77cefe9.
```

The accepted control lineage was merged without rewriting either parent as

```
commit 57afdd1f9fe497f3630f262f7febfcbdd56e4f7e,  
tree ad12554140dac65bb315dd49a0c7fc003393acb1,
```

whose parents are `8074daf...` and `fd9ca79e...`. The plan was then revised, independently reviewed for evidence integrity and runtime semantics, committed, and pushed as

```
commit dfd928cf7e861f5f828f6486730cf2836cdd3089,  
tree 1666a25b3ef188dca6c151a18209c11d57b470f9.
```

Its reviewed plan SHA-256 is

48f65aaf84d0c0ce403ed25bc9bd860c33b511c2119c28191b3e71123473e7d8.

The plan pins job 1837446’s summary, inventory, invocation, source, input, runtime, and log identities at their actual nested classifier paths. Its exact 1,281-byte root scheduler row is retained separately, without modifying the closed generation-zero directory, at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/gen0_repro_fd9ca79e697b_1837446_scheduler_record/  
sacct-root-row.txt,
```

with mode 0400, one link, and SHA-256

7b3df5231f751e1841c448cb55714c471522bcd74bbb973b577477785f07f587.

Review also closes process-RNG changes during loader read-ahead, rank-asymmetric capture failures, unexpected loader EOF/worker errors, nonfinite-forward handling, live RMSNorm autotuner provenance, and the two earlier launcher topology/cache failures. The executable scope remains the plan plus `scripts/analyses/capture_rdiag_shared_dt_trigger.py`, `scripts/analyses/run_rdiag_shared_dt_trigger_capture.sbatch`, and `scripts/tests/rdiag/test_rdiag_shared_dt_trigger_capture_host.py`. The implementation was independently reviewed, committed, and pushed as

```
commit ef57031fc2f342a0dfe9c9ef7af2d0e268e332cb,  
tree db686e18708233ecbb049cb68cbc1ace9d81d4d6.
```

The initial helper, runner, and host-test SHA-256 values were respectively

```
ec0db20eb6f0609425c5add313dab82c111b7619e8c2c0f94840ed45cc0e29d3,  
11c92b07965f69e622fe7ec0af3ae00256942eae7c9192adda7d15cf6cde95e,  
56eb9d0a04d053a0e1e19724155e56dc3f1d225ee150c4d18bbbf8c19127f8a8.
```

The initial focused suite passed 60 tests, the adjacent RDiag suite passed 170 tests, and the broad compatibility sweep passed 495 tests with three skips. Its immutable 4,253-file Git-object snapshot, excluding only the paper gitlink, is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_trigger_capture_  
ef57031fc2f342a0dfe9c9ef7af2d0e268e332cb,
```

with source-manifest SHA-256

5d3edaa28885ee012fde8887f96af827f0fe320b84eba4ea5e0b5ff46cccb907.

Both the publisher’s closed-tree verification and a fresh private-copy reconstruction from Git objects passed. The filesystem’s unavailable no-replace rename path required the publisher’s fail-closed copy fallback; the final execution root is owner-held mode 0555. Hidden staging siblings retained by publication and by a concurrent collision check are evidence only and are explicitly rejected as execution roots.

Non-authorizing two-node, 16-H200 trigger-capture job 1838085 was submitted from that snapshot at 2026-09-04 10:11:12 UTC with the exact failed-run data manifest. It ran on h200-205-088 and h200-231-137 from 10:30:02 through 10:33:51 UTC, then failed closed before capture training. Source-copy, topology, runtime-identity, and two-node exit-cleanup steps all completed 0:0; the root and batch exited 1:0. Its run root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/trigger_capture_ef57031fc2f3_1838085,
```

where `anchor` and `output` are empty and `artifacts` contains only six preflight files. There are no capture results, NCCL logs, summary, inventory, checkpoint, or completion marker. The external root and full-step scheduler rows have SHA-256 values

```
2d9eae9ab2fc00dd7e407629b9214d97efb4629fb1753f5bbac9b5f0ebf9380f,  
395f5ebda2a1ad3910b956209ead563a3de974fd80f9a72a1a27b6fec76105db,
```

under `diagnostic_runs/trigger_capture_ef57031fc2f3_1838085_scheduler_record`. Scheduler record filenames are `root-row.txt` and `all-steps.txt`. Scheduler stdout and stderr are

```
/lustre-storage/checkpoints/ngocbh/hdn/ckpts/logs/  
rdiag_sdt_trigger_1838085.log, and  
/lustre-storage/checkpoints/ngocbh/hdn/ckpts/logs/  
rdiag_sdt_trigger_1838085.err;
```

their SHA-256 values are respectively

```
158b30dd9c16f898aed9b40df8272e059506c806384c8a75359ba919fe5a5447,  
196a496ecaa4e47cba69b79bab5292c81d2006f65de9861d43c47701623cba87.
```

The defect was a strict framing mismatch rather than a scientific failure: the helper queried `sacct-P (--parsable2)`, whose 19 fields have 18 separators and no trailing `|`, but parsed it as `-p` output with a trailing separator. The retained 1,281-byte job-1837446 row and a fresh `-P` query remain byte-identical with SHA-256

```
7b3df5231f751e1841c448cb55714c471522bcd74bbb973b577477785f07f587.
```

The minimal correction keeps `-P`, requires exactly 19 fields, and uses a strict zip; it does not alter any scientific path. It was reviewed, committed, and pushed as

```
commit 4cbddbc1f715e5eb3a4177fe1557b29d7ab4f3d0,  
tree 869a5f036fe8cf2bcb085c8b4e8d38cb5e64f550.
```

The corrected helper, unchanged runner, and corrected host-test SHA-256 values are respectively

```
3d6879298bed98dfc6fa992d518d7d0ef0ae80267c4e8d5c7db3a134ba1abca6,  
11c92b07965f69e622fe7ec0af3ae00256942eae7c9192adda7d15cf6cde95e,  
8eacb854d8d3297c2e9f1eeb2a19a404f51521f900027e29af47f3ff3957602d.
```

The corrected focused suite passed 68 tests and the broad suite passed 503 tests with three skips; the full real prerequisite command also passed against the pinned checkpoint, generation-zero closure, data manifest, scheduler record, and initial snapshot. The replacement immutable snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_trigger_capture_  
4cbddbc1f715e5eb3a4177fe1557b29d7ab4f3d0,
```

with source-manifest SHA-256

```
e27fb7c3a64465ddd3b4d92cbd9d3dc0e83d0e2f9d5a48e8b69540e7f9327b6d.
```

Both publication verification and a fresh private-copy reconstruction passed. Replacement job 1838250 was submitted at 2026-09-04 10:48:12 UTC, started at 12:21:08 UTC on h200-094-010 and h200-094-088, and failed closed at 12:41:26 UTC. The two-node source-copy, topology, and runtime-identity steps completed 0:0; exact snapshot, generation-zero closure, data-manifest, and retained scheduler-evidence verification all passed, so the corrected parsable2 boundary has been exercised successfully. The intended 16-rank capture step started at 12:24:44 UTC, reconstructed the 621,284,352-parameter model, resumed the exact generation-zero state, and trained normally with 60.4 GB peak allocation per GPU. Update 835 was the last printed completed update. An eligible finite, rank-agreed trigger with norm in [3, 10] subsequently became pending before update 840, but its exact boundary and norm existed only in process memory and are not recoverable from this failed attempt. During anchor publication, all 16 ranks then failed together in `anchor-rmsnorm-tuners` before any anchor write: `inspect_rmsnorm_tuners` passed the fully qualified display strings `lit_gpt.rmsnorm._layer_norm_fwd_1pass_kernel` and `lit_gpt.rmsnorm._layer_norm_bwd_kernel` directly to module `getattr`, although the module exposes their basename attributes. Capture step .3 therefore exited 1:0 after 16:39; cache harvest/removal step .4 and private-source cleanup step .5 completed 0:0. This is an implementation failure after trigger selection, not `NO_TRIGGER_THROUGH_900` and not a scientific trigger result. Its deterministic run root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/trigger_capture_4cbddbc1f715_1838250.
```

It contains 43 mode-0400 partial artifacts, an empty anchor root, and only the expected zero-byte checkpoint placeholder under `output`. Rank capture results, `capture-phase`, post-input inventory, invocation, summary, artifact inventory, loader verification, and completion marker are all absent. The scheduler stdout/stderr and capture stdout/stderr SHA-256 values are respectively

```
2fae5175174f72c705af06fb0023a333785f447f9d83214f00e126d921a4aa1b,
028a4dacfc45125ba976c3990402c20a538e306929c352f6b8178d99351f5d8a,
1d39eebde75592b3a4fa202a37b0be5f122884f05cbfcb34b9a50b8c42d24c3,
aa7dbceeb04a817036fafc5192c1b32a7e921357e7348ce7ac2fce3c8f4567e38.
```

The externally retained root and nine-row full-step scheduler records are mode 0400, one-link files under

```
diagnostic_runs/trigger_capture_4cbddbc1f715_1838250_scheduler_record,
```

with SHA-256 values

```
3db33e5c49a94e7fcb7dfef72a19f9b4d6c67d0597a2108940e140eaf24a870e,
012202c4d97ce0ae5a7ccbcccc002e2bef202ca192cb182f8f0ea51bd1b96185.
```

The minimal repair resolves each qualified name's final component for Python lookup while retaining the qualified string in canonical policy evidence. A host regression now exercises the real qualified-symbol/module-basename shape and proves inspection leaves both tuner states unchanged; the focused suite passes 69 tests and the focused-plus-policy suite passes 91. Independent review found no blocking issue; Python compilation, shell syntax, and Git diff checks pass. A deliberately overbroad login-node sweep was not an acceptance test: its CUDA cases rejected the GPU-less host and one unrelated source-publisher test observed the already documented Lustre staging fallback. The reviewed and pushed repair is

```
commit 015ba66e07f0ec1860b188ea74e638ce955384cc,  
tree d1d59c7a08112e29ac942b0153f36552a7cc60ea.
```

The repaired helper, unchanged runner, and regression-test SHA-256 values are respectively

```
c732721ae49d4798a2ff280300ffb09f7e4f83470ee415092e8955dd7f51ca5e,  
11c92b07965f69e622fe7ec0af3ae00256942eae7c9192adda7d15cf6cde95e,  
02ec5161fbb73dbe1db0a1ab197fc9cf908844fa04c1871a48ac8e9e1ad726ef.
```

Its immutable 4,253-file snapshot, again excluding only the paper gitlink, is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/rdiag_shared_dt_trigger_capture_  
015ba66e07f0ec1860b188ea74e638ce955384cc,
```

with source-manifest SHA-256

```
bf0b5288a873f5f7c67963492239226921150b2c0f7350de6a3cf57367987dca.
```

Both publication verification and an independent fresh private-copy reconstruction passed. Replacement job 1838750 was submitted from that snapshot at 2026-09-04 13:06:42 UTC and ran on h200-094-010 and h200-094-088 from 13:09:11 through 13:45:29 UTC. The root, batch, extern, and numbered steps .0-.8 all completed 0:0; the numbered task counts were exactly 2, 2, 2, 16, 2, 16, 2, 2, 2. Its sole completion marker is present and scheduler stderr is empty. The preregistered first eligible trigger was absolute boundary 838, with rank-agreed norm 3.4058520793914795, canonical token `finite:0x1.b3f2f60000000p+1`, and zero health failures. It retained the post-update generation

```
step-000000000838-iter-000000001676-6ee7296afb63448f9863efab3454d49d
```

plus 16 RNG records and 16 future raw microbatches per rank. All 16 independent loader-replay records match every ordered joint and component-tensor hash. The continuation later stopped non-authorizingly at boundary 853 with finite norm 11.937856674194336; that postselection stop does not invalidate the retained boundary-838 anchor.

The accepted run root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/trigger_capture_015ba66e07f0_1838750.
```

Its summary, artifact inventory, invocation, byte-identical pre/post input inventory, capture phase, derived anchor manifest, and checkpoint-generation manifest SHA-256 values are respectively

```
0123887d03356ca08a0e88aa267746ac1d4680b4c17a6fb63f98800175bb9b33,  
0709972f30d9b4118e41865d8ecc03a55dda9f34bdc856d87d3ef23cbee48500,  
04f0d067c7c52cac583f9a5ca528282a5c757c560e6135a8007eedb093586509,  
729aee6bea81535c49a5c16092750d0ff603275b866fdca662abf7ca23d3231,  
05b5070b22064b56e1f3aba9403099824c45c338d3c5c6b676cb5918a89461f2,  
1c01aa9798bcf3c1d75cc016bd103cac919ca7c8296cddffb4817bede8353bd8,  
3b41e9f4590909f461b89aaec0041ff8b7f51347ae531bcd07faf18dba36d008.
```

The artifact directory has exactly 164 inventoried files and 165 physical files including the inventory, all mode 0400 and single-link. Its decision is `CAPTURE_RETAINED_NONAUTHORIZING` with `CAPTURED_LOADER_REPLAY_MATCHED`. The external scheduler-record root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/trigger_capture_015ba66e07f0_1838750_scheduler_record,
```

whose root-row, full-step, completion-marker, and scheduler-log-hash records have SHA-256 values

```
6987ed8bf54cc5b80713503dce990fad621032a4c0d2a4d3c1116c1fd0828494,  
3afc238b32b6631cba1bd64b15dd29f012c366c2f47b60789ab754fa57e8428a,  
f00be3ff84a1349b7bb29cb5a5460de217f3c932d7389d736e79809dc6d6b6ba,  
7fb33026b6196a9d9321774c1650e1d41f302c4ff60a8dcc1e3f5a9b2fc25773.
```

Scheduler stdout is fb69191ebd7bbc42e9a20222629ae4f03525badc5df4544fb59871f02c5b2b50; stderr has the empty-file digest.

A separate read-only validator, job 1838966, then ran on h200-002-213 from 13:52:43 through 14:01:15 UTC and completed 0:0. It reconstructed the invocation-bound historical path /tmp/rdiag_sdt_trigger_1838750/source, verified the immutable 4,253-file source before and after validation, revalidated source, generation zero, and anchor, returned VALID with artifact_count=164, removed both owned node-local roots, emitted one marker, and left empty stderr. Its evidence root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/trigger_capture_015ba66e07f0_1838750_independent_  
validation.
```

The retained validator runner, validation result, acceptance result, root-row, full-step, completion-marker, scheduler-log-hash, and complete evidence-list SHA-256 values are respectively

```
ce3ee70f2d5f80a9b98ab7fc692ad93d139d3d319115e4b3b889ff8478f2b600,  
2f50dd577f9b5a507280c9ec913608b22f8f56a7028dc0f852a66848b544be42,  
7e564d9f4131bbc820390cc798929739d8e291cb9535d1196e3a6e64d1511314,  
254091248607cbbd8bb476ae5350029974fdb6be1d287fe889cf91c7dece84c9,  
acde647a47074436c368fcde4cce661b49ae4bb250a53786d87a0f00e73a0444,  
0a2257b4e2b4f8c063059b52c7be11d353a2fa39a547c448d253bf6c4f247565,  
d455f9590ade96882b0cffe52132ef96977202c5e26bd559dbf5a084907647d3,  
35e2822bc0a296487522dc3f6fe8a10b17cbe196085c5227b2e095425c393cbb.
```

This closes the trigger-capture prerequisite for implementation; it remains diagnostic and cannot authorize production by itself.

The downstream matched-intervention design and implementation are isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_worktrees/rdiag_shared_dt_matched_intervention_20260904
```

on branch codex/rdiag-shared-dt-matched-intervention-20260904; its plan is docs/plans/2026-09-04-rdiag-shared-dt-matched-intervention-plan.md. The original reviewed and pushed plan-only identity was

```
commit 2b8066717dc05128a37404def1dcfa643fd0ef7d,  
tree 6aed24664104fa61833f95a7b5cc3626f1baea5d.
```

It requires one immutable post-trigger anchor and sequential fresh shared-A1, connected-zero no-rotation, and shared-A2 arms over 16 retained microbatches per rank. Exact A1/A2 equality is a precondition for interpreting the intervention. Review corrected the initial draft to compare

rank-local FSDP state only with the corresponding rank across arms, use replica-aware global gradient diagnostics, limit first-boundary DT/alpha equality to the treated layer, stop only after the eighth update is fully committed, capture nonfinite-loss exits artifact-locally, retain the ordered 32-loss boundary estimand, and bind all 256 rank/batch joint hashes. That original plan’s SHA-256 is

```
17a7d50a0349ed1038b4c703dfe25a12e791e5da40c194b5493a46ee087ca0db.
```

At that stage, a read-only merge preflight found no textual conflicts, and the branch deliberately did not absorb the plan-only trigger tip; it remained gated on a successful final trigger capture and repin before implementation.

Before observing the trigger outcome, the plan was tightened to pin accepted generation-zero job 1837446, current trigger commit 4cbddbc... and manifest e27fb7c3..., and the still-pending retry without claiming its result. It now fixes the 32-loss ordering and arithmetic, phase-dependent unavailable fields, value-exact zero semantics, first-boundary frontend scope, early-stop estimands, operational-failure continuation, nonfinite loss interception, and reconstruction of the trigger invocation’s historical source path. The reviewed and pushed revision is

```
commit 444946e6b1f588c4bcfffc083c54ac80ac1cca6d,  
tree ffa896e55eebdb48f15c4b71ae2bb89da117c2e3,
```

with plan SHA-256

```
43dc150759e293f39e810f42cc4d6dae11909d5ce28d10f4753a5fa6c2eefa51.
```

The acceptance clause explicitly requires both a structurally valid trigger closure and CAPTURE_RETAINED_NONAUTHORIZING with CAPTURED_LOADER_REPLAY_MATCHED; a valid no-anchor or loader-mismatch outcome cannot release implementation. Because revision 444946e... pins failed job 1838250, it is now a historical blocked design record. The reviewed plan was repinned without implementation to pending job 1838750, trigger commit 015ba66e..., tree d1d59c7a..., manifest bf0b5288..., the exact historical source path /tmp/rdiag_sdt_trigger_1838750/source, and the corrected retained-anchor closure of 164 inventoried files (165 physical files including the inventory). That reviewed and pushed plan is

```
commit 41de0ee4836fa492e560f0d1640f6cac73c13382,  
tree 4ebdfc9335b9f4c2f11c6e2c86e98a48eca21587,
```

with plan SHA-256

```
e7a4df5999f17cae172d9de2f454299fca38879773e225690a14c2aed2c91ff8.
```

That repin was the last pre-outcome design state. After both job 1838750 and external validator 1838966 passed, the accepted trigger source was merged without conflict as

```
commit 9d65a587be64f0a7dc9b2c3b3ed40ac0351c08ae,  
tree 53136608092429aad484efcaffed22cb021233cf.
```

The plan was then independently checked against every retained job, source, scheduler, artifact, anchor, and validator identity and pushed as

```
commit 285c36b9e75c897521e6529cd0e5766c2fcafb7,  
tree 3787abb2b4480cf73cc581e718a3558706a6dd5d,
```

with plan SHA-256

```
3b869d76eb17fafcca9ddb1364524927be4a924642f726d8a27649b076c98355.
```

The reviewed implementation is commit

```
49f06078d19f349ca9724e3bbfc08fa436e91e54,
```

with tree ebb54a1719a6bf8490e9d6997ec471b5201423f3. Its implementation paths are scripts/analyses/rdiag_shared_dt_matched_intervention.py, scripts/analyses/run_rdiag_shared_dt_matched_intervention.sbatch, and scripts/tests/rdiag/test_rdiag_shared_dt_matched_intervention_host.py. The final plan, helper, runner, and host-test SHA-256 values are respectively

```
13d5bf40a3c5f5974a64a91b4cf3ae72efd999e1bbf012f2e57fc3bac32643c3,  
0a339701716fcb5d404deecd849d36a3cbd40eee654520a229801de90204b180,  
ceaae69aacba7b12325188c7b50d2e503977968f0022b777cfc79c331b40f935,  
09002a176e6eceedf504746b774d0dcf50db08b02b616cc53c5e40c722964482.
```

All implementation steps before publication are complete. The focused host suite passed 126 tests, and the matched plus trigger, generation-zero, and soak-replay sweep passed 295 tests; Python compilation, shell syntax, diff hygiene, direct validation of all 16 retained rank-local initial states, and live scheduler revalidation of jobs 1838750 and 1838966 also passed. Independent scientific, distributed-runtime, and immutable-artifact reviews found no remaining code blocker after repairs to collective forwarding, post-update evidence, rank-local failure coordination, exact anchor binding, nonfinite-loss rederivation, contiguous-prefix estimands, and fixed-order arm continuation.

The immutable source snapshot is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
source_snapshots/  
rdiag_shared_dt_matched_intervention_  
49f06078d19f349ca9724e3bbfc08fa436e91e54.
```

Its SOURCE_MANIFEST.json SHA-256 is

```
45644d7de85703ceacfb4789925245097b00332f1c9a9c5df8da334ce8ce3a5.
```

The manifest contains 4,257 regular Git files; the physical frozen tree contains 4,258 files including the manifest and 409 directories, with directories and executables mode 0555 and other files mode 0444. The **paper** gitlink is recorded and intentionally excluded. The publisher's built-in verification and a separate reconstruction into a private source tree both matched the commit and manifest exactly. Because Lustre used the supported no-replace fallback, the complete read-only hidden staging tree is retained as publication evidence; it is not an execution source.

Exactly one two-node/16-H200 matched-intervention allocation, job 1839531, was submitted from that named snapshot at 2026-09-04 15:37 UTC. It ran on h200-071-128 and h200-074-040 from 15:59:07 through 17:06:49 UTC and completed 0:0. The root, batch, extern, and numbered steps .0-.16 all completed 0:0; the numbered task counts were exactly 2, 2, 2, 2, 2, 16, 2, 2, 2, 16, 2, 2, 2, 16, 2, 2, 2. The run executed fresh shared-A1, connected-zero no-rotation, and shared-A2 arms in that order, with no requeue and no production authority. Its run root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/  
diagnostic_runs/matched_intervention_49f06078d19f_1839531.
```

The in-job final validator returned **VALID**, revalidating the matched source, accepted trigger anchor, and all prerequisites. The artifact inventory contains 118 closed files and the directory contains 119 physical files including the inventory; every artifact is a mode-0400, single-link regular file. The summary and inventory SHA-256 values are respectively

```
140ca76485e6838c6ae3f6bfc086e0112aae5c48f809cf06adc92724efe83c,  
bdd78e22df0ac67242c36f559b9ede1c1ae3485b575bc3647d25285599d62765.
```

The validity checks passed exactly. A1 and A2 are byte-identical under the complete common-arm projection, with `a1_a2_first_difference=null`; initial state, topology, all 256 rank/microbatch inputs, and pre-forward RNG records match. All three arms committed eight updates through boundary 846. The no-rotation arm has exact-zero angle projection output, phase increment, phase, and current angle gradient while the radial gradients remain live; layer-0 DT and `alpha` match the normal arm exactly at boundary 839. Consequently the summary reports `intervention_isolation=MATCHED`, no isolation failures, and causal estimands on every boundary 839–846.

Table 2: Matched boundary-838 intervention, job 1839531. “On” is the exactly repeated normal shared-DT arm and “off” is connected-zero rotation. Only boundary 839 is an instantaneous counterfactual; later rows are paired longitudinal outcomes.

Boundary	g_{on}	g_{off}	$g_{\text{on}} - g_{\text{off}}$	Ratio	Limit-10 class
839	1.423852	0.565017	0.858835	2.5200	neither
840	1.352574	0.451292	0.901282	2.9971	neither
841	1.605586	0.371823	1.233762	4.3181	neither
842	2.048217	0.338519	1.709698	6.0505	neither
843	1.848073	0.317264	1.530809	5.8250	neither
844	2.508924	0.304979	2.203945	8.2266	neither
845	4.601322	0.301506	4.299816	15.2611	neither
846	4.129866	0.277695	3.852171	14.8720	neither

At boundary 839, enabling rotation therefore raises the pre-clip norm by 0.8588348031044006, or $2.52001478149663\times$, at this one captured state/input. The A-minus-B global-loss contrast there is -0.00017467141151428223 . All eight norm differences are positive, and the seven later signs agree with the first sign. This is persistent longitudinal separation in the observed window, but neither arm crosses 10 at any boundary. It therefore does not establish that rotation caused the historical limit crossing, does not explain the later boundary-853 event, and does not justify a threshold change. The summary remains explicitly non-authorizing.

The retained scheduler record is `diagnostic_runs/matched_intervention_49f06078d19f_1839531_scheduler_record`. Its root-row, full-step, completion-marker, and scheduler-log-hash records have SHA-256 values

```
01713739b41fbc83a04e5a449cddf702103c3b87568d8087adc1887c178c2053,  
74ec9cb245a3086a1ad7afaa0da6479259492b730befaa42f951a1565b5ba672,  
5346f5e235063d09bcf8e870fb98a19a05d2e099f99ab0ab8cd5a3735bcdad2cd,  
23b49df7cf3b72bb6de57273350ce165ce96676015f6e54cf30f70bad4905c38.
```

Scheduler stdout contains exactly one completion marker; its SHA-256 is

```
2760327de1c640203d5bc1552f0807fe298743b5f702997e9b022e51322d38ae.
```

Scheduler stderr is empty. Fresh-node independent validator job 1840069, submitted from retained runner SHA-256

9e602deac9319a7e12d1d71fbcba75bd6d52da5cdbfcfafd77d34661ff3e7c84,

ran on h200-021-171 from 18:30:14 through 18:39:40 UTC and completed 0:0. It reconstructed both invocation-bound source paths, validated all 4,257 matched-source and 4,253 trigger-source files before and after use, independently returned VALID, rederived the 118-file inventory, and removed all three marker-owned node-local roots before its sole completion marker. Its evidence root is `diagnostic_runs/matched_intervention_49f06078d19f_1839531_independent_validation/` evidence-job-1840069. The validation result, acceptance result, and complete 16-row evidence-list SHA-256 values are respectively

55d0fed3dcabcbf9a60a444cce149108f4ec891a428946269a84fcb1330fcbafe,
364d899e1a7e0b999a68917780c562d00638226a6d5640653ecf1b53ad6c3ccf,
19d802d2e8a49f93d9fd8124078f307aa31f6cdf7bb5354bf478757672cb716f.

The retained validator scheduler root-row, all-step, marker, and log-record SHA-256 values are

e186d5a86aa67c75ec0fa070dbd5fb6c78ab3bb947e1d8e51d3ade503bb1c6b1,
5f1e6be98f469a337e6915104b2bb3b02cc8bdb54f711e3108973cd1eb536fe3,
04165dfb52dfe1f9ced189fbb18ed94bdf6b518bc6e28631fa39d8453dbf9451,
974b7bdd010fb454f4063b6b13b705e9f8bdf929032a3a01ab5c4c57c1275cfb.

All retained validator files are mode-0400, single-link regular files and its scheduler stderr is empty. Remediation work is isolated from the main checkout in

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_late_angle_scale_20260904`

on branch `codex/rdiag-shared-dt-late-angle-scale-20260904`. The independently reviewed, non-authorizing design is

`docs/plans/2026-09-04-rdiag-shared-dt-late-angle-scale-design-final.md`.

It is committed and published with commit, tree, and file SHA-256 respectively

c57c9617f0256189bcc5a733789699bf32731906,
1279bb20cac06d7ee8b5ba231a273cb955b8874e,
549a7592ffe10b3c516f98d023ea67cb00e64ad02c8529df13e2af9325f427ba.

It selects the checkpoint-compatible, rotation-preserving effective angle scale $0.5H^{-1/2} = 0.125$ at $H = 16$, while retaining direct $H^{-1/2} = 0.25$ and connected-zero controls. It does not alter `shared_dt_v1`. The executable plan is

`docs/plans/
2026-09-04-rdiag-shared-dt-late-angle-scale-implementation-plan-final.
md`.

It is published with commit, tree, and file SHA-256 respectively

17d86199a0eccf91a4ad8c4e0a256d063efc1d85,
65bd0d9d3c3f3d24b9eb51e4fd42ad327c60e942,
8497caef38ea1fc4811b629704ac97604e3daea8da46830a2865329365ce20e5.

Two formal adversarial plan-review rounds found and drove fixes for exact H=16 oracle coverage, a distinct candidate-above-10 terminal, real-anchor FSDP preflight, complete later-boundary liveness, fail-closed outcome mapping, immutable H200 provenance, prospective scheduler-step counts, and four final CLI/snapshot/validator interfaces. The final plan contains those last repairs and requires their named red/green tests during implementation; the second formal review predated those repairs and therefore remained `NEEDS_REVISION`. Implementation seed commit `68932d5aed62f717b91f8a5c46b38679b70805bb` added only the seven preregistered helper, host, H200, preflight, runner, and validator paths. The reversible H=16 angle-scale override was then implemented and hardened through commits `a14168a...`, `5253faf...`, and `11e7d37...`. Fresh specification and quality reviews accepted that slice after it made rollback best-effort and fail-closed, detected mid-arm scale drift, validated exact per-rank topology evidence, and preserved RNG and state-dict identity. The matched-prerequisite slice was then implemented at `d31ba7d...`, hardened at `31962f3...`, and completed by the dependency bootstrap at `9bb7d3c...`. Adversarial review first demonstrated that an unlisted `sitecustomize.py` could forge the historical child result, JSON numeric aliases could pass Python equality, and evidence could be exchanged between hashing and parsing. The accepted implementation now verifies the complete 4,257-file historical source closure before and after the child, uses trusted-directory file descriptors and single-captured bytes for evidence, requires canonical exact-type JSON, bounds and reaps child output, and retains Python `-S` while explicitly appending only the active interpreter’s canonical, non-writable `purelib/platlib` directories. The latter was necessary because the real historical validator imports PyTorch while checking the retained anchor; a real import smoke proves that PyTorch loads without executing `sitecustomize.py`. Fresh specification and quality re-reviews accepted the slice with no remaining findings. Its five focused adversarial tests and 140 host tests pass; nine deliberately red assertions remain for later implementation slices. The extended one-chain prerequisite schema and external-preflight interfaces were then completed at commit `323ce5fc699e8611b5440cf368830a4d7098e839`. This slice fully verifies both the node-private execution tree and immutable source snapshot against the reviewed Git commit, tree, and 4,257-file manifest; binds the exact numerical and actual-anchor preflight closures; checks the root and every scheduler step, TRES, submission, working-directory, stdout, and stderr record; and uses a sealed semantic projection on the retained rank path without repeating live queries or full tree walks. Source and scheduler sub-audits found no remaining scoped blocker. A subsequent quality review initially flagged a same-timestamp transient-edit probe, but its minimized trace showed that it changed an already-consumed preflight runner which was neither reopened nor executed while changed. The reviewer therefore withdrew that finding and returned no Critical, Important, or Minor issue and **Ready:Yes**; the independent specification review returned **ACCEPTED**. Eleven newly focused adversarial cases pass, the specification review’s nine focused functions pass as 15 parameterized cases, and the complete host file now has 152 passing tests with the same nine deliberately red assertions reserved for later-boundary persistence, final artifact closure, and runner postchecks. The candidate finite-norm safety-stop half of the next slice was implemented at `838c236...`, hardened at `156098f...`, and accepted at `d72e10e5a886300af42f3cf83d95ec5ada5380d6`. It preserves exact scalar dtype through the rank norm collective, continues at equality 10, stops only candidate arms for finite norm strictly above 10, and retains the historical continuation of old-scale and no-rotation controls. Rank disagreement, finite negative values, and rank-agreed NaN, positive infinity, and negative infinity have distinct terminal reasons. Every no-step terminal retains complete preclip and named-gradient evidence, proves the full real gradient inventory was cleared, skips AdamW and counter commit, and binds unchanged model and optimizer roots to the authoritative prior state while accounting only for the scheduled learning-rate change. Review found and repaired a float32 threshold collapse, a forgeable one-leaf cleared-gradient record, an invalid cross-boundary optimizer comparison across LR

scheduling, a missing no-rotation availability converse, and an initially unbound equal optimizer-root pair. Final specification and quality reviews report no remaining finding, **ACCEPTED**, and **Ready:Yes**; 23 focused tests and 178 host tests passed at that checkpoint.

The complete liveness half was implemented through commits `d72fcd2...`, `6db3ade...`, `d2891f7...`, `de6e919...`, `47547ca...`, `35a5f31...`, `68846ec...`, `d0a19b3...`, and accepted at `0d28a268295dde0dc534230b6f62d763b400001b`. It records exact compact frontend counts for all 32 layer-microbatch observations at boundaries 839–846, exact candidate four-category gradient liveness, and replica-factor-two Adam `exp_avg/exp_avg_sq` liveness over the real sparse FSDP ownership pattern. Three coordinated optimizer phase gates prevent pre-step, raw-step, and post-health collective skew; empty shards contribute zeros, while nonempty shards require exact FP32 state, optimizer membership, and step equal to the current boundary. Both no-rotation repeats are checked across all ranks and observed boundaries, including uncommitted terminal boundaries, with exact-zero rotational activations and gradients and live radial gradients. Review exposed and repaired numeric type aliases, incomplete first-boundary metadata binding, impossible count tuples, conditional collective ordering, stale optimizer state, unfaithful empty-shard tests, a single-repeat isolation path, and a nonfinite residual-gradient false acceptance. Final reviews returned **ACCEPTED** and **Ready:Yes**; 56 focused implementation tests, 74 independent focused regressions, and 221 full-host tests pass. Eight deliberately red assertions remain for analytics/artifact closure and runner postchecks. The repeatability and historical-reproduction checkpoint was then implemented at `759b752...` and accepted after the exact-rank-type repair `94f83dc5a94e0862fe12cad2dea79b731245f9e0`. All three old-scale, candidate, and no-rotation repeat pairs now compare the complete rank record while excluding only the repeat-specific arm label. The accepted job-1839531 controls are projected from their already sealed bytes into an ordered 3×16 digest record of 3,472 canonical bytes with SHA-256 `c65a7608683b10403ffae4a6eebe106fa2b16e3d403206b434d429f95d1a54da`; the four current control arms make all 64 rankwise comparisons, while candidate arms have no historical comparator. Review found that Python equality had allowed rank zero to be represented as `False` or `0.0`; exact integer typing and paired-candidate regressions close that gap. Independent reviews now report **Ready:Yes**. The focused repeat/history and arm-result reviews pass 25 and 16 checks respectively, and the full host file has 231 passing tests. The same eight deliberately red assertions remain: seven belong to the final 226-file artifact closure and one to runner source postchecking. The boundary-839 scale and decision checkpoint was implemented at `ac0c55eb5f5cc6788f4da9d296f3346e7cb4a8d7`, completed at `f7ed6c88941820ddcf4289be5a20c0cc688a5eba`, and accepted after the final post-AdamW consensus repair `838b04566ec5347946b9e7be7f747f088af25a0f`. Candidate arms now retain the live layer-0 record of twice the raw angle at boundary 839; exact per-rank and per-microbatch tensor records prove common DT/alpha inputs and the preregistered half-angle relation without comparing layers 1–15 after their hidden states diverge. Canonical A1 estimands preserve signed zero and exact endpoint behavior, distinguish the sole instantaneous boundary from longitudinal outcomes, and report the ordered eight-ratio median/p95 together with loss, gradient-category, angle-maximum, and DT-tail descriptives. The pure outcome classifier applies `INVALID_EVIDENCE > VALID_FAIL_SAFETY > VALID_FAIL_CRITERION > VALID_PASS`, while every summary remains non-authorizing. Review found and repaired partial-prefix indexing, finite-sum overflow, omitted descriptives, inconsistent status/reason combinations, and rankwise-repeated but globally inconsistent safety norms, including the post-AdamW path. Final independent reviews report **Ready:Yes**; 247 host tests pass and the same eight deliberately red assertions remain. The artifact-sealing and invocation-identity checkpoint was then completed at `79222cf0f750a71c3d06e4506b81426f1f42fb44`. Structurally malformed or partial evidence remains exceptional and cannot produce a trusted summary, whereas structurally complete semantic

contradictions seal only the non-authorizing status `INVALID_EVIDENCE`. The main closure is now derived as 10 fixed files plus 36 files for each of six arms: 226 inventoried files and 227 physical files including `artifact-inventory.json`; the historical matched closure remains pinned at 118/119. Final validation reads sealed captures, rederives the scientific status, revalidates the closure after rederivation, and reports that status separately from validator success. Invocation and summary identity now bind the design, matched validator, both preflight closures, exact intervention constants, two-node/16-rank execution contract, retained runner, and source manifest without reopening the runner between hashing and contract validation. The three named acceptance tests and seven focused sealing tests pass; the full host file reports 257 passes and only the deliberately deferred runner source-postcheck assertion. An initial independent review found no remaining issue in this slice, but a stricter immutable review found two blockers: complete, structurally valid source/anchor/RNG/scale contradictions still escaped as structural exceptions instead of sealed `INVALID_EVIDENCE`, and final validation bracketed reopened files rather than rederiving from the exact bytes checked by the inventory. Those repairs and direct identity-mutation regressions were completed and independently accepted at `3f94176c94d8c383b0723f1e8d915ab0e10fc28c`, then integrated on top of the runner work as `bbe9868`. . . . Every semantic input is now parsed from the first bounded, inventory-verified capture; a second pass checks only post-derivation integrity. Complete source/prerequisite, anchor, first-forward RNG, scale, and CUDA-device contradictions become deterministic invalid-evidence facts, while malformed shape/count/token/hook evidence still fails externally. The integrated full host suite passes all 269 tests.

The main-runner portion of that next task was completed and independently accepted at `0ce13479ef78c854b5060609f1e4b20deb66b2ea`. It expands the diagnostic to the frozen six-arm order and derives the exact 29-step task vector from the runner’s actual successful control flow; adversarial duplicate arm, prerequisite, and cleanup calls are rejected. Scientific candidate stops remain zero-exit evidence-bearing outcomes so every arm runs, whereas the first operational failure is preserved through owned-root cleanup and suppresses summary, inventory, and the sole completion marker. Current, trigger, and matched sources are reconstructed on both nodes; the full accepted prerequisite chain and both preflight closures are checked before every arm and again in the source postcheck, while the in-job final validation deliberately uses retained mode. The focused runner/parser set passes 13 tests, the full host file passes all 264 tests, and shell syntax, Python compilation, and diff checks pass. This commit does not implement or run either preflight or the result-bearing six-arm GPU allocation. Numerical, actual-anchor, and fresh-node validator surfaces are therefore the current implementation task. No numerical or result-bearing GPU job has yet been accepted from this worktree.

The numerical-preflight host implementation was independently accepted at `417fe21644000c72f91e5f39946c0d` and integrated as `3617d40`. . . . Its one-H200 runner is source-bound, network-scrubbed, uses private node-local caches, and seals seven inventoried files plus the inventory. Exactly three H200 tests are collected at $H = 16$, key/value width 128, and rotary width 64: a bitwise direct-0.25 historical replay through forward, VJP, parameters, and AdamW state; a candidate-0.125 comparison to the independent FP32 literal recurrence; and an update/schema/liveness check for hidden, angle, DT, and Adam moments. Review found and repaired stale gradient accumulation between historical branches. The integrated host suite passes all 270 tests, the H200 module collects exactly three nodes, and shell/Python checks pass. These are host-side contracts only; the three numerical tests had not yet run on an H200 at that implementation checkpoint; their subsequent S1 execution is recorded below.

The actual-anchor preflight was independently accepted at `ae328418d406eae099e48a7a441399044d8aae8` and integrated as `25e63fb`. . . . Its two fresh 16-rank process groups use a metadata-only anchor resolver and sentinel loaders: no retained batch is opened, no loader is iterated, and no forward or

optimizer update occurs. Each rank proves one strict `Fabric.load`, exact accepted checkpoint identity, rederived model/optimizer Merkle equality, ordered state topology before installation and after restoration, all 16 layer overrides, and selected-device-only CUDA RNG preservation. Review added dynamic guards for all three file-open APIs, bound every row to the candidate source and pre-inventory digest, and rejected forged Merkle roots/leaves. The runner has the exact 13-step topology and 82/83 closure. Its standalone full host suite passes 266 tests; after integration with the semantic, runner, and numerical slices, all 272 host tests pass. The first S1 actual-anchor allocation and its fail-closed result are recorded below.

The fresh-node validator was independently accepted at `357febcb395e07e72c6b393c28b98848a27ea05` and integrated as `7d92a8f214a081c5bf3b39c07fe3db60217929`. It accepts the exact 12-argument source/data/target/preflight tuple, waits for the target job to become terminal, reconstructs all three source roots, validates the target closure in `fresh-recursive` mode, and preserves every sealed scientific status rather than requiring `VALID_PASS`. Its scheduler parser binds root and unfiltered queries, canonical task strings, job names, two-node membership, per-step task counts, raw resource allocation, timestamps, and the exact 29-step topology. Review repaired final-marker cleanup ordering, NodeList and cross-query gaps, and raw extern-step AllocTRES comparison. The validator's own mode-0400 evidence closure is checked before its sole stdout marker. Frozen host verification passes 4/4 and 13/13 focused science batches, 4/4 runner contracts, and the complete 284-test host suite; the four accepted regression suites pass 126, 69, 52, and 48 tests respectively.

The canonical branch was pushed and immutable snapshot S1 was materialized at `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_snapshots/`

`rdiag_shared_dt_late_angle_scale_7d92a8f214a081c5bf3b39c07fe3db60217929`. Its Git tree is `d060cd04a60ad201cc75238f5de3a8fd3c20ef83`; its 4,266-file `SOURCE_MANIFEST.json` has SHA-256 `1368bb987edcd3957e9b6d2cdc5a16f9268030da97f056a41829201d4672312d`. Snapshot-local checks reproduce the live counts exactly, and a fresh private-copy verification passes all 4,266 entries. Numerical attempt 1844567 passed all three H200 tests but is not accepted because its scheduler `SubmitLine` began with bare `sbatch`, rather than the exact frozen `/usr/bin/sbatch` executable. The byte-identical replacement job 1844573 completed 0:0 on one H200 in 2 minutes 18 seconds: all three tests passed with zero skips, scheduler stderr was empty, the sole completion marker was present, and all seven inventoried/eight physical artifact hashes matched. Its accepted closure is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_runs/`

`late_angle_scale_numerical_preflight_7d92a8f214a0_1844573_accepted_closure.json`, SHA-256 `fe613ec2fb146cce7f52f906fdd42cb65278f7e24ed7be12f0032e51539950bb`. Both live-query and retained-record validators pass that closure. Actual-anchor preflight job 1844581 then ran from the same immutable snapshot on nodes `h200-115-025`, `h200-126-216`. The root and batch records failed 1:0 after 12 minutes 56 seconds; extern completed, preliminary steps .0, .1, and .2 completed, and cleanup step .3 completed. No anchor arm or rank record, summary, inventory, or completion marker was published. The fail-closed exception was `LateAngleScaleError:acceptedmatchedprerequisiteprojectionchanged`. Investigation reproduced the mismatch on the host: the already validated matched source identity has six fields, including integer `file_count=4257`, while the strict downstream expected projection listed only the other five. This is an internal provenance-schema defect rather than a numerical, checkpoint, kernel, or optimization observation. The minimal remediation adds the pinned file count to both expected matched-source projections and adds a realistic regression asserting exact six-field equality; it does not change either preflight runner, the intervention, the optimizer, the acceptance criteria, or the scheduler topology. The red regression reproduced

the exact projection failure, affected focused tests pass, and the complete host file passes all 284 tests with ten warnings. Independent review found no scoped issue and returned **READY**. The repair is published as commit `17e82ca154f0c22543cb5424fe876d1d832284b8`, tree `394d9b82910c1bc830561f325b9540d45c10a5a8`. Per the frozen plan's remediation rule, this accepted source change makes S1 and its numerical closure ineligible for later gates. The never-before-existing immutable S2 snapshot is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_snapshots/ rdiag_shared_dt_late_angle_scale_17e82ca154f0c22543cb5424fe876d1d832284b8`; its 4,266-file manifest has SHA-256 `5f5cf4ee079c69f37e1ecb01fed9f1a04d79677f215edaf1eeb612b5f94b4e8b`. The canonical worktree and S2 independently reproduce focused counts 4 and 13, four runner-contract passes, all 284 host tests with ten warnings, and the four accepted regression counts 126, 69, 52, and 48. Python compilation and all four shell syntax checks pass; the live source diff/scope checks are clean, and a fresh private copy verifies every one of S2's 4,266 manifest entries. Numerical preflight job `1844729` completed `0:0` on `h200-246-014` in 2 minutes 44 seconds. All three tests passed with zero skips; root, batch, and extern are the only scheduler rows and all completed `0:0`; stderr is empty; the sole marker is present; and all seven inventoried/eight physical artifact hashes match. Its canonical mode-0400 accepted closure is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_runs/ late_angle_scale_numerical_preflight_17e82ca154f0_1844729_accepted_closure.json`, SHA-256 `4e22001f66832c1f4d150992732ea98670e84442ae48ea6c81426243d1bd47c2`. Independent retained, live-query, scheduler-byte, live-log, mode, and artifact audits return **APPROVED**. The first S2 actual-anchor attempt, job `1844730`, correctly passed the repaired prerequisite projection and began its first per-node recursive revalidation on `h200-132-171`, `h200-134-017`. During monitoring, however, an unauthorized diagnostic `srun` added scheduler step `.4`; because this makes the preregistered exact 13-step topology unattainable, the attempt was cancelled after 18 minutes 20 seconds and is ineligible regardless of its payload. It produced no accepted closure and contributes no scientific observation. Clean replacement job `1844962` was submitted from the byte-identical S2 tuple and ran on `h200-152-154`, `h200-186-088`. It passed immutable-source reconstruction, runtime identity, the repaired initial prerequisite projection, and the old-scale per-node recursive prerequisite recheck. Its first 16-rank strict restore then failed collectively before any forward, update, retained-batch read, or rank evidence publication with `LateAngleScaleError:shared-DTscale-layercountchanged`; root and batch ended `1:0`, extern and cleanup completed, and no result, inventory, marker, or accepted closure was emitted. Strict `Fabric.load` and the model/optimizer Merkle checks had already succeeded. The cause is deterministic wrapper aliasing: Lightning's `_FabricModule` registers the same GPT below both `_forward_module` and `_original_module`, while the helper scanned `named_modules(remove_duplicate=False)` over the wrapper. It therefore observed 32 paths for the 16 logical mixers. Existing forward-determinism code already establishes the correct contract: resolve the semantic GPT from `fabric_model.module`, canonicalize the exact FSDP wrapper components, and retain a bijection between 16 semantic paths and 16 object identities. A faithful host reproduction confirms 32 raw paths and 16 unique objects. C2/S2 and its otherwise valid numerical closure are now stale for authorization. The test-first repair resolves and binds the semantic `fabric_model.module` root, strips only exact `_fsdp_wrapped_module` path components, and requires a bijection between the ordered forward and semantic inventories. Regressions cover the installed Lightning wrapper, nested FSDP blocks, same-object aliases, canonical-name collisions, incomplete/mismatched/duplicated roots, decoy owner graphs, and missing layers. Two independent re-reviews returned **READY**; 17 focused tests and all 288 host tests pass. The change is published as commit `711155e9a685dfbdd5a4c11954017e4b6b583ee6`, tree `7e3dd18b8baa9a522e86a574a0ebf74fd625a59b`. Immutable S3 is `/lustre-storage/`

checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_snapshots/ rdiag_shared_dt_late_angle_scale_711155e9a685dfbdd5a4c11954017e4b6b583ee6; its 4,266-file manifest SHA-256 is 8eafc98de7ed172ec0142a9a3160df6d076755f4f291bef140ca40b730130223. The canonical worktree and S3 reproduce focused counts 4 and 13, four runner tests, all 288 host tests with ten warnings, and regression counts 126, 69, 52, and 48; compilation, shell syntax, scope, and a fresh 4,266-file private-copy verification pass. S3 numerical preflight 1845249 completed 0:0 on h200-246-014 in 2 minutes 44 seconds: all three tests passed with zero skips, only root/batch/extern were present, stderr was empty, and all seven inventoried/eight physical artifacts matched. Its canonical accepted closure is /lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_runs/ late_angle_scale_numerical_preflight_711155e9a685_1845249_accepted_closure.json, SHA-256 6d98106493f257e94ac8c466b79e7106c4a2cd70a1434ba0664eaf2ff291ebd5; live, retained, and record validators pass, and an independent byte-level audit returns APPROVED. S3 anchor preflight 1845250 ran on h200-005-176,h200-246-014 for 23 minutes 18 seconds. Its source reconstruction, two-node topology, runtime identity, initial prerequisite, old-scale recursive prerequisite, and cache-preparation steps all completed, but the first 16-rank no-forward arm failed on every rank; root and batch ended FAILED1:0, extern and both cleanup steps completed, and scheduler stderr contained only the resulting task-exit reports. No rank payload, result, inventory, completion marker, or accepted closure exists. The fail-closed exception was LateAngleScaleError:anchorpreflightstate-dicttopologyisincomplete. Strict anchor load, model/optimizer Merkle equality, and the repaired 16-owner angle-scale discovery had already succeeded, so this remains a pre-forward infrastructure failure rather than a numerical result.

The exact mismatch is now independently evidenced and reproduced with Lightning's installed `\_FabricModule`. Its inherited `named\_parameters()` traverses the execution graph and emits names such as `\_forward\_module.\_fsdp\_wrapped\_module.transformer.h.0.\_fsdp\_wrapped\_module.attn.rdiag\_angle\_proj.weight`, whereas its overridden `state\_dict()` delegates to the original module and the accepted checkpoint key is `transformer.h.0.attn.rdiag\_angle\_proj.weight`. The topology recorder mixed those two namespaces, leaving their name sets disjoint; its validator correctly rejected them. The host suite already contained a faithful Fabric/FSDP alias fixture, but exercised it only through scale-owner discovery; the topology tests used either a bare layer or hand-built matching names. The C4 repair resolves the same validated semantic root used by the state dictionary, removes only path components exactly equal to `\_fsdp\_wrapped\_module`, rejects every canonical-name collision, and uses the resulting names consistently for parameters, buffers, state dictionary, optimizer groups, and initialized optimizer slots. The forward graph must bind the semantic root exactly once and expose the identical ordered canonical object inventory; non-deduplicating enumeration rejects parameter/buffer aliases, optimizer groups cannot duplicate or introduce parameters, and optimizer state cannot survive outside a known group. The generic collector still permits a deliberate partial optimizer used by the numerical oracle, while the anchor-record validator separately retains its exact full-partition requirement. New regressions exercise an installed Fabric wrapper with a persistent buffer, two AdamW groups and sparse live slots, the actual nested-FSDP namespace difference, canonical collisions, aliases, missing and duplicate group membership, orphan state, mismatched roots, and forward decoys. Ten focused topology/owner/numerical checks pass, and two independent adversarial re-reviews return READY. Live-worktree verification reports 293 host tests with ten warnings and the accepted regression counts 126, 69, 52, and 48; both changed paths pass the scope and diff checks. The repair is published as commit `f869d5e6db5f9f8863c6ffae03bc4814bbf2bd1`, tree `b442e6570d9fb962875888a608394c96ff5cd565`. Its never-before-used immutable S4 root is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_`

snapshots/`rdiag_shared_dt_late_angle_scale_f869d5e6db5f9f8863c6ffaeb03bc4814bbf2bd1`;
the 4,266-file source-manifest SHA-256 is `a14f398377ea72c3f9674887fe59c04c408d796262784f2a9bce773cfb7ca`;
Snapshot-local focused and static checks pass; its complete suites reproduce 293 host tests
with ten warnings and regression counts 126, 69, 52, and 48. A fresh private execution
tree rebuilt from Git independently verifies all 4,266 manifest entries. C4 invalidates S3
and its otherwise valid numerical closure, so fresh S4 numerical job 1845517 and actual-
anchor job 1845518 were submitted through the frozen absolute `/usr/bin/sbatch` interfaces.
Numerical job 1845517 completed 0:0 on `h200-246-014` in 2 minutes 45 seconds: its
only scheduler rows are root, batch, and extern; all three tests passed with zero skips,
seven warnings, one completion marker, empty scheduler stderr, and an exact seven-file
inventory/eight-file physical closure. Its independently requeried, mode-0400 accepted closure is
`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_`
`runs/late_angle_scale_numerical_preflight_f869d5e6db5f_1845517_accepted_closure.`
`json`, SHA-256 `ae422425c386787c639adb37d608d6396418676f6e62cc52ba7b8fc06a88fb0c`; both
retained and live-query validators pass. Actual-anchor job 1845518 then completed 0:0 on
`h200-005-176`, `h200-186-088` in 42 minutes 11 seconds. Its root, batch, extern, and all 13 numbered
steps completed 0:0; the exact numbered-step task vector is `2,2,2,2,2,16,2,2,2,16,2,2,2`.
Both 16-rank arms strictly loaded the accepted checkpoint without opening a retained batch
or executing a forward or update. All 32 rank records prove equal loaded/restored model
and optimizer Merkle roots, stable canonical parameter/buffer/optimizer topology, exact
16-layer scale installation and restoration, unchanged CUDA RNG, and complete cleanup.
The closure contains 82 inventoried/83 physical artifacts, exactly one completion marker,
and empty scheduler stderr. Its independently requeried mode-0400 accepted closure is
`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_`
`runs/late_angle_scale_anchor_preflight_f869d5e6db5f_1845518_accepted_closure.`
`json`, SHA-256 `d381c5770cbe2f545b67cb2bb1432ecc537640d4ad3846f76c3c1a001edc9f79`; re-
tained, live-query, and scheduler-record validators pass, and independent scientific and
operational reviews both return APPROVED. A second private-tree revalidation of S4
then reproduced all 4,266 source-manifest entries. The frozen launch tuple is retained
as `diagnostic_runs/late_angle_scale_launch_f869d5e6db5f_operator_record.json`,
SHA-256 `8f09aa03fcb23238cb10ac5b4f98e8153adfb56a9111701e41ef7e8183a6e93c`. Ex-
actly one result-bearing six-arm allocation, job 1845784, was submitted through the
frozen absolute `/usr/bin/sbatch` interface from the clean canonical worktree. It re-
quests two nodes and 16 H200 ranks. It ran on `h200-186-088`, `h200-193-042` for 1 hour
44 minutes 13 seconds. All 29 numbered steps completed 0:0 with exact task vector
`2,2,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2`, including all six
16-rank arms, their prerequisite rechecks and cache cleanups, the final two-node source/prerequisite
postcheck, and private-root cleanup. All 96 rank records were emitted. Both candidate repeats
completed all eight updates through boundary 846, were exactly repeat-equal, and had finite live
angle/DT/decay gradients and Adam moments. Both no-rotation repeats were exactly repeat-equal,
with all rotational activations and gradients zero and radial gradients live. Both old-scale repeats
were exactly equal, and all 64 current control records exactly reproduced the accepted historical
semantic projections.

The root job nevertheless ended **FAILED1:0** during local post-step summarization, so these
payload observations are not an accepted result. The producer writes the real full 18-field
`RDiaqCheckpointIdentity.as_dict()` into each initial-state record, while the structural summary
validator and its synthetic fixture still required an obsolete five-field subset. Consequently rank
zero of `old_scale_a1` raised `LateAngleScaleError:initialanchoridentityschemachanged`. An

independent audit confirms that all 96 full identities are identical, equal the authoritative trigger identity, and that all 96 complete initial states equal the corresponding retained anchor state. Fail-closed handling suppressed `summary.json`, `artifact-inventory.json`, and the completion marker; the run retained 225 partial top-level artifacts, and no fresh-node validator was submitted. This is classified `INVALID_EVIDENCE` due to a diagnostic-validator schema defect, not as an infrastructure or scientific failure. Twice-queried scheduler root and all-step records were byte-stable with SHA-256 values `f0df12e06291b684c3507389c7c6fd62b722c99f3b4937d8b16b4c94dd83ba67` and `81bec94ea18ae59a5f3e60292242b77257b7465e8d1ed7f53ba6e10b67286d01`. Live stdout and stderr SHA-256 values are respectively `6f7722c57b87d6d2b856fb55cbb531034bb37034ce34b6f3c5aeef0383647` and `6baf9c9b5af503ee95eed862edc3d151b6eff79b8e5c1aacf9692db771c58b7d`; the pre/post prerequisite inventories are byte-identical at SHA-256 `7f869aaf25307e73546f14e709f6bab4aab03de846d2adf8151`. No conventional accepted scheduler record was invented for this failed topology.

The test-first remediation replaces the stale fixture with the real 18-field schema, adds a producer-to-validator round trip plus missing/extra/type/hash-alias regressions, and leaves semantic equality to the later retained-anchor comparison. Its RED test reproduced the exact failure; the repaired focused set passes 29 tests and all 96 retained initial states validate. Two independent exact-diff reviews return `READY`; the final live-worktree suite has 299 passes with ten warnings, the four accepted regression suites pass 126, 69, 52, and 48 tests, and the four runner contracts, Python compilation, shell syntax, diff, and scope checks pass. The repair is published as commit `9f5ec8c806f79731f2f1cd5339cde93eaf5fbf20`, tree `e9bcd490bd96e55e0b6ddc640c430844ffef8ec2`. Immutable S5 is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_snapshots/rdiag_shared_dt_late_angle_scale_9f5ec8c806f79731f2f1cd5339cde93eaf5fbf20`; its 4,266-file manifest SHA-256 is `edcfd78431d8d4c152768f905834a5872bb0a8bf549d0839daa5681a7636b061`. A private tree rebuilt from Git verifies every S5 manifest entry. This source change requires fresh numerical and actual-anchor preflights and a fresh candidate allocation; S4 and job 1845784 cannot authorize a later stage. Fresh S5 numerical preflight 1846328 and actual-anchor preflight 1846329 were submitted through the frozen absolute `/usr/bin/sbatch` interfaces. Numerical job 1846328 completed 0:0 on `h200-112-045` in 2 minutes 35 seconds: all three H200 tests passed with zero skips, only root/batch/extern were present, stderr was empty, and all seven inventoried/eight physical artifacts matched. Its accepted closure is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_runs/late_angle_scale_numerical_preflight_9f5ec8c806f7_1846328_accepted_closure.json`, SHA-256 `44390184226902be6a4fd7488f7752555e1fdefbf45d097a828ae2e798b20fc9`; live-query, offline, and retained-record validators pass. Actual-anchor job 1846329 completed 0:0 on `h200-101-019`, `h200-186-088` in 43 minutes 30 seconds. Its root, batch, extern, and all 13 numbered steps completed 0:0 with exact task vector 2, 2, 2, 2, 2, 16, 2, 2, 2, 16, 2, 2, 2. Across both arms, all 32 ranks strictly restored the accepted boundary-838 checkpoint without opening data, executing a forward, or applying an update. The candidate changed only the installed 16-layer multiplier and effective scale from 1/0.25 to 0.5/0.125; model and optimizer Merkle roots, state topology, and device RNG were restored exactly, and cleanup completed. The pre/post prerequisite inventories are byte-identical at SHA-256 `fa0943d2f98c6c0dede5b6e89e5f6c30a63fdb57dd996c899711a0123bb0d669`; the closure has 82 inventoried/83 physical mode-0400 single-link artifacts, exactly one completion marker, and empty scheduler stderr. Its accepted closure is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_runs/late_angle_scale_anchor_preflight_9f5ec8c806f7_1846329_accepted_closure.json`, SHA-256 `63f4e61d8c531ad1b0702185bf49a1485bea9c24941f89a8622aaf477cc22970`; live-query, offline, and retained-record validation all pass. Independent scientific and operational reviews

return APPROVED, including a replay of all 96 S4 rank records through the S5 repair. A final private Git reconstruction again verified all 4,266 S5 manifest entries. The one-shot operator record is `diagnostic_runs/late_angle_scale_launch_9f5ec8c806f7_operator_record.json`, SHA-256 `d3d0347bb47117e179a42d05692ed01e7e891ab26a4969e82d3ed22cb04e3dfd`. Exactly one S5 six-arm allocation, job 1846561, was then submitted through its recorded 12-token absolute `/usr/bin/sbatch` interface from the clean canonical worktree. It completed 0:0 on `h200-101-019,h200-186-088` in 1 hour 46 minutes 18 seconds. Root, batch, extern, and all 29 numbered steps completed 0:0 with exact task vector `2,2,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2`. All six arms completed eight updates through boundary 846; all three A1/A2 repeat projections are exact, all four control records reproduce their accepted historical projections, intervention isolation passes, and all candidate gradient and Adam-moment liveness observations are live. The pre-registered classifier returns non-authorizing `VALID_PASS`. At the sole primary instantaneous boundary, candidate/no-rotation norm ratio is `1.2335551366597468`, below the locked `1.25` maximum. Candidate norm is below old-scale norm at every boundary and never exceeds the unchanged limit of 10; the boundary-839 candidate and old-scale norms are respectively `0.6969801187515259` and `1.4238522052764893`. This remains one captured state/window: only boundary 839 is an instantaneous counterfactual, while later boundaries compare paired longitudinal branches. The exact artifact closure is 226 inventoried/227 physical mode-0400 single-link files. Its inventory and summary SHA-256 values are `7d9e81dc67e0f1e1610135fb8ab994697afdf289b0ebd48f12b4677490f83938` and `7ea9caa4d6b3ecf731df73cd8fe0aaaebdbf3c5cd9e43129fd4175ecbde0672b`; scheduler stderr is empty and stdout SHA-256 is `0a5915dec9377b9745c9d75523d7380d52cea8f10dc976e4f7f4a24c3580f4d3`. Fresh-node validator job 1846728 was submitted with `h200-101-019,h200-186-088` explicitly excluded. It correctly landed on the fresh node `h200-052-125`, but failed 2:0 after one second before creating a validation root or numbered step. Its sole stderr line is `targetrunrootdoesnotbindcommitandjobID`; stdout is empty. The submitted target path and job ID are correct. The frozen runner instead placed `'\_'` inside one outer double-quoted expected-path expansion, so Bash retained the apostrophes literally and every valid target path fails that comparison. This is a validator control-plane defect, not contrary scientific evidence, but it leaves job 1846561 unaccepted. The repair must add a behavioral shell regression, replace that expression with braced adjacent expansions, freeze a new source version, and repeat the source-bound preflights, six-arm allocation, and fresh-node validator. No production implementation or authorization is claimed.

The validator repair is now frozen as S6 on branch `codex/rdiag-shared-dt-late-angle-validator-fix-2026` at commit `3d9c20f4e1019d6e1117604e94622ed46bc06084` (tree `bcba0c6c7f1780326cf018f70c1811c8ba8c5ee2`). Relative to S5, the diff contains only the one-line braced target-path expansion and a behavioral host regression that extracts and executes the actual two-line guard. The S5 expression rejects its valid target with status 2; the repaired expression accepts that target while retaining rejection of wrong commit, job, and suffix values. Numerical-preflight, anchor-preflight, six-arm, helper, and H200-test bytes are unchanged. The immutable S6 snapshot is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/ source_snapshots/rdiag_shared_dt_late_angle_scale_3d9c20f4e1019d6e1117604e94622ed46bc06084`; its 4,266-entry manifest has SHA-256 `85d1a7f8de3de4274df8904cce2f119ee88b900f5063fa6bf14efff515d325c8`. Independent private Git reconstructions before and after testing verified every entry. The frozen snapshot passed 595 host regressions, helper compilation, and all four runner syntax checks. Independent scientific and operational launch reviews both returned APPROVED; a reviewer's discarded test attempt wrote temporary pytest state into the live worktree only because the host temporary filesystems were full, then correctly failed the source-closure guard. Those artifacts were moved intact to a

recoverable diagnostic quarantine, the canonical worktree was restored clean, and the post-test private reconstruction passed. The first S6 submission attempt mistakenly added the operator-only `--parsable` option to both `sbatch` commands. Because the frozen scheduler validator requires exact equality with the registered six-token numerical and eight-token anchor command lines, jobs 1846846 and 1846847 were immediately classified ineligible and canceled; the former never allocated, while the latter stopped after 81 seconds during preliminary step 2, before any result, inventory, completion marker, or accepted closure. The four retained preliminary anchor files do not constitute result evidence. The exact registered interfaces, with no extra option, were then submitted as replacement numerical job 1846871 and actual-anchor job 1846872. Their scheduler `SubmitLine` fields exactly match the frozen contracts. Numerical job 1846871 completed 0:0 on h200-067-061 in 2 minutes 34 seconds with the expected root/batch/extern-only topology, three passing H200 tests, its sole completion marker, and empty `stderr`. Its eight physical/seven inventoried artifacts are mode-0400 single-link files, and the result is non-authorizing `VALID`. Live-query, offline, and retained-record validation all accept its source-bound closure `diagnostic_runs/late_angle_scale_numerical_preflight_3d9c20f4e101_1846871_accepted_closure.json`, SHA-256 004867fe85bbce178b861095dde71b9a68f5c8873214c1c095093ff227fc8093. Anchor job 1846872 completed 0:0 on h200-067-061,h200-128-233 in 44 minutes 28 seconds. Root, batch, extern, and all 13 numbered steps completed cleanly with exact task vector 2,2,2,2,2,16,2,2,2,16,2,2,2. Both 16-rank arms completed their strict restore-only checks; the pre/post prerequisite inventories are byte-identical at SHA-256 f8801e1163bf859bf420bb9dc25f0565e4a74e034fd92998300db03781e2eccd. Its 82 inventoried/83 physical artifacts are all mode-0400 single-link files, its sole completion marker is present, and scheduler `stderr` is empty. Live-query, offline, and retained-record validation all accept its source-bound closure `diagnostic_runs/late_angle_scale_anchor_preflight_3d9c20f4e101_1846872_accepted_closure.json`, SHA-256 46d5ef721929457f3ccfbb3ad814461d06d8b3d24350442430498018c. No S5 closure was reused. Final scientific and operational Action89 reviews both returned `APPROVED`. The mode-0400, single-link one-shot operator record is `diagnostic_runs/late_angle_scale_launch_3d9c20f4e101_operator_record.json`, SHA-256 4c68809091d51cecf35344e32b5a31c0946b5c8166eca1. A final private Git reconstruction again verified all 4,266 S6 manifest entries. The exact 12-token main interface, with no `sbatch` option, was then submitted once as job 1847039; its scheduler-recorded command matches the operator record. It completed 0:0 on h200-129-161,h200-130-169 in 1 hour 47 minutes 8 seconds. Root, batch, extern, and exactly numbered steps 0–28 completed cleanly with task vector 2,2,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2. The retained non-authorizing summary reports `VALID_PASS`: all six arms completed eight updates, A1/A2 repeatability matched, topology and boundary-839 intervention isolation passed, all locked criteria are true, causal interpretation is allowed, and every failure-reason list is empty. At the instantaneous boundary 839, candidate, historical-scale, and no-rotation norms are respectively 0.6969801187515259, 1.4238522052764893, and 0.5650174021720886; the candidate/no-rotation ratio is 1.2335551366597468, below its preregistered 1.25 ceiling. The candidate is below the historical scale and below the norm-10 ceiling at all boundaries 839–846. Its 226 inventoried members plus the inventory are exact mode-0400, single-link files; inventory, summary, invocation, and unchanged pre/post input-inventory SHA-256 values are respectively 373f6c6d20d50e5e616403d4963433a6d6f7f1630fdeca291fa5de0fd2dd6adf, 99e35928cb6e7d510b55c431747ad445241aad59793d1774f9a42a9d5bf7de00, 3f085cbd42ec6125a4fbb2f76b1f4, and e28296916695c549b2ecb75f94fd1bf491e6be380338a530023c8378df6b063b. The completion marker occurs once and scheduler `stderr` is empty. Fresh-node validator job 1847464 was then submitted with both main-job nodes explicitly excluded and ran on h200-017-170. Its recursive scientific validation completed: the retained `validation-result.json` reports

VALID and revalidates all 226 artifacts, prerequisites, anchor, and source; the subsequently written `acceptance-result.json` reports the target's `VALID_PASS`. Their SHA-256 values are respectively `84ff545fb479a089f547e8fec9c654b66c975ba6d2627a8685cfeed88e6c8c83` and `71ee18f943f3c121dbe9f0d2a899839a8be48c4c2bf69ad0218434e92baae6c6`. The validator nevertheless ended `FAILED1:0` after 11 minutes 35 seconds during final sealing. The frozen runner invoked `/usr/bin/printf-v; -v` is a Bash-builtin option, not a `coreutils` option. External `printf` therefore returned zero while writing the two bytes `-v` to `stdout` and a warning to `stderr`, did not assign `COMPLETION_MARKER`, and the next expansion failed as an unbound variable. The partial root contains six of the required nine files and lacks its completion marker, scheduler-log hash record, and evidence hash closure. It is consequently `INVALID_EVIDENCE`; the two pre-seal JSON files are forensic corroboration only, not an accepted closure. A focused executable regression reproduces the exact warning and unbound-variable failure on S6. The S7 repair replaces only that command with `builtinprintf-v` and adds the regression; the focused test, surrounding validator-contract tests, Python compilation, and Bash syntax check pass. The repair is committed on branch `codex/rdiag-shared-dt-validator-seal-fix-20260905` as `0a87fe50a47228c90a72d233615bee422fd2fe7a` (tree `a911270714350b23987e13ffb44070fbe45d1e8d`) and is pushed at that exact remote SHA. The corresponding clean worktree is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_worktrees/rdiag_shared_dt_validator_seal_fix_20260905`. Its immutable 4,266-file snapshot is `source_snapshots/rdiag_shared_dt_late_angle_scale_0a87fe50a47228c90a72d233615bee422fd2fe7a`; the source-manifest SHA-256 is `065f6b6094b064f66975acdcb41745a4d94a8c58ee92a5ae51aca0be941ea02d`. An independent private Git reconstruction verified the entire snapshot, and both the live worktree and snapshot passed the frozen 596-test host suite plus helper compilation and all four runner syntax checks. Independent specification, code-quality, scientific, and operational reviews found no remaining issue in the frozen S7 diagnostic/validator contract and approved the strict full S7 repetition. No S6 closure is reused. The exact registered commands, without extra `sbatch` options, were submitted from the clean S7 worktree as actual-anchor preflight job 1847596 and numerical preflight job 1847597; their scheduler `SubmitLine` values bind the S7 snapshot, commit, manifest, worktree, and, for the anchor, the unchanged data manifest. Anchor job 1847596 completed `0:0` on `h200-112-045,h200-128-233` in 42 minutes 57 seconds. Root, batch, extern, and all 13 numbered steps completed with exact task vector `2,2,2,2,2,16,2,2,2,16,2,2,2`; both 16-rank arms proved strict boundary-838 restore, zero data/forward/update activity, unchanged model and optimizer Merkle roots, preserved RNG, and exact historical/candidate scale behavior `0.25 → 0.25` and `0.25 → 0.125 → 0.25`. Its 82 inventoried/83 physical artifacts are sealed, its result is `VALID`, and its accepted closure is `diagnostic_runs/late_angle_scale_anchor_preflight_0a87fe50a472_1847596_accepted_closure.json`, SHA-256 `171d985217f5e119f7037a6742531838f608e65a1d502a4a7a40db46e39f4fc6`. Numerical job 1847597 completed `0:0` on `h200-128-233` in 2 minutes 19 seconds; all three H=16 historical-identity, candidate forward/VJP, and one-step live-moment tests passed. Its seven inventoried/eight physical artifacts are sealed, its result is `VALID`, and its accepted closure is `diagnostic_runs/late_angle_scale_numerical_preflight_0a87fe50a472_1847597_accepted_closure.json`, SHA-256 `d4e33bdc0f5093fbc893323be31669470f32b358d0f3f9002a2a1dd0fc9844ec`. Both closures pass live-query, offline, and retained-record validation with identical projections. They were sealed serially with the reviewed operator-side tool `operator_tools/rdiag_preflight_closure_b6aeaa289545a0d8d7b4fd27d447ae9dfc4b96001748b34fbb94ee882bebf11a.py`, whose SHA-256 is encoded in its filename; the same frozen bytes first reproduced both accepted S6 closure digests and then reproduced each S7 closure without `--seal` after publication. Independent scientific and operational Action-89 reviews returned `APPROVED`. A fresh 4,266-file private

reconstruction then reverified the S7 snapshot. The mode-0400, single-link operator record is `diagnostic_runs/late_angle_scale_launch_0a87fe50a472_operator_record.json`, SHA-256 `1aa59e8e0b0487c49636d2271f9abca567746219a594e6db014678bd41087957`. The two accepted preflight closures and this operator record all explicitly carry `authorizing=false`: they authorize only the frozen diagnostic repetition, never a production-training launch. The record’s exact 12-token main command, with no `sbatch` option, was submitted once as job 1847784; the scheduler-recorded `SubmitLine`, `worktree`, `source`, `data`, and both closure pins match the operator record. It ran from 00:40:07 UTC for 1 hour 43 minutes 44 seconds on `h200-053-020`, `h200-140-109` and completed 0:0. Root, batch, extern, and all 29 numbered steps completed with exact task vector `2,2,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2,16,2,2,2`; the completion marker occurs exactly once and scheduler stderr is empty. Its run root is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_runs/late_angle_scale_0a87fe50a472_1847784`. All 96 rank records report eight committed updates at boundaries 839–846, terminal completion, and no global non-finite loss decision. The non-authorizing summary reports `VALID_PASS`, exact A1/A2 repeatability, valid boundary-839 scale isolation, matched topology, causal interpretation allowed, and empty criterion/evidence failure lists. Its SHA-256 is `ba980ce6045b0ec84ab31ca4b37a4ecc56414240b82388397afb19584668d438`; the 226-member artifact inventory SHA-256 is `774fb297aec553de0e0ab3e335d5f835626a99c636c387c31c71357f3501ccfc`, and the pre/post input inventories are byte-identical at SHA-256 `62ce767d73abd2b8528444db5f11215a8eb1bda5c4`. After an independent terminal audit recomputed every inventory member, fresh-node validator job 1848162 was submitted with the main nodes as its sole scheduler exclusion. It completed 0:0 in 11 minutes 46 seconds on the disjoint node `h200-178-069`. The exact nine-file validation root is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/diagnostic_runs/late_angle_scale_0a87fe50a472_1847784_independent_validation`; all members are mode-0400, single-link files, and its evidence list recomputes exactly (list-file SHA-256 `779d58728154d0e12de6315bd6196be64e697d1d5fc712858f813d647b16afc7`). The validation result is `VALID` at SHA-256 `6cc0a46b2f801bf40eaa315fad49f9e3788911bf50b0b6b1125cdcc38a96c036`; the acceptance result is `ACCEPTED` at SHA-256 `d33a913fa67915812e5f548b790fe20d24ad672fe29bf4fef422971fb`. Validator stdout is exactly the sole completion-marker line and stderr is empty. Independent replay therefore accepts S7 commit `0a87fe50a47228c90a72d233615bee422fd2fe7a` as the implementation base. The closure remains explicitly `authorizing=false`: it permits v2 implementation work, not production training.

A broader preparatory 539-test host baseline also exposed one pre-existing, platform-sensitive test-harness defect: 531 tests pass and seven skip, while `test_retention_journal_resumes_partially_deleted_generation` fails on Lustre. The test globally intercepts every directory-relative deletion of `model.pth`; because Lustre rejects `renameat2(RENAMENOREPLACE)` with `EINVAL`, the injection fires during the claim-and-copy staging cleanup before pointer commit rather than during the intended post-commit retention cleanup. The resulting ordinary checkpoint error is therefore correct; the future v2 branch must scope the injection to the retained victim directory’s device/inode and assert the new pointer before the resume check. This host-test repair is separate from the frozen S7 diagnostic source and does not alter the completed and externally validated experiment.

The completed prerequisites and diagnostics are the forward localization in job 1836996, the reviewed shared-DT-only warp-8 RMSNorm-forward control, its passing bounded smoke in job 1837393, and the passing replacement generation-zero gate in job 1837446, followed by accepted trigger capture 1838750 and independent validator 1838966, the valid matched intervention 1839531, and its fresh-node validator 1840069. The remaining implementation surfaces for the additive version are `lit_gpt/rdiag_shared_dt_kla.py`, `lit_gpt/config.py`, `lit_gpt/model.py`,

lit_gpt/rdiag_shared_dt_rmsnorm.py, lit_gpt/rdiag_shared_dt_training_health.py, lit_gpt/rdiag_checkpoint.py, pretrain.py, scripts/train/rdiag_kla/rdiag_recipe.py, and the adjacent campaign, source-gate, qualification, authorization, job-control, run-guard, supervisor, and evaluation modules. These changes were made in a new worktree based on the validator-accepted S7 commit; the frozen S7 worktree above remains untouched. That worktree was created on branch codex/rdiag-shared-dt-v2-20260906 at /lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v2/source_worktrees/rdiag_shared_dt_v2_implementation_20260906, with initial HEAD exactly 0a87fe50a47228c90a72d233615bee422fd2fe7a. The implementation plan is docs/plans/2026-09-06-rdiag-shared-dt-v2-production-plan.md in that worktree. The live namespace-aware execution ledger is /lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v2/releases/v2_b37b3c4d4205ccb04ade7000cb7f3b16c42ca547/OPERATOR_STATUS.md; its empirically corrected allocation count supersedes the static estimate frozen in the source plan. All implementation and host-verification slices are frozen in reviewed source commit b37b3c4d4205ccb04ade7000cb7f3b16c42ca547, tree 4c2b2bd8cb645d342c0e605a926ccb474c961024, and the exact pushed branch above. The immutable qualification snapshot is /lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v2/releases/v2_b37b3c4d4205ccb04ade7000cb7f3b16c42ca547; its 4290-file source manifest has SHA-256 94207969a5281d42064006e7b9b177438bfba6589eb27eb4841e78a24362821f. Independent reopening revalidated the commit, manifest digest, and complete file closure. Lustre used the documented retained-complete-read-only publication fallback, so this qualification snapshot is immutable but is not itself a production snapshot or authorization. The earlier Lustre-only checkpoint-test failure was repaired by forcing the `renameat2(RENAMENOREPLACE)` fallback, scoping the injected unlink failure to the retained victim generation's device/inode, and proving that the new pointer resolves before retention reconciliation. The complete checkpoint file then reported 81 passes and three skips; after the versioned identity integration, the combined checkpoint, generation, and pretrain suite reported 128 passes and three skips. The additive layer accepts only the exact `shared_dt_v1` and `shared_dt_v2` strings, retains the v1 scale $H^{-1/2}$, and assigns v2 the sole changed scalar $0.5H^{-1/2}$ at the existing raw-angle multiplication site before the unchanged FP32 `tanh`. Four field-identical v2 configs, apart from name and clock mode, dispatch to the same layer; tests establish identical parameters, buffers, RNG consumption, optimizer topology, and parameter counts. The RMSNorm policy, first-update health, checkpoint-run, and first-update-marker identities now have explicit v2-family schemas. Crossed model-name/mode, schema/mode, v1/v2 resume, and string-subclass aliases fail closed before Lightning payload loading, while pinned v1 canonical checkpoint and health bytes remain unchanged. Independent core/health and checkpoint/pretrain reviews both returned `APPROVED`; no file below `lit_gpt/kla_ops/` changed. Parallel v2 recipe/catalog identities, separate H200 numerical and train-smoke harnesses, all four qualification gates, six-seal aggregation, exact authorization, run guard, supervisor, dedicated v2 submitter, and full seven-mode evaluation path are implemented. Exact family maps choose one v1 or v2 runner and reject crossed schemas, model names, checkpoints, manifests, seals, and string subclasses before side effects. The existing v1 submitter and runner remain byte-identical to the accepted base.

The first full broad run on Lustre reported 1125 passes, seven skips, and three failures. Root-cause tracing found test assumptions rather than production-code defects: one source-snapshot assertion assumed atomic no-replace rename instead of the documented retained-read-only Lustre fallback, a data-manifest test requested a 1 ms mtime mutation on a whole-second-resolution mount, and a pretraining source assertion still expected the retired cross-family recipe lookup. The tests now validate both snapshot publication success dispositions, force and verify a distinct

filesystem timestamp, and require manifest-schema-to-family-to-recipe lookup. The merged regressions pass 3/3 on Lustre; the clean full rerun reports 1132 passes and three skips. All 55 changed or new Python files compile, the new JSON parses, eight shell or sbatch files pass syntax checks, `gitdiff--check` is clean, and `lit_gpt/kla_ops/` still has no diff. An independent whole-diff review found no critical or important issue and returned `READY`. The 266.7 GiB corpus/tokenizer closure was first hashed inside the Codex UID namespace into data-manifest SHA-256 `adc7ad730a1d4dfcf16721733bcfca93d70c8108d7a7ec7f6eabafd3d66e3466`. Production-H200 campaign `b76e18cce1d842618b910ed164dfa9b2`, execution digest `3cb5cab2c3b671e2532222fb3f50de4bbff445a` was submitted as one-H200 job 1849816; it failed closed in 31 seconds before the scientific test. Diagnostic job 1849835 established the exact namespace crossing: the manifest recorded data and tokenizer owner UID/GID 0, whereas the compute node correctly observed UID/GID 741838. No data bytes or source identity differed. Compute-side manifest regeneration job 1849849 completed in 44m35s and produced corrected manifest SHA-256 `47e8650e20025ce8dc3d2f261b25f1493b991b34f466e644b7eb7795659641e7`. Compute-side preparer 1850489 then completed and created fresh campaign `4234b79ab0a8440f9e1532ae0baf59c8`, execution digest `0fad65d66b54fc5d618378a1ff6c686f564f684cc32eac5683826c98639a2e53`. Its exact emitted submission became workload 1850491, which completed in 29m09s on `h200-046-019`; both numerical cases passed, and all source, runtime, cleanup, and result checks returned zero. Fresh external attestor 1850765, submitted with that workload node excluded, completed in 46m26s on the disjoint node `h200-042-061`. It re-hashed the full corpus and published the read-only production-H200 seal with SHA-256 `65fbe8abc741b2f2f08e4f3d5536e28178b161bd219c6a29b6e284a1529cc463`. Independent third-node control job 1851164 then reopened the seal on `h200-004-175` and returned `VALID_PASS`. The production-H200 gate is therefore accepted. Compute-side preparers 1851167–1851171 all completed 0:0 and emitted immutable execution manifests for the same campaign. Their exact submission vectors became train-smoke, checkpoint-fault, authenticated-resume, gradient-A/B, and two-node 3300-update-soak workloads 1851172–1851176. Train-smoke job 1851172 and checkpoint-fault job 1851173 completed 0:0 in 8m42s and 4m52s, respectively, and each published a mode-0400 passing result. Their node-excluding external attestors 1851298 and 1851291 completed 0:0 on `h200-094-088` and published the train-smoke and checkpoint-fault seals with SHA-256 `fa8812b932666aa3c654c5e724236c31ff057c844d7c71b497034156842ad830` and `3e15913d33b459ae293f1dbcd0542ede550b778deb211b2fcc3e755da60eada4`, respectively. The two-node soak 1851176 ran on `h200-053-020`, `h200-124-110`; it subsequently failed closed 1:0 after 22m33s. Its authenticated journal contains 904 sequential finite updates through iteration 1808 with zero health failures; the next update was rejected because its pre-clip global gradient exceeded the declared limit 10. The sole retained scientific artifact is that mode-0400 journal, with SHA-256 `ddefb5c6d8d15eab251ed84523f021dc19a6a5b1c7e4198eee6e5657ea0fcd0e`. Cleanup completed, but no soak result or seal was published. This is a scientific gate rejection, not a scheduler failure, and the attempt will neither be attested nor reused. Resume 1851174 and gradient-A/B 1851175 started together at 2026-09-06T09:23:02Z on `h200-094-088` and `h200-052-125`, respectively; both first reverified the immutable b37 source snapshot. Gradient-A/B completed 0:0 in 6m21s with passing result SHA-256 `68734fcfcf78dfcbecc74b179a35d0fc3a79b5cbdc5edb835107c8bc00fd007c`; its disjoint-node attestor 1851656 then completed 0:0 on `h200-053-020` and published seal SHA-256 `85ea2ced80e2e43bedff943edd2f40e5160ff8676b07bc8304c484be65a7eac9`. Authenticated resume completed 0:0 in 12m10s after both its fresh and authenticated resumed segments passed, publishing result SHA-256 `05ae105d9e81e13439cf77b2d1fdacf9a8c8f7e571e580dd01317fc1247e4f86`; its node-excluding attestor 1851744 completed 0:0 on `h200-048-224` and published seal SHA-256 `6b57da131dcc81a89f45ddcf34ac46d3e6ec930a14088bb88a475b5da54160aa`. Thus five of six

external gate seals are accepted; only a scientifically acceptable replacement soak lacks a seal. No production training job is submitted; the state remains non-authorizing.

One production-launch integration risk is now explicit. Final `prepare-fresh` must run in the namespace that observes the compute-authored numeric UID/GID data witness, while its frozen source verifier also performs an authenticated `gitls-remote`. That Git subprocess intentionally removes every proxy variable. A read-only reproduction of its exact sanitized query on the login side timed out after 35 seconds, whereas the earlier proxy-routed remote query passed. Compute-side exact-verifier probe 1851318 then failed 1:0 after 25 seconds on h200-094-088 at the sanitized authenticated `gitls-remote`, with empty stdout. Together with the login-side timeout, this closes the environment search: no tested namespace can both reobserve the compute-owned data witness and satisfy the frozen network check. The qualification workloads remain valid evidence, but production launch now requires a reviewed source correction and complete requalification; no verifier bypass is allowed. A separate correction worktree developed a short-lived Ed25519 SSHSIG witness that binds the authenticated remote ref to the exact commit, tree, source manifest, campaign, aggregate, execution manifest, and launch mode, with an immutable one-use nonce claim. Independent review first found two pre-submission races: freshness after nonce burn and binding the held claim directory to the same control-root inode through submission. The first repair passed 64 focused and 240 adjacent host tests, and an explicit-config, proxy-scoped read-only query against the exact b37 branch succeeded in pueue task 4531. A second independent review correctly rejected that revision because public-repository `gitls-remote` need not invoke the credential helper, the validated GitHub configuration pathname remained mutable before child execution, and freshness was not yet the last check at the actual `sbatch` spawn boundary. A direct one-call GitHub GraphQL prototype in pueue task 4533 did prove authenticated viewer `ngocbh` and returned the exact b37 branch, commit, and tree. The current repair therefore replaces `ls-remote` with that authenticated query, supplies it from a private stable snapshot of `hosts.yml`, isolates GitHub CLI state/cache writes in the same transient directory, and moves freshness into the final pre-spawn callback. Final independent review returned `READY`: 69 focused and 286 focused-plus-adjacent tests pass, compilation and diff checks are clean, and post-fix authenticated pueue task 4540 returned the exact pushed branch, commit, and tree without leaving a caller-directory state artifact. The correction is committed and pushed on branch `codex/rdiag-shared-dt-v2-remote-witness-20260906` at commit `eea823e3dcdcb051ff256299a196a888f39b6ea0`, tree `e5a75ea5f2fa93442795f340bc9251fc35bb1e1a`. A second reviewed commit then closed the pre-freeze and launch-control gaps: canonical per-gate submit evidence, exact argv/comment/digest rederivation, all-user exact-name scheduler checks, active-lease-bound v2 adoption, protected path lineages, aggregate-output race closure, and planner/submitter parity. That 12-file change is committed and remotely verified at `c721817db9fbf8f7dff4c48850c15be34ef45549`, tree `c6cbf2620f6124ebf130a256befa6325ff001eaf`; 122 focused/bounded/remote tests and 147 legacy-adjacent tests pass, with one expected skip. This is not a production authorization: the new source still requires an evidence-supported stability correction and a completely fresh qualification campaign rather than an amendment to b37 evidence.

The non-authorizing update-905 capture implementation is independently reviewed and frozen on branch `codex/rdiag-shared-dt-v2-soak905-diagnostic-20260906` at commit `b1b46f491f274a349728cf77255cce9a9028054e`, tree `263c9dab0e0fc4cccc9000abfba34a536801a779`. Its four diagnostic-only files passed 22 focused and 166 focused-plus-adjacent host tests; production model, pretraining, launcher, and soak gate files have zero diff from b37. The immutable 4,294-file snapshot is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v2/diagnostic_source_snapshots/soak905_b1b46f491f274a349728cf77255cce9a9028054e`, with source-manifest SHA-256 `c97683c8c23c980d4fd6bfa1a22d7c3629a8c38ecc8dbd6a99bcca50da99816d`.

It retains every boundary, the exact crossing microbatches and their pre-forward RNG states, gradient categories and top parameters, a pre-update anchor, and eight committed successors while preserving clipping at 1.0. Two-node 16-H200 job 1852006 completed 0:0 in 20m16s on h200-130-169, h200-231-137, with immutable-source verification on both nodes before and after capture. A field-by-field comparison of boundary, iteration, global pre-clip norm, health-failure count, and alpha counts found diagnostic boundaries 1-904 bit-for-bit identical to all 904 retained rows of failed production soak 1851176. Boundary 905 then reproduced the finite pre-clip norm 10.604082107543945. Clipping kept every parameter and health observation finite, but the eight-successor window rules out a harmless one-batch excursion: boundaries 908-913 crossed again and boundary 913 reached 2458.377197265625; seven of boundaries 905-913 would fail the declared limit 10. At the first crossing, reconstructed category norms were 6.36818 for the rotational-angle weights, 4.34821 for DT weights, and 7.27711 for all other parameters, with the layer-9 angle projection the largest retained parameter contribution. This is reproducible escalating training instability, not scheduler failure; whether its trigger is data-specific or v2-scale-specific is not established by this run alone.

The completed non-authorizing closure contains 141 mode-0400 inventoried files totaling 15,555,765,453 bytes. Its journal SHA-256 is 37572adf7e3594238ab87fde27318d2ca20f011dd14b7cc399e4678ade, its anchor-manifest SHA-256 is e30b4432d0e8e1b53fa41b83048be173c29dec4e5b1c2b5d8dcd2ec0bc040abf, its summary SHA-256 is 6e882b60495cfd1131365a280022156a43e8faf178fafef68fc481eca27ab1a41, and its closed inventory SHA-256 is e3276e7489e0798509077e87c4822b89e260ef372a08efbfa1d7c727575faae3. The summary is explicitly `authorizing:false` with status `completed_crossing_window`; it cannot issue a production result or seal. Exact anchor-restored v2-A1, v1-scale, and v2-A2 replay was therefore run to isolate the effect of the rotational scalar.

That replay is now implemented, independently reviewed, committed, and pushed on the same diagnostic branch at commit 016a5ab9018774fae6ddd303c946b832e2a188a6, tree 01b74f1ad623c3c58762e5f9f6066d2acbc4c638. Relative to capture commit b1b46f491f274a349728cf77255cce9 it adds exactly a replay plan, helper, isolated Slurm runner, and host test; no production source changes. The 33 combined capture/replay host tests pass, and independent review returned `READY` after an initial capture-only writer schema mismatch was repaired. Its immutable 4,298-file snapshot is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v2/diagnostic_source_snapshots/soak905_replay_016a5ab9018774fae6ddd303c946b832e2a188a6`, with source-manifest SHA-256 0abe22cf6c395902dccb41bedff97486d9d36e6babd6366a3d03e400498db6b4. External-bootstrap source verification passed, as did all 31 snapshot-applicable tests; the two deselected tests require a live Git worktree. Two-node, 16-H200 replay job 1852750 completed 0:0 in 5m22s on h200-153-168, h200-252-145, with both capture nodes excluded. Both v2 references were bit-exact to each other and to every rank of the captured boundary-905 backward, and no optimizer step was executed. At effective angle scale 0.125, the reconstructed global norm was 10.604082390132142; changing only to the older 0.25 scale raised it to 12.423034238808185, a factor of 1.17153. The corresponding angle, decay, DT-bias, DT-weight, and other category norms changed from 6.36818, 0.08549, 0.14930, 4.34821, and 7.27711 to 6.55957, 0.10176, 0.21774, 5.37835, and 9.07301. Thus the larger v1 scale causally worsens this boundary, but the present v2 scale still violates the production limit and is not acceptable. The non-authorizing summary, validation, and closed-inventory SHA-256 values are respectively 82c22cee330756b2527bbc77e6fa39a2ad40f417042ec548a16039a2cc8914ed, 1a575a7c565bbae0896402be7e6d9affb70040f9f0d6bc8de0782b4d538dd7ee, and 5ae2c8adaff6d531c4a7491923.

The next non-authorizing discriminator is implemented, reviewed, committed, and pushed, but is not a source correction. It is isolated in worktree `rdiag_shared_dt_v2_stability_screen_20260906` on branch `codex/rdiag-shared-dt-v2-stability-screen-20260906`,

based exactly on replay commit 016a5ab9018774fae6ddd303c946b832e2a188a6. Its four-file diagnostic-only delta is frozen at commit 470d3af6126ea45181c5e8b6e4bb1caa98e1ec34, tree e4b201f25748d9d81a5647c1d1f096d1f33ca22b; independent review returned READY with no Critical or Important findings. The final adjacent suite passed 43 tests with one known base-inherent deselection. The immutable 4,302-file source snapshot is `diagnostic_source_snapshots/stability_screen_470d3af6126ea45181c5e8b6e4bb1caa98e1ec34`, with source-manifest SHA-256 377efb243daf0fc86610654009a75d998833067f62b2926670f4e56b4b86546d; an independent commit-derived verifier and its 42 snapshot-applicable tests passed, with only two tests requiring a live Git worktree deselected. All 16 loader snapshots and 144 retained pre-forward RNG states reopen successfully. After an all-user exact-name duplicate check returned no rows, two-node, 16-H200 job 1853781 was submitted once at 12:28 UTC under `h200_dev`, with both capture hosts excluded independently in the submission and frozen runner. It started on `h200-046-153`, `h200-094-010` and failed 15:0 after 8m05s. All 16 ranks first reproduced boundary 905 exactly at reconstructed global norm 10.604082390132142, but that boundary uses the retained tensor bundle. On the first live successor fetch, boundary 906, global rank 1 reported that DataLoader child PID 815552 died with SIGSEGV; `--kill-on-bad-exit=1` then terminated the other ranks. Each A1 journal therefore contains exactly one valid boundary-905 row, while no boundary-906 row, rank result, summary, output validation, closed inventory, or terminal marker exists. Job 1853781 is partial non-authorizing infrastructure-failure evidence and cannot select an intervention.

The failure is localized to the deep four-worker `StatefulDataLoader`/HF-Parquet state restore, not teardown, FSDP, the captured RNG, or the boundary-905 computation. The capture used one fresh loader continuously and the exact replay used retained tensors with no loader; only the screen called `load\_state\_dict` on the saved 1,810-yield multiworker cursor. The production source itself documents that this restore path can segfault a worker. Retrying the same code or merely excluding the host is therefore rejected.

The repaired diagnostic is complete, pushed, and independently accepted at commit c0aa93389e4d28ee9021bf3e0dff573817d39afe, tree bd2d32c32e65e7cc7e059fdd869835538a3589ae, in the same worktree and branch. Its focused suite passed 19 tests; the combined scoped/adjacent suite passed 51 tests with one historical cumulative-branch-only test deselected, and both specification and quality review reported no findings. Before CUDA/NCCL, all 16 ranks now construct fresh pinned four-worker loaders, discard the first 1,808 batches, materialize batches 1,808–2,415 (boundaries 905–1208), require all 18 batches for boundaries 905–913 to match the retained capture byte-for-byte, synchronously close every worker, and seal one rank-local read-only tape. The FSDP phase independently rehashes that tape and creates no loader. Retained RNG is restored through boundary 913, while every later shared arm boundary carries cross-arm batch and RNG witnesses. Exact rank/node mapping, isolation on entry and exit, cache separation, cleanup-before-completion, and the 126-file/125-record terminal closure are all fail-closed. Its immutable 4,302-file snapshot is `diagnostic_source_snapshots/stability_screen_c0aa93389e4d28ee9021bf3e0dff573817d39afe`, with source-manifest SHA-256 fd03d23ae4b4650f11ee9db194af937ed5088b49fa154bb2e7fc2fe18f177b7a; a separate commit-derived private rematerialization verified every file and mode. The first probe allocation, 1854865, was cancelled before start after its unnecessary GPU reservation was noticed; it produced no artifacts. CPU-only H200-node replacement 1855110 started on `h200-060-199`, constructed the fresh four-worker stream, skipped 1,808 batches, and fetched the 18-batch comparison prefix, but failed closed during worker teardown before executing the retained-byte comparison. Torchdata 0.11 waits five seconds per worker, sends SIGTERM to survivors, and does not reap them afterward. Two PyArrow workers emitted “terminate

called without an active exception” and remained in kernel `do\_coredump`; because this cluster pipes dumps to `systemd-coredump`, the job was killed after 4m59 to release the node. It has only the capture-validation artifact and cannot pass the probe. The first reviewed cleanup repair was committed and pushed as `b5bfe19bf00f8a6602addc03b152cacdd327337`, tree `02207725432936cda53370311300d744342a3f86`. Its immutable 4,302-file snapshot is `diagnostic_source_snapshots/stability_screen_b5bfe19bf00f8a6602addc03b152cacdd327337`, with source-manifest SHA-256 `3f4e661d94496f5da4f38554bfd34b3792bc476149a3a78bf8e5552e2f1700fc`; an independent private rematerialization was byte-identical and 25 snapshot-applicable tests passed. Probe 1855634 then failed in one second before creating a run root because its `ad-hoc sbatch--wrap` used `Bash pipefail` under `/bin/sh`. Corrected wrapper probe 1855635 ran on `h200-128-202`, rebuilt the four-worker stream, and reached teardown, but exposed an earlier failure than the timeout fallback: `torchdata` first sends a queue sentinel, and unwinding the live `PyArrow` `parquet` readers made three workers abort before `terminate()` was reached. The probe failed closed in 1m57s, retained only the exact capture-validation document (SHA-256 `116b4cb5adf6cb20f2f6181ae118cdf701a8437caaf68557b35cd64952d28bb2`), and produced no probe result or marker. The final repair removes the iterator from `PyTorch`’s `SIGCHLD` registry while all workers are live, sends `SIGKILL` to and reaps all four, and only then lets the parent cancel/close its index and result queues directly. It never invokes `torchdata`’s graceful routine or touches the process-shared done event after a forced kill, avoiding the documented dead-child event-lock hazard. Independent review returned `READY` after 27 focused tests, a 59-test scoped suite (one historical cumulative-branch test deselected), registered-child stress, and explicit ineffective-kill fallback. The reviewed source is pushed at commit `7ea1735b3fed99b6a24e7ac07753771defdec944`, tree `2a9b111657f0d8b68a6839fbcea7a2c2a593e6b6`. Its immutable 4,302-file snapshot is `diagnostic_source_snapshots/stability_screen_7ea1735b3fed99b6a24e7ac07753771defdec944`, with source-manifest SHA-256 `4a5caf3db17da8be95660f78074f972765df624f49a8c1dd31b53c74fe8eb769`; private rematerialization was byte-identical and 26 snapshot-applicable tests passed. Fresh CPU-only H200 probe 1855869 completed 0:0 in 1m49s on `h200-126-216`. Its 18 retained batches matched, `CUDA`/`distributed` remained uninitialized, and all four worker PIDs were reaped with exit code `-9`. Its probe-record SHA-256 is `0a481a5313de98c2f48715d69b4f887997b2e1335751e4de804835e9df5cebe8`; `stderr` is empty. After an all-user exact-name conflict check, exactly one full two-node/16-H200 non-authorizing screen, job 1855985, was submitted from this snapshot. It completed 0:0 in 16m54s on `h200-017-170`, `h200-046-153`; every root and step row is `COMPLETED`. References A1 and A2 are exact to each other and to retained capture boundaries 905–913, reaching the same maximum norm 2458.377197265625. The quarter-head arm ($0.25/\sqrt{H}$, effective scale 0.0625) completed boundaries 905–1200 with zero crossings, maximum 2.5316383838653564, and nearest-rank `p99 0.5541900992393494`; it is the sole direct-promotion-eligible arm. Connected-zero rotation was also stable through 1200 (maximum 0.4557827115058899, `p99 0.3324354290962219`) but remains mechanistic-only by declaration. Half-LR crossed again at boundary 911, reached 28.53986930847168, and closed its successor window at 919, so it is rejected. The read-only closure has exactly 126 files and 125 inventory rows. Summary, validation, and inventory SHA-256 values are respectively `19f0962e04cc45b5df8d9df6e39025cecf54b48423d6e0290cd60a6180d95ec5`, `4683c0609bccbde088b9c41ad1ea32f509346826fbc255ce0c87229048699776`, and `4c9180ab8b49d5b206a5dabe56`. Independent re-execution of the immutable source, output, and inventory validators passed; all 125 inventoried payloads rehashed exactly, the terminal marker occurs once, and cleanup precedes that marker. The retained 98,660-byte screen `stderr` contains only `FSDP`/`NCCL` deprecation and process- group teardown warnings, with no traceback or error. Cleanup proof is the fail-closed runner ordering plus completed scheduler steps and marker, rather than a standalone per-node certificate. The screen remains explicitly `authorizing=false`: it selects quarter-head for a new

production-candidate source and fresh soak, not for direct launch.

This failure also exposes a separate production-readiness gap. The intended 50B job is budgeted for 70 training hours inside a 72-hour allocation and, at the measured 31.08K tokens/s/GPU on 16 GPUs, nominally needs about 27.9 hours. Nevertheless the launcher explicitly enables requeue, the production loader uses four persistent workers, and the first ordinary checkpoint cursor is iter 2,000, deeper than the failed 1,810-yield cursor. The accepted resume job 1851174 covered only a 220M, two-GPU, one-worker restore from iter 16 to iter 24; it does not test this failure class. The bounded control plane forbids HDN_SKIP_DATA_RESUME, and a worker SIGSEGV becomes terminal rather than safely requeued. Production therefore remains blocked until the loader-resume mechanism is fixed or replaced and the mandatory resume gate is upgraded to the real two-node/16-rank, four-worker HF-Parquet path at iter at least 2,000, with byte-exact successor batches against an uninterrupted control and successful post-restore optimizer progress. The diagnostic batch tape fixes D1 only; it is not a production-resume correction. The selected production design is instead a snapshot-free deterministic replay cursor: set `snapshot\_every\_n\_steps=0`, prohibit managed-production loader `state\_dict/load\_state\_dict`, checkpoint the yielded-batch count and a rolling SHA-256 chain over canonical rank-local raw batches plus the exact loader/data/runtime identities, and on resume fresh-build the same four-worker stream, replay and rehash the saved prefix, restore the saved training RNG, and continue from that iterator. Opaque v1 sidecars must fail closed rather than migrate silently. The replacement gate will use the exact 2-by-8, 750M, four-worker geometry, checkpoint at iter 4,000/update 2,000, compare the next two batches and the resulting update against an uninterrupted control in a separate allocation, deliberately fail if either loader restore API is called, and require a projected full-horizon replay time of at most eight hours. A sealed rank-local token tape remains the fallback only if deterministic replay misses that throughput gate.

Implementation is isolated from both the diagnostic branch and dirty main check-out in worktree `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v2/source_worktrees/rdiag_shared_dt_v2_replay_cursor_20260906`, branch `codex/rdiag-shared-dt-v2-replay-cursor-20260906`, based on reviewed launch-control tip `c721817db9fbf8f7dff4c48850c15be34ef45549`. The cursor/data core is committed and pushed as `4a2002b`; v2-only trainer integration is committed and pushed as `cc18784afe16ee760af596de3c24461e283a34e5`, tree `87181e1d47edbfcc27ad294bb6b659aeb8c82d1c8`. It uses one iterator for prefix replay and live continuation, validated cursor/RNG sidecars, no v2 loader state API, coordinated asymmetric failure handling, and four-worker hard-stop cleanup before any terminal checkpoint or PASS evidence. Independent review returned READY after 573 tests (three skips), a two-rank asymmetric-collective probe, and a real four-child cleanup test. A separate reviewed hardening commit `7b5808a177d03dd1526d1e0c2dc7a83a0f816217`, tree `ba0a884d48aedba42ce673bc1c7a0c089a316401`, installs an opt-in, spawn-picklable worker initializer that verifies `RLIMIT_CORE=(0,0)` and Linux dumpability zero before any PyArrow read; 158 focused/adjacent tests and a real subprocess check pass, while legacy loaders are unchanged. This is a reviewed implementation slice, not a launch authorization: the v2 control-plane/evidence schemas, uninterrupted-control gate, compute-node StatefulDataLoader/PyArrow validation, and final executable plan remain open. Its immutable 4,303-file snapshot is `diagnostic_source_snapshots/replay_cursor_7b5808a177d03dd1526d1e0c2dc7a83a0f816217`, with source-manifest SHA-256 `e657f2617e98b25b4c413120bd92b3bf59b0d636e1888f3ec85cc63a449bf445`; private commit-derived rematerialization and 101 snapshot-local tests passed.

The implementation audit also identified work that is still genuinely prospective at this tip. The RNG payload currently contains Python tuples and a NumPy array, and its sidecar reader still uses `torch.load(..., weights_only=False)`; the current v2 resume evidence class also retains

the old pending-loader-state attestation. The final implementation must replace these with the specified primitive/tensor-only RNG representation, restricted weights-only loading, and the new replay evidence object before the source can be frozen as a production candidate.

A separately reviewed, non-authorizing compute probe then exercised the actual production StatefulDataLoader/HF-Parquet path. CPU-only H200 job 1856719 completed 0:0 in 3m19s on the previously unused node h200-171-121. Two wholly fresh logical-rank-1-of-16 loaders, each with four persistent workers, consumed 1,826 batches and agreed on the boundary-1,808 cursor digest and every one of the following 18 cursor digests. All eight distinct workers reported both core limit zero and `CoreDumping:0`, were reaped at exit code `-9`, and left no visible `/proc` entry; CUDA and distributed remained uninitialized, post-run data/tokenizer identity validation passed, and stderr is empty. The slower transcript took 58,317,465,752 ns, giving the integer-ceiling full-190,734-batch projection 6,091,524,377,187 ns, or 1.6921 hours, below the proposed eight-hour gate. The mode-0400 result has SHA-256 `ab308edc368c7920cd8f338efa200819609a1e4d1887ba46b20931500dd3502c` and remains explicitly `authorizes_production=false`; it validates one-rank loader determinism, throughput, and teardown, not the required two-node training/resume equivalence. Independent post-run audit returned `VALID` after rehashing all 4,303 snapshot files, the scheduler rows and artifact closure, both transcript chains, all worker records, and the integer projection.

Plan revisions v1-v3 are retained in the worktree. Revision v4 was also rejected by independent review: it selected the invalid login-namespace data witness for compute work, put full-data attestation on the login node, omitted node-local source/cache setup for the second node, had contradictory timing cardinalities, tried to construct all four family manifests despite singleton 750M authorization, under-specified canonical recipe and detached-state mechanics, and merely printed rather than machine-asserted the final Slurm lifecycle. Revision v5 then exposed remaining compute-namespace, staged-publication, descriptor-bound loading, and cleanup-handshake gaps. Revision v6 is complete at `docs/plans/2026-09-06-rdiag-shared-dt-v2-replay-cursor-plan-v6.md`, SHA-256 `9f4fec8f3c5e208a0b33af05656283975c0c44dc103c56f8f83040391e675094`. Independent review returned `NEEDS\_REVISION`: its nominally allowed U07a-U45d tranche still rewired active v2 sidecars, evidence, runners, and seals; the v2 retirement was prose rather than an executable guard; dependencies placed publication tests before staged APIs; final verification was blocked; and the prepared-final handle, bounded cleanup acknowledgement, signal conversion, and mapped-reader lifetime were incomplete. No v6 implementation began. Revision v7 was then frozen at SHA-256 `2c2851f6e2db2ae4a84aaee0e09dccf965894156fd4b5c6a3913b14b4f7fccbe`, but independent review again returned `NEEDS\_REVISION`: retirement missed qualification preparers and occurred after some CLI side effects; it bound the legacy final-model path instead of the actual `rdiag-final-committed.json` pointer; post-rename failures were modeled as rollbackable; cleanup and detached publication lacked exact collective APIs; and several contracts/dependencies remained underspecified. No v7 implementation began. Revision v8 was frozen at SHA-256 `dcf4ea6294c901b3d4dbcb2a6362af303274b1ff034c2f134c278de95c4b0105`. It expands retirement to 25 supported v2 construction, publication, attestation, witness-generation, and submission boundaries; preserves historical read-only validation; specifies separate stable directory and closed-file identities; uses the actual `rdiag-final-committed.json` pointer; models irreversible and uncertain commit outcomes; and defines family-neutral restricted-load, Merkle-evidence, collective cleanup, and staged publication APIs. Independent review of that exact revision is still pending, so no v8 implementation has begun.

The first additive `shared\_dt\_v3` quarter-head plan was separately frozen at SHA-256 `ae51f621f5e66aa15b94a369b2cf61b3a79f4b28bcd8d3f1769ee399ba398e` and rejected by independent review. It retained a stale eleven-boundary retirement check, disagreed with v8's sidecar API/schema, omitted the required 48-rank-timing authorization calculation, placed

the final pointer directly under the output root instead of the manifest's `out\_dir`, allowed a TERM-before-KILL cleanup path, omitted exact attestor provenance and a v3 nonce-claim contract, and contained dependency inversions. Revision v2 is in progress. It must preserve v1/v2 identities, add exactly the singleton `rdiag\_kla\_shared\_dt\_v3\_750M`, and own every new model/schema/runner/seal binding before any production operation.

The completed matched screen crossed the exact warmup peak at boundary 954 with one model resident at a time and satisfied every declared comparability invariant before interpretation. Under the predeclared policy, quarter-head is therefore selected over half-LR as the production-source candidate; the radial-only result strengthens the causal attribution to the rotational path but is not itself eligible. This selection is still only a design decision. The new versioned clock must pass a fresh generation-zero 3,300-update soak and disjoint-node attestation; the diagnostic cannot replace that qualification gate.

The remaining sequence is:

1. preserve failed infrastructure run 1853781, failed cleanup probes 1855110/1855635, and pre-Python wrapper failure 1855634; preserve accepted non-authorizing screen 1855985, and implement its sole promotion-eligible quarter-head correction in a new versioned source without weakening or blindly retrying the gate;
2. complete the control-plane/evidence integration around reviewed snapshot-free replay tip `7b5808a177d03dd1526d1e0c2dc7a83a0f816217`, then run the replacement authenticated gate at exact production topology through iter 4,000, proving byte-exact successor batches, a committed post-resume update against an uninterrupted control, and the eight-hour maximum projected full-horizon replay time;
3. combine only an evidence-supported soak and resume correction with reviewed launch-control tip `c721817db9fbf8f7dff4c48850c15be34ef45549`, freeze one final production candidate, and do not aggregate the superseded b37 seals;
4. reuse only the recursively revalidated compute-authored data manifest, then repeat all six preparations, workloads, and disjoint-node attestors for the final source, run the independent derived-gates control, build a fresh six-seal aggregate, and independently reopen and recursively validate that aggregate while keeping every existing v1 canonical byte sequence and pin unchanged; and
5. generate a short-lived signed remote-ref witness for that exact aggregate and execution manifest, create the separate required production snapshot, and invoke the **prepare-fresh** mode of a new exact-family v3 submitter; the existing v1 and retired v2 submitters must not be reused. Submit only the additive singleton `rdiag_kla_shared_dt_v3_750M` with train configuration `tsz128x4k_50B`; retain the supervisor-enforced `standard`, `niah_single_1`, `niah_single_1_8k`, `niah_single_2`, `niah_single_2_8k`, `niah_single_3`, and `niah_multikey_1` evaluation modes.

Job 1835824's scientific failure invalidated the prior nine-workload lower bound; jobs 1836996, 1837446, 1838750, 1839531, and 1847039 completed the localization, replacement generation-zero, trigger-capture, matched-intervention, and first source-matched replacement-candidate workloads; S7 job 1847784 has now completed their strictly repeated six-arm diagnostic. Jobs 1846561 and 1847039 returned provisional valid passes, but validators 1846728 and 1847464 respectively failed before validation and after scientific validation but before evidence sealing; neither cycle is externally accepted. From this point, S7 sealing and the v2 production-H200 gate are complete, and the matched screen has selected quarter-head as the correction candidate. It still requires implementation, a new acceptable soak, and then a complete fresh-source campaign before production launch; all four short

b37 qualification workloads and their external attestors have already completed. Production training would follow the new campaign. Because data identities include numeric UID/GID, the final-source hard successful-path minimum is 21 compute allocations before production: six gate preparers, six scientific workloads, six fresh external attestors, one independent prepared/derived-gates control, one aggregate builder, and one independent recursive aggregate validator. The existing compute-authored data manifest is reusable only after full compute-side verification folded into the first H200 preparation; regeneration would add an allocation. A compute-side guarded-submit controller and production itself add two more, while retaining the earlier optional third-node H200-seal control adds one. On the historical b37 path, seventeen of its original 21 outcomes have completed: the manifest builder, all six preparers, the H200 workload and attestor, and the train-smoke and checkpoint-fault workloads and their attestors, plus the gradient-A/B workload and attestor and the resume workload. Its attestor is also accepted. The failed soak still leaves four original hard-minimum outcomes: its successful replacement and attestor, the aggregate builder, and the aggregate validator. At least one new soak preparation is additional to that lower bound. Including the new soak preparation and preserving the earlier independent pre-submit-control convention adds two allocations, leaving at least six outcomes before complete b37 scientific closure; retries and diagnostic allocations are additional. More importantly, the required remote-ref correction changes source identity, so the b37 seals cannot authorize that source: after review, all six preparations, all six workloads, all six attestors, the prepared-gates control, aggregate construction, and independent aggregate validation must be rerun in a fresh campaign. The compute-authored data manifest remains reusable subject to recursive revalidation. The six immutable runners contain 13 distinct `sruntime` call sites and execute 15 steps on a normal success path because the soak reuses its source-copy and cache actions during cleanup. One separate production `sbatch` follows only after aggregate acceptance. A held-out replication would increase that count. Failed attempts 1838085 and 1838250 did not satisfy the trigger-capture stage. This count excludes retries, additional discriminator arms, implementation-time GPU checks, held-out causal replication, and production requeues. Until that sequence closes, neither a production authorization nor a replacement training result is claimed.

A Reproducibility audit

RDiag implementation baseline. The sealed E2 baseline implementation statements in this note were audited against branch `rdiag` at Git commit `c3346feaecae1486cdbc71f82b4edeacc4198950` (tree `88e423d2926d8ae81945c08393a59861d285907c`). This pin identifies the sealed implementation snapshot; the present documentation-only working-tree update is intentionally outside that snapshot. Relevant regions are:

- `lit_gpt/rdiag_kla.py`, construction and buffers around lines 137–193, phase construction around 222–293, and the fused layer route around 409–516;
- `lit_gpt/kla_ops/rdiag_clock.py`, anchor and eager oracle around 72–119 and 190–270, exact fused clock around 283–390, and native-order custom backward beginning around line 515;
- `lit_gpt/kla_ops/rdiag_direct.py`, the unchanged direct-pair forward around line 162, guarded direct-pair backward around line 338, precise clock materializer around line 300, and radial wrapper around line 920;
- `lit_gpt/kla_ops/rdiag_phase.py`, negative scan contract and reference around 1–19 and 294–328;
- `lit_gpt/kla_ops/rdiag_bounded_phase.py`, bounded fail-visible phase contract around 1–11

and 41–172;

- `lit_gpt/kla_ops/rdiag_norm.py`, public fused normalization/rotation contract around 10–90;
- `lit_gpt/kla_ops/rdiag_norm_reference.py`, normalization, BF16 boundary, and adjacent-pair rotation around 24–96;
- `lit_gpt/kla_ops/diag_naive.py`, KLA equations around 14–32, 144–199, and 451–491;
- `lit_gpt/diag_kla.py`, process-noise and isotropic-output initialization around 280–300;
- `lit_gpt/config.py`, bounded RDiag recipe fields around 349–369 and 742–763;
- `lit_gpt/rdiag_training_health.py`, health thresholds and observed counters around 24–37 and 121–255, and fail-closed decision logic around 579–612;
- `lit_gpt/fsdp_strategy.py`, the FP32-buffer mixed-precision override beginning around line 131;
- `scripts/tests/rdiag/test_rdiag_bounded_h200.py`, exact radial, actual-logit, exceptional-value, full-layer parity, and A/B performance gates;
- `sanboxes/kernels/rotational_diag_kla_v1/DESIGN.md`, moving-frame semantics and the pair-tied literal model; and
- `sanboxes/kernels/rotational_diag_kla_v1/RESULTS.md`, the correctness and v1 performance evidence around lines 128–166.

The whole tree is sealed by the snapshot manifest identified below, which is the authoritative byte-level inventory. Three central files in the final parity implementation and gate have SHA-256 values

- `lit_gpt/kla_ops/rdiag_clock.py`,
d02a3f0d0994067afc174f1a77f7ebd8fced73c3a18956e72ab79ab49afc966;
- `lit_gpt/kla_ops/rdiag_direct.py`,
9fa3e29102bfa512ecbc57fc04ce6453e3a33e7aed61e7049026b71fe81faace;
- `scripts/tests/rdiag/test_rdiag_bounded_h200.py`,
68d7db9b6793a33ef2574c7a39527306c3b3b500a03e1807790ce59889be8517.

The final result followed three deliberately narrow diagnoses. Commit `fe1b1258` made the radial clock mean bitwise exact but left three full-layer angle-weight mismatches in job 1777082. Job 1777229 then proved that the first remaining forward divergence occurred at sigmoid output despite a bitwise-identical input logit. Commit `4ce6e19` matched the CUDA-eager sigmoid and angle VJP, and commit `c3346fe` added the missing-delta compile-time guard that makes the detached radial path immune to nonfinite discarded angles.

Installed Mamba implementation. The source audit used `mamba-ssm 2.3.2.post1` installed under the active `hdn` environment; those bytes match upstream tag `v2.3.2.post1` at commit `a14b1dff0454a3bc27d9eb31355dc01e4b2490ec`. Paths and SHA-256 identities are:

- `mamba_ssm/modules/mamba2.py`,
605e4439ff0baec8d8acaf4a191d9f0570eea9900065a065909124c472b08707; inspect roughly lines 95–136, 210–260, and 309–318;
- `mamba_ssm/ops/triton/ssd_combined.py`,
39c19c1e4c8e36982847079bc12b4f07ed3af03ac68f6a6653200ef80df5515d; this is the fused Mamba-2 SSD scan;

- `mamba_ssm/ops/triton/ssd_chunk_state.py`,
6d82771b2f62a7bf84c381c8ef8e4b579a95e0352021d117507a1072e4d71dcd; inspect roughly lines 40–86 and 102–174;
- `mamba_ssm/modules/ssd_minimal.py`,
6442a9fe25a6792258df657392ab1fab4dd873ee777d9dd5783cc76030218e07; this is the readable Mamba-2 SSD reference;
- `mamba_ssm/modules/mamba3.py`,
de1b104865c941eb6eeac63c3a75d5913fc2e7a119d52991590ef19c381b05ac; inspect roughly lines 76–87, 150–178, and 220–237;
- `mamba_ssm/ops/triton/mamba3/angle_dt.py`,
7d99f8cff37371cfbfe9d4dd36f17a16135fb07f2c6e14add3ad4dd5c0034e2e; inspect roughly lines 83–118 and 298–337;
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_combined.py`,
410ad57555b5841fba7bd76fdafdc3ddf82741176489c3e3493f5c6f3c0c6012; inspect roughly lines 90–105 and 247–259;
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_fwd.py`,
68d8d0dba0e3913f2e996862603289fd0d7157e7dc57c872e04234be8bb82695; inspect roughly lines 256–332, 348–352, and 386–425; and
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_bwd.py`,
aa1b0aa8f8f7950503adf55220d3feebfea2fe006dc418b5a4e75e1e6842eadc; inspect roughly lines 978–1051 and 1475–1550.

The hashes, package version, paths, and upstream commit are the durable identity; line numbers are navigation hints. Repository Mamba-3 SISO dispatch goes through the FLA wrapper. Its executable baseline uses the patched `hdn_mamba` environment; the local compatibility/decay patches do not alter the angle scan or adjacent-pair rotation formulas. The unpatched `hdn` FLA wrapper is not the runnable decay-correct baseline and was not used to infer those formulas.

Qualification evidence and present status. The lower-tail rejection is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/probe_v1_1774423/probe_trace.json`. The redesigned E2 decision is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/qualification_bounded_de_20260828_1145/D_E_DECISION.json`. The precision experiment is recorded in `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/radial_clock_precision_1776573/result.json`; focused VJP evidence is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_vjp_diag_k64_1776509.log`, and the five-case composed check is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_angle_order_1776788.log`. These artifacts support the specific debugging observations in Section 8.

The sealed implementation snapshot used by the one-H200 gate is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/snapshots/production_e2_20260828_2344`. Its `SOURCE_MANIFEST.json` SHA-256 is

05db76db18b8f4fe170a102e2e5df28bb54a4e36ca7ebe4565a1d813b44530a3.

The authoritative qualification log is `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_h200_1777777.log`. SLURM job 1777777 completed with status `COMPLETED`, exit code 0:0, 117 passing core tests, and 54 passing E2 tests. This closes the one-H200 kernel/layer gate. Distributed training-system checks and the multi-node soak/attestation remain before submission; training completion and the full post-training evaluation are subsequent empirical deliverables.